Factorization Algebras

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MA822

Abstract

MA822: Topics in Geometry and Topology was taught my Professor Matiej Szczesny in the spring 2026 semester (my junior year). The topic for this semester was an introduction to the factorization algebras of Kevin Costello and Owen Gwilliam, and we followed their book [CG16, CG20], focusing on the first volume. The course was designed to be accessible to a wide range of students, and the main prerequisites were basic topology, the theory of manifolds, and Lie theory.

Contents

Chapter 1

Introduction

Lecture 1: Syllabus day

Tue 20 Jan 2026 14:06

We will move the thursday class to 2-3:30 friday. Maybe cds424?

A factorization algebra is essentially an algebraic formulation of observables in a QFT. Examples are numerous: associative algebras, vertex algebras, $\mathbb{E}_n$-algebras over the little n-cubes operad, etc.

FAs are of course motivated by QFT, and more generally a classical field theory. A classical field theory is a space $\mathcal{E}$ of classical fields, typically the space $\Gamma(E)$ of sections of some bundle E over a manifold M, usually a fiber bundle; and an action functional $S : \mathcal{E} \to \mathbb{R}$ or $\mathbb{C}$. Write CLFT for classical field theory. In CLFT, we look for the *critical locus*

$$\mathrm{Crit}(S) = \left\{ \phi \in \mathcal{E} \mid \frac{\delta S}{\delta \phi} = 0, \mathrm{d}S|_\phi = 0 \right\}.$$

These are solved by the Euler-Lagrange equations.

For example, classical mechanics on $\mathbb{R}^n$. Take $\mathcal{E} = \mathrm{Hom}(I, \mathbb{R}^n)$ for $I = [0,1]$. The category can be any function, smooth maps, piecewise linear, etc. The action functional can be defined as follows. Given $q \in \mathcal{E}$, take

$$S(q) = \frac{1}{2} \int_0^1 \langle \dot{q}(t), \dot{q}(t) \rangle \, \mathrm{d}t = \frac{1}{2} \int_0^1 E(t) \, \mathrm{d}t,$$

where E is the kinetic energy. Take the functional derivative of S to extremize S locally. These are of course just geodesics in $\mathbb{R}^n$, meaning they satisfy

$$\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} = 0$$

since $\Gamma_{ij}^k = 0$ on $\mathbb{R}^n$. Of course these are functions of the form $at + b$, $a, b \in \mathbb{R}$ fixed. We also obtain another constraint from the IBP (see my ma721 project): $\langle \dot{q}, \delta q \rangle |_0^1 =$ for any variation δq of q. If we dont fix the endpoints, then $\delta q(0) \neq \delta q(1)$ in general. This forces $\dot{q} = 0$ at $0, 1$, and since $\dot{q} = a$ we have $a = 0$. Thus $q = b \in \mathbb{R}$ is constant. This is boring. Hence why we fix straight endpoints of q. We thus define

$$\frac{\delta S}{\delta q} = \left. \frac{\mathrm{d}}{\mathrm{d}\varepsilon} \right|_{\varepsilon = 0} S(q + \varepsilon \delta q)$$

for a variation δq of q fixing endpoints. In this case, $\mathrm{Crit}(S) \subseteq \mathcal{E}$ is a $2n$-dimensional space when $\mathcal{E}$ is functions on $\mathbb{R}^n$.

Definition 1.0.1. Define the *classical observavles* $\mathcal{O}^{cl}$ as the algebra of functions on $\mathrm{Crit}(S)$. This is a fairly informal definition, since we have not specified what types of functions we are considering. Sometimes we will consider all functions $\mathcal{S}\mathrm{et}(E, \mathbb{R})$, but since E is rarely finite-dimensional we often restrict to one of the two following important subalgebras:

- Smooth functions $C^\infty(\mathrm{Crit}(S), \mathbb{R})$;
- Polynomials in the coefficients of $\mathrm{Crit}(S)$ elements.

We have another example: scalar FT, which we will call scalar ϕ^m theory on $\mathbb{R}^n$. Take $\mathcal{E} = C^\infty(\mathbb{R}^n)$, and

$$S(\phi) = \int_{\mathbb{R}^n} \frac{1}{2}\phi\Delta\phi + \frac{g}{m!}\phi^m \,\mathrm{dVol}.$$

Here Δ is the laplacian, which exists on any Riemannian manifold. However we are integrating over something noncompact, so we will later get around this issue. Intuitively ϕ must decay sufficiently rapidly at ∞. We get around it by taking the derivative of the action. We also only consider $\delta\phi$ that are compactly supported. Thus the functional derivative wrt $\delta\phi$ always converges. Extremitization is more complex but we obtain

$$\Delta\phi + \frac{g}{(m-1)!}\phi^{m-1} = 0.$$

If $m = 1, 2$ then this is a linear PDE, and we call it a *free field theory.* Once m is higher, the ϕ^{m-1} term is nonlinear in ϕ. If $g = 0$ then ϕ is harmonic. These are the various characterizations of solutions to special cases of this PDE, and most critical loci are not finite dimensional.

Finally, just note that even if S is not well-defined, if we only vary wrt compactly supported variations then the critical locus is well-defined.

Summary of first day: A field theory is a set $\mathcal{E}$ of fields and an action functional $S : \mathcal{E} \to \mathbb{R}$. Usually $\mathcal{E}$ is the vector space $\Gamma(E)$ of a vector bundle $\pi : E \to M$ for a manifold M, generally a fiber bundle defined below. The *critical locus* $\mathrm{Crit}(S)$ is the set of extrema for S, also denoted $EL(S)$ for Euler-Lagrange. These are the equations of motion. Classical observables are then defined as the algebra of functions $\mathcal{S}\mathrm{et}(\mathrm{Crit}(S), \mathbb{R})$, smooth functions $C^\infty(\mathrm{Crit}(S), \mathbb{R})$, or polynomials in $\mathrm{Crit}(S)$. The way to interpret the final one is less formal. For an example, let $\mathcal{E} = C^\infty(I, \mathbb{R}^n)$ such that $\mathrm{Crit}(S) = \{q(t) = at + b \mid a, b \in \mathbb{R}^n\}$. Then the example is $\mathbb{R}[a^i, b^i]$ – that is, polynomials in components of the coefficients of the extrema.

Often times, $\mathcal{E}$ is not a finite-dimensional vector space, meaning to give it the structure of a smooth manifold we would need to pass to infinite-dimensional differential geometry. This is possible, but quite difficult, and we ignore this point for now, since this is not an important point.

Lecture 2: Finishing classical motivation and beginning quantum motivation

Fri 23 Jan 2026 13:59

Recall that the issue of boundary conditions vanishes by restricting to compactly supported variations.

Definition 1.0.2. Let $U \subseteq M$ open. Define $EL(U)$ as solutions to E-L eqns on U via restricting $S|U$. Define $\mathcal{O}^{cl}(U)$ to be functions on $EL(U)$.

We are intentionally vague with what "functions on $EL(U)$" means. Generally the critical locus is at least topologized since we can endow $\mathcal{E}$ with the compact-open topology and consider $EL(U)$ as a subspace. The point is that we are thinking of $\mathcal{O}^{cl}$ as some sheaf-like construction. We will then axiomatize factorization algebras as such structures.

Note that $U \subseteq V$ induces $EL(V) \to EL(U)$ via restriction, which then induces $\mathcal{O}^{cl}(U) \to \mathcal{O}^{cl}(V)$ via precomopsition. Thus $\mathcal{O}^{cl}$ is a *precosheaf.* We are thus interested in how $\mathcal{O}^{cl}$ interacts with open sets.

If $U_1, U_2 \subseteq M$ are open and disjoint, then specifying a solution to the Euler Lagrange eqns on $U_1 \cup U_2$ is the same on specifying a solution on U_1 then U_2, since the solution on one region does not impact the other. Thus we have

$$EL(U_1 \amalg U_2) \cong EL(U_1) \times EL(U_2).$$

We want to see how this changes when we take $\mathcal{O}^{cl}$. To take functions on a product $EL(U_1) \times EL(U_2) \to \mathbb{R}$, we should be able to pass to the tensor product. For example, if we consider polynomials, then we can view polynomials on U_1 as polynomials in x and polynomials on U_2 as polynomials in y. We then simply note that $\mathbb{C}[x] \otimes_{\mathbb{C}} \mathbb{C}[y] \cong \mathbb{C}[x, y]$, and similarly for other fields. There is a bit of a more rigorous argument that I will not get into. The point is that for classes of functions we are considering, we should have

$$\mathcal{O}^{cl}(U_1 \amalg U_2) \cong \mathcal{O}^{cl}(U_1) \otimes \mathcal{O}^{cl}(U_2).$$

This isomorphism need not hold in maximal generality, but there are various tools we can use to make it hold more generally, such as completing a ring with respect to a topology. This is apparently discussed further in [CG16].

We should expect $\mathcal{O}$ to pass a notion of functoriality to these tensor products, meaning that given a collection $\{U_i \subseteq M\}_{i \in n}$ of pairwise disjoint open sets and an open set $V \subseteq M$, there should be a *structure map*

$$\bigotimes_{i \in n} U_i \to V.$$

It may be useful to think of this like a functor of colored operads.

In classical theory, when $U_1 \cap U_2$ is nonempty, it is straightforward to just say we consider solutions that agree on the overlap. So we can consider overlaps. However in a quantum theory, it doesn't make sense to make meaurements at the same time (recall our manifold represents space*time*), since observables don't commute and thus measuring two things at once would not just be measuring the particle, but the other measurement would be interfering. Hence we cannot consider overlaps in quantum theory. Thus we do not have a cosheaf.

Chapter 2

From gaussian measures to factorization algebras

2.1 Gaussian integrals in finite dimensions

We now consider another motivation via Feynman's path integral and the BV formalism. Let (X, μ) a probabiliy space (measure space with a specific measure that behaves like you would expect) and random variables $\{f_i : X \to \mathbb{C}\}_{i \in r}$. The expectation value is defined by $\langle \{f_i\} \rangle = \int_X \prod_i f_i \, \mathrm{d}\mu$. For quantum mechanics, we replace classical variables with classical observables $\mathcal{E} \to \mathbb{C}$ and we rplace μ with $e^{-S(\phi)} D\phi$. This yields the path integral

$$\left\langle \left\{ \mathcal{O}_i^{cl} \right\}_{i \in r} \right\rangle \text{``}=\text{''} \int_{\mathcal{E}} \prod_{i \in r} \mathcal{O}_i(\phi) e^{-S(\phi)} \, \mathrm{D}\phi.$$

There is a problem introduced by the $\mathrm{D}\phi$ term. This should be a measure on the space $\mathcal{E}$ of fields, but such a meaure either doesn't exist, or is sufficiently poorly behaved to be wrong. For example, on topological vector spaces such a meaure is not translation invariant.

Thus it's hard (for mathematicians) to figure out what this means. We can make sense of this in the perturbative case, but physicists and mathematicians agree this is not the whole picture. This is a serious problem, because physicists use the path integral to obtain correct physical answers, so there must be some rigorous mathematical definition of the path integral. This would be quite useful to find, since computing expectation values van yield nearly all the information in quantum physics.

Let's reframe integrals in homological terms. Let M a compact orientable n-manifold (compactness not required, but simpler) without boundary. Recall the de Rham complex $(\Omega^\bullet(M), \mathrm{d})$ is truncated at n, and yields de Rham cohomology. The top cohomology $H^n_{\mathrm{dR}}(M) = \Omega^n(M)/\mathrm{d}\Omega^{n-1}(M)$ is isomorphic to $\mathbb{R}$ since it is compact. Since we integrate n-forms, we can view integration as a map

$$\int : \Omega^n(M) \to \mathbb{R}.$$

By Stokes' theorem, the kernel of $\int$ is exact forms since $\partial M = \varnothing$, so it lowers by universality to

$$\int : H^n(M) \cong \mathbb{R}.$$

The issue is when passing to infinite dimensional manifolds with $M = \mathcal{E}$, where top forms don't make sense. To make sense of this, we review another characterization of the de Rham complex in the finite case.

Recall that given a k-vector field $X \in \Gamma(TM^{\otimes k})$ and an ℓ-form $\omega \in \Omega^\ell(M)$, then the interior product $X \lrcorner\, \omega$ is defined as the $(\ell - k)$-form defined by

$$(X \lrcorner\, \omega)_p = \omega_p(X_p, -).$$

This makes sense since X_p is a k-tensor, and up to multilinearity represents k vectors. Thus $\omega(X, -)$ is a map in $\ell - k$ variables.

Recall also that a *volume form* for a Riemannian manifold is a form ω such that for every oriented orthonormal local frame $\{E_i\}_{i=1}^n$, we have $\omega(E_1, \dots, E_n) = 1$. In smooth coordinates,

$$\omega_g = \sqrt{\det g_{ij}}\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

Take $\omega \in \Omega^n(M)$ a volume form. This is nondegenerate, so the interior product $- \lrcorner\, \omega$ yields isomorhisms $\Lambda^k TM \cong \Lambda^{n-k}T^*M$. Thus contracting with ω yields by functoriality of taking sections

$$\Gamma(\Lambda^{n-k}TM) \cong \Omega^k(M).$$

We call these *polyvector fields*. We can induce a differential on $\Lambda^\bullet TM$ to promote this to a map of cochain complexes:

$$\begin{tikzcd}
\cdots \arrow[r, "\mathrm{d}"] & \Omega^{n-1}(M) \arrow[r, "\mathrm{d}"] & \Omega^n(M) \arrow[r, "\mathrm{d}"] & 0 \\
\cdots \arrow[r, "?"'] & \mathfrak{X}(M) \arrow[u, "\cong", "\lrcorner\omega"'] \arrow[r, "?"'] \arrow[d, equal] & C^\infty(M) \arrow[u, "\cong", "\lrcorner\omega"'] \arrow[r, "?"'] \arrow[d, equal] & 0 \\
\cdots & \Gamma(\Lambda^1 TM) & \Gamma(\Lambda^0 TM) &
\end{tikzcd}$$

We can compute the map $\mathfrak{X}(M) \to C^\infty(M)$ using the fact that contraction with ω is an isomorphism. Let $X \in \mathfrak{X}(M)$. Then the image of X in $\Omega^n(M)$ is $\mathrm{d}i_X\omega$. We then use Cartan's magic formula ([Lee00] 14.35)

$$\mathcal{L}_X\omega = i_X\mathrm{d}\omega + \mathrm{d}i_X\omega$$

to compute

$$\mathcal{L}_X\mathrm{d}\omega = i_X\mathrm{d}^2\omega + \mathrm{d}i_X\omega = \mathrm{d}i_X\omega.$$

Hence the image of X in $C^\infty(M)$ must contract with ω to yield $\mathcal{L}_X\omega$. Since contracting with a function is simply pointwise multiplication and ω is nonvanishing, it makes sense to divide by it, yielding the map

$$X \mapsto \frac{\mathcal{L}_X\mathrm{d}\omega}{\omega} = \frac{\mathrm{d}\mathcal{L}_X\omega}{\omega}.$$

This map turns out to be Div_ω. We call the complex the *divergence complex* and in finite dimensions it is isomorphic to the de Rham complex. The relevance is that it can pass to infinite dimensions by rephrasing integration as a map

$$\int : \Omega^n(M) \cong \Gamma(\Lambda^{n-n=0}TM) = C^\infty(M) \longrightarrow \mathbb{R}$$

and remarking that in infinite dimensions, the de Rham complex is concentrated in positive degree, whereas the divergence complex must be concentrated in negative degree. We can hope to make sense of div_ω, called the BV laplacian Δ, in ∞ dim when $M = \mathcal{E}$.

Some remarks from [CG16] that I find important. Perturbative theory corresponds to working over the ring $\mathbb{C}[[\hbar]]$ of formal power series in $\hbar$ (so like stuff is just power series). At $\hbar = 0$, we recover classical theory. In perturbative theory, the vector space $\mathcal{O}(U)$ of observables on an open set U will form a flat $\mathbb{C}[[\hbar]]$-module.

Lecture 3: The quantum motivation

Tue 27 Jan 2026 14:01

So far we have been fairly imprecise in our motivation of (pre)factorization algebras using the classical reasining for preFAs. Now we are exploring the *quantum* motivation for prefactorization algebras via the path integral. We will for now think of observables $\mathcal{O}^{cl}$ as functions on *all fields*, not just the critical locus. Recall that we want something like

$$\left\langle \mathcal{O}_1^{cl}, \ldots, \mathcal{O}_r^{cl} \right\rangle = \int_{\phi \in \mathcal{E}} \mathcal{O}_1^{cl}(\phi) \cdots \mathcal{O}_r^{cl}(\phi) e^{-\frac{1}{\hbar}S(\phi)} \, \mathrm{D}\phi.$$

Of course this should be some kind of probability measure, but it doesn't really make mathematical sense. Recall that we invoked the isomorphism of forms and polyvector fields to construct the *divergence complex*, provided we have a volume form. The maps $\Gamma(\Lambda^{n-k}TM) \to \Gamma(\Lambda^{n-k-1}TM)$ are the divergence maps. In de Rham, integration lowers to

$$\int : H^n(M) \cong \mathbb{R}.$$

Thus we can identify along the chain isomorphism to think of the projection $C^\infty(M) \to C^\infty(M)/\operatorname{Im} \operatorname{div}_\omega$ as $f \mapsto \int_M f\omega = \langle f \rangle$.

Let's derive prefactorization algebras from this. Consider the free scalar field theory $\mathcal{E} = C^\infty(\mathbb{R}^n)$,

$$S(\phi) = \int_{\mathbb{R}^n} \phi(\Delta + m^2)\phi \, \mathrm{d}^n x.$$

Although this need not make sense, recall that provided our variations are compactly supported, extremizing S does indeed make sense. Fix A a positive definite $n \times n$ symmetric matrix. Write $\mathbb{P}(\mathbb{R}^n) = \mathbb{R}[x_1, \ldots, x_n]$. Then recall a Gaussian integral is of the form $\langle - \rangle : \mathbb{P}(\mathbb{R}^n) \to \mathbb{R}$ by

$$f \mapsto \int_{\mathbb{R}^n} f(x) \exp\left(-\frac{1}{2}x^t A x\right).$$

This makes sense since A is positive definite. We claim that path integrals

$$\int_{\mathcal{E}} \mathcal{O}_1^{cl}(\phi) \cdots \mathcal{O}_r^{cl} \exp\left(\int -\frac{1}{2}\phi(\Delta + m^2)\phi \mathrm{d}^n x\right)$$

are like an infinite dimensional version of the first problem. This follows since

$$\phi \otimes \psi \mapsto \int \frac{1}{2}\phi(\Delta + m^2)\psi \, \mathrm{d}^n x$$

is positive definite and symmetric. Oh btw m is mass.

Now let's try to generalize. Consider $\mathbb{R}^n$ which is noncompact. We have $\operatorname{div}_\omega : \mathbb{P}\mathfrak{X}(\mathbb{R}^n) \to \mathbb{P}(\mathbb{R}^n)$ and then integation maps to $\mathbb{R}$. We cannot consider all of C^∞ since $\mathbb{R}^n$ is contractible so its top de Rham cohomology is 0. bla bla see picture i aint typin allat rn

Now we want to integrate over $C^\infty(\mathbb{R}^n)$ rather than just $\mathbb{R}^n$. We should start by defining polynomials on $V = C^\infty(\mathbb{R}^n)$. Given a finite vector space W, a polynomial on W is a symmetric algebra on W^*. This is since stuff like x_i are projections (lin functionals) so $x_1^2 x_2 \cdots$ something like that is a symmetric product of those functionals. Hence motivating a polynomial.

We don't want the entire algebraic dual of V, since it would be massive. It would make more sense to at least restrict to *continuous* functionals with respect to some natural topolgy on $C^\infty(\mathbb{R}^n)$, like the compact-open topology. Certaintly we want the continuous functions ev_x evaluating at x. We can rewrite them as distributions using the dirac delta (a distribution)

$$\mathrm{ev}_x f = \int_{\mathbb{R}^n} f(z)\delta(z-x)\,\mathrm{d}^n z.$$

Thus the correct analogue of polynomials here is something like the symmetric algebra on the space of distributions on $\mathbb{R}^n$. Integrating against compactly supported functions yields an inclusion of compactly supported C^∞ functions into distributions. We will restrict to those for our discussion. Delta functions and evaluation functionals are not in the image of this inclusion. We will write this as $C_c^\infty(\mathbb{R}^n)$, and note that this is dense in the space of distributions. For our purposes, distributions are defined as dual to a space of functions.

Apparently distributions generalize functions and it makes sense to consider diffeqs of distributions, so if you can't find a function that satisfies your PDE then you can look for a distribution.

We define $\mathbb{P}(C^\infty(\mathbb{R}^n)) \coloneqq \operatorname{Sym} C_c^\infty(\mathbb{R}^n)$.

Lecture 4: Formalizing Gaussian integrals over infinite dimensional manifolds

Fri 30 Jan 2026 14:03

For now we will continue to meet in psy b33 pending further room changes.

First recall some stuff from the previous lecture. A gaussian integral is of the form

$$\int_{x\in\mathbb{R}^n} \exp\left(-\frac{1}{2}\sum_{i,j} x_i A_{ij} x_j\right) \mathrm{d}^n x$$

for A positive-definite and symmetric. Here positive definite means $x^t A x > 0$ for all nonzero $x \in \mathbb{R}^n$. These integrals are analogous to our definition of the path integral in the following way. By recognizing the exponential as $-1/2x^t Ax$, by changing $x \to \phi$ and $A \to (\Delta + m^2)$, we obtain something like the free ϕ-field theory path integral. In quantum mechanics, it is standard to consider a combinatorial approach to this integral, and to generalize it to infinite dimensions. We instead focus on the divergence operator we previously defined.

Let $\omega = \exp\left(-\frac{1}{2}\sum_{i,j} x_i A_{ij} x_j\right) \mathrm{d}^n x$ be a volume form (aka a measure) on $M = \mathbb{R}^n$. Recall that we are defining the divergence operator abstractly via

$$\operatorname{Div}_\omega X\omega = \mathcal{L}_X \omega$$

for any $X \in \mathfrak{X}(M)$. Then for $X = \sum_i f_i \partial_i$, we can compute using Cartan's magic formula that

$$\mathrm{Div}_\omega X = \mathrm{Div}_\omega\left(\sum_i f_i\partial_i\right) = -\sum_{i,j} f_i A_{ij} x_j + \sum_i \partial_i f_i.$$

It's convenient to restrict only to polynomial $P(\mathbb{R}^n)$ on $\mathbb{R}^n$. By diagonalizing A, it follows that the kernel of Div_ω has codimension 1 in $P(\mathbb{R}^n)$, and thus it lowers to an isomorphism with $\mathbb{R}$, as desired. This justifies working in polynomials, since we may once again identify integration with computing the divergence operator in homology.

Now we pass to infinite dimensional manifolds. Such a measure ω no longer exists,so our task is to generalize our definition of divergence in such a way that makes sense of a "measure" of the form

$$\widetilde{\omega} = \exp\left(\int_{\mathbb{R}^n} -\frac{1}{2}\phi(\Delta + m^2)\phi\,\mathrm{d}^n x\right) \mathrm{D}\phi.$$

Taking inspiration from $\mathbb{R}^n$, we try to work in polynomials once again. It is useful also to consider the set $\mathrm{Vect}(\mathbb{R}^n)$ of *polynomial vector fields* on $\mathbb{R}^n$, meaning for $X \in \mathrm{Vect}(\mathbb{R}^n)$, there is a coordiante representation of X as

$$X = \sum_i P_i(x) \frac{\partial}{\partial x_i},$$

for $P_i \in \mathbb{R}[x_1, \dots, x_n]$ a polynomial on $\mathbb{R}^n$.

In order to begin replicating this process in the infinite dimensional case of $C^\infty(U)$, we need to define polynomials in $C^\infty(U)$. We can do this slightly more generally. Let V a vector space, and note that $C^\infty(U)$ is a vector space, hence this generalizes our situation. We cannot directly define polynomials on V, but we *can* pass to the symmetric algebra $\mathrm{Sym}(V)$. However, this is a somewhat naïve approach. A *monomial* on $\mathbb{R}^n$ is a linear functional on $\mathbb{R}^n$; it is precisely an element of $\mathcal{V}\mathrm{ect}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R})$. Hence we would expect that polynomials would instead be defined as something like $\mathrm{Sym}(V^*)$.

This is overkill for our approach. If we only want to look at $V = C^\infty(U)$, then the dual space of this can be identified with the set of *compactly supported distributions* $\mathcal{D}_c(U)$ in U. This is a bit more than we need, so we note that we can identify the set $C_c^\infty(U)$ of compactly supported smooth functions on U with a subspace of $\mathcal{D}_c(U)$ via, for all $g \in C_c^\infty(U)$, considering $g \in \mathcal{D}_c(U)$ as the linear functional

$$g(\phi) = \int_U g\phi \, \mathrm{d}^n x.$$

It can be shown $C_c^\infty(U)$ is dense in the set of distributions, so functions such as the Dirac delta admit approximations in $C_c^\infty(U)$. Hence we work with $C_c^\infty(U)$, since it is convenient for carrying out the calculus of variations.

With this understanding, a priori we might take polynomials to be $\widetilde{P}(C^\infty(U)) = \mathrm{Sym}(C_c^\infty(U))$ the symmetric algebra. This makes intuitive sense, since a monomial in n variables would be some coefficient times an n-fold product in linear functionals, hence an element of the nth symmetric power. A full polynomial would then be a finite linear combination of elements of the symmetric algebra. We will see issues with this definition, hence the tilde. However it is good motivation.

From this, we note that given V a vector space, there is an identification $T_pV \cong V$. Since a polynomial vector field is a linear combination in tangent vectors with polynomial coefficients, we might a priori take

$$\widetilde{\mathrm{Vect}}(C^\infty(U)) \coloneqq \mathrm{Sym}(C_c^\infty(U)) \otimes C^\infty(U) \cong \mathrm{Sym}(C_c^\infty(U)) \otimes T_pC^\infty(U).$$

Here we recall that $\widetilde{P}(C^\infty(U)) \coloneqq \mathrm{Sym}(C_c^\infty(U))$.

Let $\{f_i\}_{1 \le i \le r} \subseteq C_c^\infty(V)$ such that their product $\prod_i f_i$ is a monomial in $P(C^\infty(U))$. Then such a monomial acts on $C^\infty(U)$ via

$$\begin{aligned}\left(\prod_i f_i\right)(\phi) &= \int_{x \in U^r} \prod_i f_i(x_i)\phi(x_i) \, \mathrm{d}vol(x_1) \wedge \cdots \wedge \mathrm{d}vol(x_r) \\ &= \prod_i \int_U f_i \phi \, \mathrm{d}vol.\end{aligned}$$

Note that this is the same as the usual action, simply generalized in the natural way for symmetric products. Imposing linearity allows arbitrary polynomials to act on $C^\infty(U)$.

An element of $\widetilde{\mathrm{Vect}}(C^\infty(U))$ is a finite linear combination of the form $X = f_1 \cdots f_r \partial_\phi$ with $f_i, \phi \in C_c^\infty(U)$. [**What is** ∂_ϕ?] Now let $g_1 \cdots g_m$ be a monomial in $P(C^\infty(U))$. Then X can act on

$g_1 \cdots g_s$ by imposing linearity, the product rule, and applying the action of polynomials on $C^\infty(U)$:

$$f_1 \cdots f_r \frac{\partial}{\partial \phi}(g_1 \cdots g_s) = f_1 \cdots f_r \sum_{i=1}^{s} g_1 \cdots \widehat{g_i} \cdots g_s \int_U g_i \phi \, \mathrm{d}vol.$$

Definition 2.1.1. The divergence operator associated to the action functional

$$S(\phi) = \int_U \phi(\Delta + m^2)\phi \, \mathrm{d}vol$$

is the linear map

$$\widetilde{\mathrm{Div}}_\omega : \widetilde{\mathrm{Vect}}(C^\infty(U)) \to \widetilde{P}(C^\infty(U))$$

defined by, for $f_i, \phi \in C_c^\infty(U)$,

$$\widetilde{\mathrm{Div}}_\omega \left(f_1 \cdots f_r \frac{\partial}{\partial \phi} \right) = -f_1 \cdots f_r \left(\Delta + m^2 \right) \phi + \sum_{i=1}^{r} f_1 \cdots \widehat{f_i} \cdots f_r \int_U \phi f_i \, \mathrm{d}vol.$$

Now this is generally not sufficient. Later in the text, we will see an issue with making homological algebra work with topological vector spaces. It is generally difficult to do homological algebra without breaking the topological properties of the vector spaces. The solution we will employ is working with a certian category that we will not discuss in class, but is in the book. It sounds like some kind of Grothendieck construction. To work in this category, we need to consider a *completed* tensor product.

Note that there is a canonical map $C^\infty(M) \otimes C^\infty(M) \to C^\infty(M \times M)$ by $f \otimes g \mapsto f(x)g(y)$; x coordinates on the first M and y on the second. This is injective but not surjective, but the image is dense. We define $C^\infty(M \times M)$ to then be the *completed* tensor product of $C^\infty(M)$ with itself. Moreover note that C^∞ is lax monoidal.

In a similar manner, we define the *completed* symmetric power $\mathrm{Sym}^n(C^\infty(M))$ to be $C^\infty(M^n)^{S_n}$, which comes with an action of S_n, which it also injects into. This means we need to modify our previous definitions to introuce the completed tensor and symmetric power. Then

$$\mathbb{P}(C^\infty(V)) = \bigoplus_{n \geq 0} C_c^\infty(V^n)^{S_n}$$

$$\mathrm{Vect}(C^\infty(V)) = \bigoplus_{n \geq 0} C_c^\infty(V^{n+1})^{S_n}.$$

Here the $n+1$ is because we want an extra component for the vector part. So now $\widetilde{\mathrm{Div}}$ extends to the completion

$$\mathrm{Vect}_C(C^\infty(V)) \mapsto F(\Delta + m^2)_{x_{n+1}} F(x_1, \ldots, x_{n+1}) - \sum_{i=1}^{n} \int_{x_i \in U} F(x_1, \ldots, x_i, \ldots, x_n, x_i) \, \mathrm{dvol}.$$

Definition 2.1.2. Quantum observables $\mathrm{Obs}^q(U)$ on $U \subseteq M$ are defined as the cokernel of divergence:

$$\mathrm{Obs}^q(U) \coloneqq \frac{\mathbb{P}(C^\infty(U))}{\mathrm{Im}\,\mathrm{Div}} \cong \mathrm{Coker}\,\mathrm{Div}.$$

We think of quantum observables as path integrals of fields supported in U. This makes sense due to our recharacterization of integration via de Rham cohomology and the divergence complex. We now want to know what the structure of this is for different open sets $U, V \subseteq M$?

Polynomials, even in infinite dimensions, form a commutative ring. Thus $\mathbb{P}(C^\infty(U))$ has a commutative ring structure. Write $\mathbb{P}^n(C^\infty(U)) \coloneqq C_c^\infty(U^n)^{S_n}$ such that $\mathbb{P}(C^\infty(U)) = \bigoplus \mathbb{P}^n(C^\infty(U))$. Then we can define

$$\mathbb{P}^n(C^\infty(U)) \otimes \mathbb{P}^m(C^\infty(U)) \to \mathbb{P}^{n+m}(C^\infty(U)).$$

This follows by symmetrizing the product of two polynomials

$$F(x_1, \ldots, x_n) \otimes G(x_{n+1}, \ldots, x_{n+m}) \mapsto \frac{1}{|S_{m+n}| = (m+n)!} \sum_{\sigma \in S_{n+m}} F(x_{\sigma(1)}, \ldots, x_{\sigma(n)}) G(x_{\sigma(n+1)}, \ldots, x_{\sigma(n+m)}).$$

Note that under inclusions $U \subseteq V$ of open sets, a compactly supported function on U is compactly supported on V, so C_c^∞ is a cosheaf. So given $F \in P(C^\infty(U))$, we can extend it to a polynomial on $P(C^\infty(V))$.

Lemma 2.1.3 ([CG16] 2.3.1)**.** *The product map*

$$\mathbb{P}(C^\infty(U)) \otimes \mathbb{P}(C^\infty(U)) \to \mathbb{P}(C^\infty(U))$$

does not *descend to the quotient, meaning it does not induce* $\mathrm{Obs}^q(U) \otimes \mathrm{Obs}^q(U) \to \mathrm{Obs}^q(U)$*. However, if* U_1, U_2 *are disjoint open subsets of an open set* V*, then the product map*

$$\mathbb{P}(C^\infty(U_1)) \otimes \mathbb{P}(C^\infty(U_2)) \to \mathbb{P}(C^\infty(V))$$

defined by composing the inclusions into $\mathbb{P}(C^\infty(V))$ *with the product map* does *descent to such a map on quantum observables.*

Recall $\mathrm{Obs}^q(U) = \mathbb{P}(C^\infty(U))/\operatorname{Im}\mathrm{Div}$. The non-existence follows since Div need not be a ring homomorphism, so the image need not be an ideal. We complete the proof next time, showing we obtain a prefactorization structure.

Lecture 5: Concluding motivation with the divergence complex

Tue 03 Feb 2026 13:58

Proof. Suppose we have vector spaces M, N, and subspaces $K \subseteq M$ and $L \subseteq N$. Recall that $F : M \to N$ descends to $M/K \to N/L$ whenever $F(K) \subseteq L$. We apply this to the case $P(C^\infty(U_1)) \otimes P(C^\infty(U_2)) \to P(C^\infty(V))$. The tensor product admits a projection q_{12} to $\mathrm{Obs}^q(U_1) \otimes \mathrm{Obs}^q(U_2)$. We can then write

$$\operatorname{Ker} q_{12} = (\operatorname{Im}(\mathrm{Div}_{U_1}) \otimes P(C^\infty(U_2)) + (P(C^\infty(U_1)) \otimes (\operatorname{Im}\mathrm{Div}_{U_2})).$$

So we want to induce a map $\widetilde{\pi}$ in the diagram

$$\begin{tikzcd}
\operatorname{Ker} q_{12} \arrow[r, hook] & P(C^\infty(U_1)) \otimes P(C^\infty(U_2)) \arrow[r, "q_{12}"] \arrow[d, "\pi"] & \mathrm{Obs}^q(U_1) \otimes \mathrm{Obs}^q(U_2) \arrow[d, dashed, "\pi"] \\
\operatorname{Im}\mathrm{Div}_V \arrow[r, hook] & P(C^\infty(V)) \arrow[r, "q"'] & \mathrm{Obs}^q(V)
\end{tikzcd}$$

It's enough to check that π identifies the kernel into the image of $\operatorname{Im}\operatorname{Div}_V$; we may freely split the kernel at the $+$.

There is a key identity. Let X a vector field on N, f a function on N, and ω a measure. Then $\operatorname{Div}_\omega(fX) - f\operatorname{Div}_\omega(X) = Xf$. To prove this (in finite dimensions), recall that $\operatorname{Div}_\omega X = \mathcal{L}_X\omega/\omega$. Then Cartan's magic formula and the product rule yield (since ω is a top form and $\mathrm{d}\omega = 0$)

$$\begin{aligned}\operatorname{Div}_\omega X &= \frac{\mathrm{d}\iota_{fX}\omega}{\omega} - f\frac{\mathrm{d}\iota_X\omega}{\omega} = \frac{\mathrm{d}(f\iota_X\omega)}{\omega} - f\frac{\mathrm{d}\iota_X\omega}{\omega}\\ &= \frac{\mathrm{d}f\cdot\iota_X\omega}{\omega} + \frac{f\cdot\mathrm{d}\iota_X\omega}{\omega} - f\frac{\mathrm{d}\iota_X\omega}{\omega} = \frac{\mathrm{d}f\cdot\iota_X\omega}{\omega}.\end{aligned}$$

Passing to coordinates, let $\omega = g(x_i)\mathrm{d}x^i$ and $X = h_j\partial_j$. Then

$$\iota_X\omega = \sum_{j=1}^n (-1)^j h_j g \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^j} \wedge \cdots \wedge \mathrm{d}x^n.$$

Then

$$\mathrm{d}f \wedge \iota_X\omega = \sum h_j g \frac{\partial f}{\partial x_j}\mathrm{d}^n x.$$

Then $\mathrm{d}f\cdot\iota_X\omega/\omega$ is simply $\sum_j h_j\partial_j f = X(f)$. This proof used finite dimensions, but the statement still follows for infinite dimensional manifolds. Hence if $X \in \operatorname{Vect}(C^\infty(U))$ and $F \in P(C^\infty(U))$, then $\operatorname{Div}(FX) - F\operatorname{Div}(X) = X(F)$.

We will prove the statement in the dense subspaces $\widetilde{P}$ and $\widetilde{\operatorname{Vect}}$ since it is moer clear why this follows. Take $F = f_1\cdots f_r$ and $X = g_1\cdots g_s\frac{\partial}{\partial\phi}$, with $f_i \in C_c^\infty(U_1)$ and $g_i, \phi \in C_c^\infty(U_2)$. Then

$$X(F) = \sum_{j=1}^r g_1\cdots g_s f_1\cdots\widehat{f_j}\cdots f_r \int_V f_j\phi\,\mathrm{d}vol.$$

But since f_j, ϕ have disjoint supports, the integral vanishes. Thus $X(F) = 0$, showing that $\operatorname{Div}(FX) = F\operatorname{Div}(X)$. This implies that if something is a divergence then it all is???

If divergence were an ideal in general, then Xf would always be a divergence, which need not hold. ■

This gives Obs^q the structure of a precosheaf, together with maps

$$\{U_i \subseteq V\}_{1\le i\le r} \text{ disjoint}: \qquad \bigotimes_{i=1}^r \mathrm{Obs}^q(U_i) \to V.$$

Hence Obs^q is a prefactorization algebra.

In physics, there is a parameter $\hbar$ which determines the strength of quantum effects. When $\hbar = 0$, the system is classical, but for nonzero $\hbar$ (as is reality) we see quantum effects. In finite dimensions (purely for illustration), we can take a Gaussian measure

$$e^{-S/\hbar}\mathrm{d}^n x.$$

Then

$$\operatorname{Div}_\omega\left(\sum_i f_i\partial_i\right) = -\frac{1}{\hbar}\sum_{i=1}^n f_i\frac{\partial S}{\partial x_i} + \sum_{i=1}^n \frac{\partial f_i}{\partial x_i}.$$

Hence polynomials will simply be polynomials mod the image of this divergence, and when $\hbar \neq 0$, this is the same as the image of $\hbar\,\mathrm{Div}$. This allows us to clear the denominator and take $\hbar \to 0$:

$$\mathrm{Div}_\omega\left(\sum_i f_i\partial_i\right) = -\sum_{i=1}^n f_i\frac{\partial S}{\partial x_i} + \hbar\sum_{i=1}^n \frac{\partial f_i}{\partial x_i} \xrightarrow{\hbar\to 0} -\sum_{i=1}^n f_i\frac{\partial S}{\partial x_i}.$$

The set of all $-\sum_i f_i\partial_i S$ is the ideal of $P(\mathbb{R}^n)$ generated by partials of S. Hence in classical mechanics in finite dimenisons, quantum observables go to

$$P(\mathbb{R}^n)/\left\langle \partial_i S\right\rangle_{1\leq i\leq n}.$$

This is called the Jacobian ideal. Recall that although quantum mechanics is about path integrals (polynomials modulo divergences), classical mechanics was about critical loci. Note that by setting $\partial_i S = 0$, we obtain only elements of the critical locus of S. Hence $P(\mathbb{R}^n)/\left\langle \partial_i S\right\rangle$ is polynomial functions on the critical locus of S, meaning we recover $\mathcal{O}^{cl}(\mathrm{Crit}(S))$. The same follows for infinite dimensional free field theories. I will not explain it because I'm tired of typing, but the various analogies of the infinite-dimensional setting to Gaussian measures should make the analogue clear.

This is a very simple example, and oftentimes what we will get is not very nice. Later on we will improve our construction by taking a differential graded object as a resolution of obserables, usually a chain complex, whose zeroth cohomology is observables. In the next chapter, we will properly define prefactorization algebras, and show that there is a family $\mathcal{O}_\hbar$ of flat prefactorization algebras over $\mathbb{C}[\hbar]$, such that when $\hbar = 1$ we get quantum observables, and in $h = 0$ we get classical observables.

Our permanent friday room (except one weird day in March) is CDS 548. Hopefully we have

Chapter 3

Prefactorization algebras and examples

3.1 Prefactorization algebras

Lecture 6: Prefactorization algebras

Fri 06 Feb 2026 14:05

Definition 3.1.1. A prefactorization algebra on a topological space X (usually a manifold) valued in vector spaces is a functor $\mathcal{F} : \mathcal{O}\mathrm{pen}(X) \to \mathcal{V}\mathrm{ect}_k$ such that for any collection $\{U_i \subseteq V\}_{1 \le i \le n} \subseteq \mathcal{O}\mathrm{pen}(X)$ of pairwise disjoint open sets, where V is open, there is a map

$$\bigotimes_{i=1}^{n} \mathcal{F}(U_i) \to \mathcal{F}(V)$$

which are compatible in the obvious way: let $\left\{U_i^j \subseteq V^j \subseteq W\right\}_{1 \le i \le n}^{1 \le j \le m}$ be collections of opens. Then the diagram

$$\begin{tikzcd}
\bigotimes_{i=1}^{n} \bigotimes_{j=1}^{m} \mathcal{F}(U_i^j) \arrow[rr] \arrow[dr] & & \bigotimes_{j=1}^{m} \mathcal{F}(V^j) \arrow[dl] \\
& W &
\end{tikzcd}$$

commutes.

Example 3.1.2. Any (unital) associative algebra A yields a prefactorization algebra on $\mathbb{R}$ denoted A^{fact}. Note that any open subset of the reals is at most a *countable* union of open intervals (use countability and density of $\mathbb{Q}$ to prove). To each open interval (a, b), define $A^{fact}(a, b) = A$ with compatibility on intervals given by the identity. To any open set $U = \amalg_j I_j$, we want to set $A^{fact}(U) = \bigotimes_j A_j$, where $A_j = A$. The structure maps are simply multiplication. The issue is when the tensor is countable, this need not be defined. To do this, we require that in a countable tensor, all but finitely many elements of each tensor are the unit 1 of the algebra. Thus multiplication only needs to be performed finitely many times. The compatibility condition comes from the strict associativity of the multiplication maps. Since A need not be commutative, the ordering of the intervals in $\mathbb{R}$ yields the ordering of our elements. Raising the dimension of $\mathbb{R}^n$ can yield the notion of an $\mathbb{E}_n$ algebra here.

We can also consider the other direction, but we need to require our factorization algebra be locally constant. In our example, inclusions of smaller intervals into bigger ones yield the identity map, but not all prefactorization algebras have this property. For example taking the symmetric algebra of a precosheaf does not yield this: $\mathrm{Sym}(\mathcal{F}(U_1) \oplus \mathcal{F}(U_2)) \cong \mathrm{Sym}(\mathcal{F}(U_1)) \otimes \mathrm{Sym}(\mathcal{F}(U_2))$.

Definition 3.1.3. Let $\mathcal{F}$ be a prefactorization algebra on a manifold X with values in an ∞-category or model category. Then $\mathcal{F}$ is *locally constant* if inclusions $D \subseteq D'$ of open disks which are bijections on connected components induce equivalences $\mathcal{F}(D) \to \mathcal{F}(D')$.

Lemma 3.1.4. *Let $\mathcal{F}$ be a locally constant prefactorization algebra on $\mathbb{R}$ valued in $\mathcal{V}\mathrm{ect}_k$. Set $A := \mathcal{F}(\mathbb{R})$. Then $\mathcal{F} \cong A^{fact}$.*

Proof. Let $I = (a, b) \subseteq \mathbb{R}$ an interval. Then $\mathcal{F}(I) \to \mathcal{F}(\mathbb{R})$ is an isomorphism. Hence all open intervals map under $\mathcal{F}$ to be isomorphic to A. We define a multiplication map for A by taking intervals $(a, b), (c, d) \subseteq I$ and considering the structure map $m : \mathcal{F}(a, b) \otimes \mathcal{F}(c, d) \to \mathcal{F}(I) \cong A$. To show A is associative, we simply apply the compatibility condition for 3 intervals. ■

We can see how prefactorization algebras can give rise to much less trivial algebras.

Lecture 7: Further examples and a crash course in Lie theory

Tue 10 Feb 2026 14:00

MA822 will be on thursday this week from 2-3:15, and the room is still to be found; probably CDS 424.

Remark 3.1.5. If we instead take preFAs valued in chain complexes and define a locally constant preFA to preserve weak equivalences, then we get A_∞-algebras.

Example 3.1.6. Let A, B be R-modules and M an (A, B)-bimodule, meaning it has compatible left- and right-actions by A and B, respectively. We can encode this data into a prefactorization algebra as follows. Fix $p \in \mathbb{R}$, and let U_1 denote any interval contained fully within $\mathbb{R}_{<p}$; U_2 denote any interval contained within $\mathbb{R}_{>p}$; and U_3 any interval containing p. Define $\mathcal{F}(U_1) = A$, $\mathcal{F}(U_2) = B$, and $\mathcal{F}(U_3) = M$. Let V be any interval containing the relevant subsets. The map $\mathcal{F}(U_1) \otimes_R \mathcal{F}(U_2) \to V$ is the A-multipliction $A \otimes_R M \to M$. Similarly, $\mathcal{F}(U_2) \otimes_R \mathcal{F}(U_3) \to V$ is the action $M \otimes_R B \to M$.

Note that if V contains U_1 and p, then we get a map $A \to M$. We take this map to be A-linear, and thus fully determined by how it maps $1 \mapsto m_0$. This yields M the structure of a *pointed* bimodule, so the question is how to choose the base point m_0. We use the empty set $\varnothing \subseteq V$ to define a map $R \to M$ which chooses the basepoint. We similarly get a basepoint determined by the B-action; this must agree with that of A.

All other relevant maps are the identity. Coherence arises via compatibility and associativity properties. This is a special case of a *stratified* factorization algebra. In physics, this is a simple example of a *defect*: some notion of two different theories (like different potentials) that are somehow coupled at some point p.

Remark 3.1.7. The prefactorization algebra axioms imply that $\mathcal{F}(\varnothing)$ is a unital commutative algebra.

Final example. Let A, B, C unital R-algebras, M an (A, B)-bimodule, and N a (B, C)-bimodule. We can do the same thing with $M \sim p$ and $N \sim q$, but we need interaction between M and N. If $p, q \in V$, then we take $\mathcal{F}(V) = M \otimes_B N$. Here recall that algebras are rings together with an action of another ring, so this makes sense. There exist maps $M \hookrightarrow M \otimes_B N$ and $N \hookrightarrow M \otimes_B N$ by making use of the pointed-ness of M, N and mapping $m \mapsto m \otimes n_0$ and $n \mapsto m_0 \otimes n$. Most generally, $(a, m, b, n, c) \mapsto amb \otimes nc$; the b can travel between the two tensors since we tensor over $B \ni b$.

3.2 A crash course in Lie theory

Definition 3.2.1. Let $\mathfrak{g}$ a Lie $\mathbb{K}$-algebra. It's universal enveloping algebra $U\mathfrak{g}$ is an associative algebra which is universal in the following sense: let A be a (unital) associative $\mathbb{K}$-algebra and $\mathfrak{g} \to A$ a Lie algebra homomorphism, where A is viewed as a Lie $\mathbb{K}$-algebra by taking the bracket to be the commutator. Then there is a unique unital algebra homomorphism $U\mathfrak{g} \to A$ making the relevant diagram commute.

Remark 3.2.2. Let $\mathrm{Lie} : \mathcal{A}\mathrm{lg}_{\mathbb{K}} \to \mathrm{Lie}_{\mathbb{K}}$ be the functor which views associative $\mathbb{K}$-algebras as Lie algebras by taking the Lie bracket to be the commutator. Let $U : \mathrm{Lie}_{\mathbb{K}} \to \mathcal{A}\mathrm{lg}_{\mathbb{K}}$ be the universal enveloping algebra. Then by construction, $U \dashv \mathrm{Lie}$.

Construction 3.2.3. We construct U explicitly using the tensor algebra $T(\mathfrak{g})$:

$$\mathcal{U}(\mathfrak{g}) = \frac{T(\mathfrak{g})}{\langle x \otimes y - y \otimes x - [x, y] \rangle}.$$

In particular, $U\mathfrak{g}$ is the free associative $\mathbb{K}$-algebra on a Lie $\mathbb{K}$-algebra which respects the Lie bracket, and contains $\mathfrak{g}$ in a universal way.

Definition 3.2.4. Let $\mathfrak{g}$ be a Lie $\mathbb{K}$-algebra. A $\mathfrak{g}$-module is a $\mathbb{K}$-module M together with a $\mathbb{K}$-linear map $\mathfrak{g} \otimes_{\mathbb{K}} M \to M$ such that $[x, y]m = x(ym) - y(xm)$.

Remark 3.2.5. A $\mathfrak{g}$-module M is precisely a $U\mathfrak{g}$-module, viewed as a module over an algebra, where we define $(x_1 \dots x_k)m = x_1(\dots(x_k m)\dots)$. In particular, we obtain an isomorphism $\mathcal{M}\mathrm{od}_{U\mathfrak{g}} \cong \mathcal{M}\mathrm{od}_{\mathfrak{g}}$. Hence $\mathcal{M}\mathrm{od}_{\mathfrak{g}}$ is an abelian category with enough projectives and injectives, so we can do homological algebra.

Definition 3.2.6. Let $\mathfrak{g}$ be a Lie $\mathbb{K}$-algebra. Define the *invariants* $M^{\mathfrak{g}}$ of a $\mathfrak{g}$-module M as the largest subspace which is annihilated by $\mathfrak{g}$:

$$M^{\mathfrak{g}} := \{m \in M \mid xm = 0 \forall x \in \mathfrak{g}\}.$$

Define the *coinvariants* $M_{\mathfrak{g}}$ to be the smallest quotient which identifies the $\mathfrak{g}$-action to 0:

$$M_{\mathfrak{g}} := M/\mathfrak{g}M.$$

The invariants and coinvariants determine functors. We can rephrame them in terms of the universal enveloping algebra to show they are not exact.

Remark 3.2.7. We regard $\mathbb{K}$ as a trivial $\mathfrak{g}$-module. Hence a $\mathfrak{g}$-linear map $f : \mathbb{K} \to M$ must first off be $\mathbb{K}$-linear, meaning it is fully determined by the image of 1; but since it must be $\mathfrak{g}$-linear, we must have $xf(1) = f(x1) = 0$ due to the trivial $\mathfrak{g}$-action on $\mathbb{K}$. Hence $M^{\mathfrak{g}} \cong \mathrm{Hom}_{\mathfrak{g}}(\mathbb{K}, M) \cong \mathrm{Hom}_{U\mathfrak{g}}(\mathbb{K}, M)$. Thus the invariants are only left exact.

Remark 3.2.8. We now claim that $M_{\mathfrak{g}} \cong \mathbb{K} \otimes_{U\mathfrak{g}} M$. We do this by showing that $-^{\mathfrak{g}} \dashv -_{\mathfrak{g}}$ and using uniqueness fo adjoint functors up to natural isomorphism. A $\mathfrak{g}$-linear map $f : M/\mathfrak{g}M \to N$ must fall fully into $N^{\mathfrak{g}}$ since $M/\mathfrak{g}M$ has trivial $\mathfrak{g}$-action. Hence we can define a map $M \to N^{\mathfrak{g}}$ by precomopsing f by the projection $M \to M/\mathfrak{g}M$. For the converse, let $f : M \to N^{\mathfrak{g}}$. Then $f(\mathfrak{g}M) = \mathfrak{g}f(M) \subseteq \mathfrak{g}N^{\mathfrak{g}} = 0$, so f factors uniquely through the quotient. Uniqueness of adjunction shows that $-_{\mathfrak{g}} \cong \mathbb{K} \otimes_{U\mathfrak{g}} -$.

Definition 3.2.9. We define Lie $\mathbb{K}$-algebra cohomology $H^n_{\mathrm{Lie}}(\mathfrak{g}, M) = R^n \mathrm{Hom}_{\mathcal{U}(\mathfrak{g})}(\mathbb{K}, M) = \mathrm{Ext}^n_{U\mathfrak{g}}(\mathbb{K}, M)$. We similarly define Lie $\mathbb{K}$-algebra homology $H^{\mathrm{Lie}}_n(\mathfrak{g}, M) = L_n(\mathbb{K} \otimes_{U\mathfrak{g}} M) = \mathrm{Tor}^{U\mathfrak{g}}_n(\mathbb{K}, M)$.

Lecture 8: Lol brian taught class tday

Thu 12 Feb 2026 14:01

To compute the cohomology $H^i_{\mathrm{Lie}}(\mathfrak{g}, M)$, we need to compute $\mathrm{Ext}^n_{U\mathfrak{g}}(\mathbb{K}, M)$. Since $\mathbb{K}$ never changes, it might be easiest to find a free resolution of $\mathbb{K}$ and then to tensor with M over $U\mathfrak{g}$ and take the homology of that. We proceed by doing just this using the Chevalley-Eilenberg resolution. To do this, we work with dg Lie algebras.

3.3 dg Lie $\mathbb{K}$-algebras

Recall that a graded $\mathbb{K}$-vector space is a collection $\{V^i\}_{i\in\mathbb{Z}}$ of $\mathbb{K}$-vector spaces. It is differential graded if it has a differential $d^i : V^i \to V^{i+1}$ for all i with $d^2 = 0$. The graded tensor product of graded $\mathbb{K}$-modules is

$$M \otimes_{\mathbb{K}} N \coloneqq \bigoplus_{i+j=k} M^i \otimes_{\mathbb{K}} N^j.$$

Definition 3.3.1 (dg Lie $\mathbb{K}$-algebra)**.** A differential graded Lie $\mathbb{K}$-algebra is a differential graded $\mathbb{K}$-vector space $\mathfrak{g}$ together with a bracket $[-,-] : \mathfrak{g} \otimes_{\mathbb{K}} \mathfrak{g} \to \mathfrak{g}$ such that

- The bracket respects degrees:
$$[\mathfrak{g}_i, \mathfrak{g}_j] \subseteq \mathfrak{g}_{i+j};$$
- the bracket satisfies *graded anticommutivity*:
$$[x, y] = -(-1)^{|x||y|}[y, x];$$
- the differential is a derivation for the bracket:
$$d[x, y] = [dx, y] + (-1)^{|x|}[x, dy];$$
- the bracket satisfies the graded Jacobi identity:
$$[x, [y, z]] = [[x, y], z] + (-1)^{|x||y|}[y, [x, z].$$

Here if $x \in \mathfrak{g}$ is a homogeneous element, then $|x|$ is the *degree* of x, meaning the number i such that $x \in \mathfrak{g}_i$.

Example 3.3.2. Any ordinary Lie $\mathbb{K}$-algebra can be viewed as a dg Lie $\mathbb{K}$-algebra concentrated in degree 0. More interestingly, given a Lie $\mathbb{K}$-algebra $\mathfrak{g}$, define the *de Rham* dg Lie $\mathbb{K}$-algebra $\mathfrak{g}_{\mathrm{dR}}$ which is concentrated in degrees 0 and -1, and has $\mathfrak{g} = \mathfrak{g}^0_{\mathrm{dR}} = \mathfrak{g}^{-1}_{\mathrm{dR}}$. Let $x, y \in \mathfrak{g}^0$ and $x', y' \in \mathfrak{g}^1$. Then the bracket is defined as

$$[x + x', y + y']_{\mathfrak{g}_{\mathrm{dR}}} = [x, y]_{\mathfrak{g}} + [x, y']_{\mathfrak{g}} + [x', y]_{\mathfrak{g}}.$$

The differential is given by the identity $1 : \mathfrak{g}^{-1}_{\mathrm{dR}} \cong \mathfrak{g}^0_{\mathrm{dR}}$.

Remark 3.3.3. There is a quasi-isomorphism $\mathfrak{g}_{\mathrm{dR}} \simeq 0$.

Definition 3.3.4. A *differential graded commutative $\mathbb{K}$-algebra A* is a differential graded $\mathbb{K}$-module which is

- a graded $\mathbb{K}$-algebra, meaning it has a multiplication map $C \otimes_{\mathbb{K}} C \to C$ which satisfies $A^i A^j \subseteq A^{i+j}$ and graded commutivity:

$$xy = (-1)^{|x||y|} yx;$$

- a differential graded associative $\mathbb{K}$-algebra, meaning that the differential acts as a derivation:

$$d(xy) = d(x)y + (-1)^{|x|} x d(y).$$

Remark 3.3.5. Let A be a dg commutative $\mathbb{K}$-algebra. Then A is a dg Lie $\mathbb{K}$-algebra by taking the Lie bracket to be a sort of graded commutator. Define

$$[a, b] := ab - (-1)^{|a||b|} ba.$$

We show this satisfies the axioms. Graded antisymmetry:

$$[b, a] = ba - (-1)^{|a||b|} ab = -(-1)^{|||b|} \left(ab - (-1)^{|a||b|} ba \right) = (-1)^{|a||b|} [a, b].$$

The graded Jacobi identity is more than we would like to type, so we omit it. Compatibility with the differential:

$$\begin{aligned} d[a, b] = d(ab) - (-1)^{|a||b|} d(ba) &= d(a)b + (-1)^{|a|} a d(b) - (-1)^{|a||b|} \left(d(b)a + (-1)^{|b|} b d(a) \right) \\ &= \left(d(a)b - (-1)^{(|a|+1)|b|} b d(a) \right) + (-1)^{|a|} \left(a d(b) - (-1)^{|a|(|b|+1)} d(b) a \right) \\ = \left(d(a)b - (-1)^{|da||b|} b(da) \right) &+ (-1)^{|a|} \left(a d(b) - (-1)^{|a||db|} d(b) a \right) = [da, b] + (-1)^{|a|} [a, db]. \end{aligned}$$

Here we used that $|a| = |a|^2$. Given any dg commutative algebra A, we denote the corresponding dg Lie $\mathbb{K}$-algebra by $\mathrm{Lie}(A)$. Note that Lie is a functor $\mathcal{DG}\mathcal{A}\mathrm{lg}_{\mathbb{K}} \to \mathcal{DG}\mathrm{Lie}_{\mathbb{K}}$.

Definition 3.3.6. Let $\mathfrak{g}$ be a dg Lie $\mathbb{K}$-algebra and A a dg commutative $\mathbb{K}$-algebra. Then the *universal enveloping algebra* $U\mathfrak{g}$ is universal in the sense that there are isomorphisms

$$\mathcal{DG}\mathrm{Lie}_{\mathbb{K}}(\mathfrak{g}, \mathrm{Lie}(A)) \cong \mathcal{DG}\mathcal{A}\mathrm{lg}_{\mathbb{K}}(U\mathfrak{g}, A)$$

natural in $\mathfrak{g}$ and A. By construction, $U \dashv \mathrm{Lie}$.

Definition 3.3.7. Let M be a dg $\mathbb{K}$-module. Then the *graded symmetric* $\mathbb{K}$*-algebra* is the free graded commutative $\mathbb{K}$-algebra on M, and is given by formal symmetric products of elements in M. Explicitly, if $T(M)$ is the tensor algebra,

$$\mathrm{Sym}(M) := \frac{T(M)}{\langle x \otimes y - y \otimes x \rangle}.$$

Remark 3.3.8. The symmetric algebra of a dg $\mathbb{K}$-module is canonically a dg commutative $\mathbb{K}$-algebra with the following differential. First, the tensor algebra $T(M)$ on a dg $\mathbb{K}$-module (M, d) is a dg associative $\mathbb{K}$-algebra with differential

$$d(v_1 \otimes \cdots \otimes v_n) = \sum_{i=1}^{n} (-1)^{|v_1| + \cdots + |v_{i-1}|} v_1 \otimes \cdots \otimes dv_i \otimes \cdots \otimes v_n.$$

An annoying check shows that the differential indeed preserves the quotient, meaning we obtain a differential on $\mathrm{Sym}(M)$ given by

$$d(v_1 \cdots v_n) = \sum_{i=1}^{n} (-1)^{|v_1| + \cdots + |v_{i-1}|} v_1 \cdots dv_i \cdots v_n.$$

Theorem 3.3.9 (PBW)**.** *Let $\mathfrak{g}$ be a dg Lie algebra. Then $U\mathfrak{g} \cong \mathrm{Sym}(\mathfrak{g})$.*

Lemma 3.3.10. *$\mathcal{U}(\mathfrak{g}_{\mathrm{dR}})$ is a free $\mathcal{U}(\mathfrak{g})$-module.*

Proof. The universal enveloping algebra is monoidal $(\mathcal{M}\mathrm{od}_{\mathfrak{g}}, \oplus) \to (\mathcal{M}\mathrm{od}_{\mathcal{U}(\mathfrak{g})}, \otimes_{\mathbb{C}})$. Additionally, $\mathcal{U}(\mathfrak{g} \oplus \mathfrak{h}) \cong \mathcal{U}(\mathfrak{g}) \otimes_{\mathbb{C}} \mathcal{U}(\mathfrak{h})$ is free (take a basis for $\mathfrak{h}$). Apply this to $\mathfrak{g}_{\mathrm{dR}}$ to obtain via PBW that

$$\mathcal{U}(\mathfrak{g}_{\mathrm{dR}}) \cong \mathcal{U}(\mathfrak{g}) \otimes_{\mathbb{C}} \mathrm{Sym}(\mathfrak{g}[1])$$

as $\mathcal{U}(\mathfrak{g})$-modules. This proves it. ■

Since $\mathcal{U}(\mathfrak{g}_{\mathrm{dR}}) \simeq \mathbb{C}$, we have a free resolution as a dg Lie algebra. Hence

$$\mathrm{Ext}^i_{\mathcal{U}(\mathfrak{g})}(\mathbb{C}, M) \cong H^i\left(\mathcal{M}\mathrm{od}_{\mathcal{U}(\mathfrak{g})}(\mathcal{U}(\mathfrak{g}_{\mathrm{dR}}), M)\right).$$

If A is a free $\mathbb{C}$-algebra, then $\mathcal{M}\mathrm{od}_A(A \otimes F, M) \cong \mathcal{M}\mathrm{od}_{\mathbb{C}}(F, M)$. Applying this here yields

$$\mathcal{M}\mathrm{od}_{\mathcal{U}(\mathfrak{g})}(\mathcal{U}(\mathfrak{g}_{\mathrm{dR}}), M) \cong \mathcal{M}\mathrm{od}_{\mathcal{U}(\mathfrak{g})}(\mathcal{U}(\mathfrak{g}) \otimes_{\mathbb{C}} \mathrm{Sym}(\mathfrak{g}[1]), M) \cong \mathcal{V}\mathrm{ect}_{\mathbb{C}}(\mathrm{Sym}(\mathfrak{g}[1]), M).$$

We often call this complex $C^\bullet(\mathfrak{g}; M)$. In degree 0, then this is $\mathcal{V}\mathrm{ect}_{\mathbb{C}}(\mathbb{C}, M) \cong M$. In degree 1, we take the degree -1 generators $\mathcal{V}\mathrm{ect}_{\mathbb{C}}(\mathfrak{g}, M)$. In degree 2, its $\mathcal{V}\mathrm{ect}_{\mathbb{C}}(\Lambda^2\mathfrak{g}, M)$. This generalizes.

$$C^i(\mathfrak{g}; M) = \mathcal{V}\mathrm{ect}_{\mathbb{C}}(\Lambda^i\mathfrak{g}, M).$$

Cheveleakslky-Eilenberg's theorem si that if G is a connected compact Lie group, then the cohomology of the invariants of the de Rham complex $\Omega(G)^G$ are that of $C^\bullet(\mathfrak{g}; \mathbb{C})$. More powerfully, $\Omega(G)^G \cong C^\bullet(\mathfrak{g}; \mathbb{C})$. This follows because your tangent bundle for Lie groups is trivial so ur forms are constant coefficient. Hence we obtain a differential on $C^\bullet(\mathfrak{g}; \mathbb{C})$ via the exterior derivative.

We claim for any M, then $C^\bullet(\mathfrak{g}; M)$ is a $C^\bullet(\mathfrak{g}; \mathbb{C})$ module. Take $C^\bullet(\mathfrak{g}; M) \cong C^\bullet(\mathfrak{g}) \otimes M$. We may now define the differential. For trivial coefficients, d being a derivation implies we need only define it on generators:

$$C^\bullet(\mathfrak{g}) \cong \mathrm{Sym}(\mathfrak{g}[1]^*) = \Lambda^\bullet\mathfrak{g}^*$$

has it as $0 : \mathbb{C} \to \mathfrak{g}$, and it suffices now only to define it on $\mathfrak{g}^* \to \Lambda^2\mathfrak{g}^*$ since all others are generated on it. We define this as the adjunct of $[-,-] : \Lambda^2\mathfrak{g} \to \mathfrak{g}$. Check wikipedia for a full explicit definition.

For $C^\bullet(\mathfrak{g}; M)$, we only need it on degree 0 since it is a $C^\bullet(\mathfrak{g})$-module. The map $M \to \mathrm{Hom}(\mathfrak{g}, M)$ is given by $m \mapsto (x \mapsto xm)$. Wikipedia also computes the first few cohomology groups. H2 corresponds to "central extensions" modulo some sort of isomorphism classes. These are extensions of $\mathfrak{g}$ as a Lie algebra by $\mathbb{C}$, meaning Lie algebras $\mathfrak{h}$ which extend $\mathfrak{g}$ in the sense taht

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathfrak{h} \longrightarrow \mathfrak{g} \longrightarrow 0$$

is exact. Short exact sequences of vector spaces always split, so $\mathfrak{h} \cong \mathfrak{g} \oplus \mathbb{C}$. Thus its bracket has to be $[x, y] + \varphi(x, y)c$ for some basis $c \in \mathbb{C}$ and some $\varphi : \Lambda^2\mathfrak{g} \to \mathbb{C}$. What in all of the fucks is happening????? review these notes later. see wikipedia page on Lie group cohomology, nLab page on dg Lie algebras, [CG16] A.3, and check the paper they cite at the end and apparently Lurie treats this for ∞-categories somewhere so maybe I can find that.

Lecture 9: sdfgh

Fri 20 Feb 2026 14:01

Now we want to return to our goal of constructing $\mathcal{U}(\mathfrak{g})$ via prefactorization algebras. Might be useful for reviewing notes: $\mathcal{U}$ adjoint implies it commutes with $\oplus$, so $\mathcal{U}(\mathfrak{g})$ is graded if $\mathfrak{g}$ is graded.

Recall the *homology* is computed by $C^\bullet(\mathfrak{g}, M) \cong \mathrm{Sym}(\mathfrak{g}[1]) \otimes M$ (dualize $\mathfrak{g}$ and $1 \to -1$ for cohomology), with differential $d_{\mathrm{CE}} + d_{\mathfrak{g}} + d_M$, which acts by (up to the grading signs)

$$\begin{aligned} d(x_1 \cdot x_2 \cdots x_r \cdot m) &= \sum (-1)^{?}[x_i, x_j] x_1 \cdots \hat{x}_i \cdots \hat{x}_j \cdots x_r m \\ &+ (-1)^{?} \sum_{r=1}^{n} x_1 \cdots \hat{x}_i \cdots x_r (x_i m) \\ &+ (-1)^{?} \sum_{i=1}^{r} x_1 \cdots d_{\mathfrak{g}}(x_i) \cdots x_r m \\ &+ (-1)^{?} x_1 \cdots x_r d_M(m). \end{aligned}$$

Remark 3.3.11. Replaing normal commutativity with Kozkul's sign rulle yields

$$\mathrm{Sym}(V) = \mathbb{T}(V) / \left\langle x \otimes y - (-1)^{|x||y|} y \otimes x \right\rangle.$$

Here $\mathbb{T}(V)$ is the tensor algebra.

For example, if V is concentrated in odd degree, then it becomes the exterior algebra. If concentrated in even degree, we get the usual symmetric algebra. The gading takes care of all relevant variations on the rules.

Definition 3.3.12. A differential graded algebra (DGA), (A, d_A) is a graded algebra such that

$$d_A(a \cdot b) = d_A(a)b + (-1)^{|a|} a \cdot d_A(b).$$

For example, de Rham's complex. In fact, de Rham's complex is *commutative* in the graded sense (sometimes called supercommutative???) meaning it is commutative up to Kozkul's sign rule.

Typically we will have $M = \mathbb{C}$. Then we write $C^\bullet(\mathfrak{g})$ and $C_\bullet(\mathfrak{g})$ for simplicity, and note that d_M has no contribution to the boundary morphism.

Warning 3.3.13. Although *cohomology* $C^\bullet(\mathfrak{g})$ is a DGA, *homology* $C_\bullet(\mathfrak{g})$ is NOT a DGA. It would be a "differential graded coalgebra".

Definition 3.3.14. Let $(\mathfrak{g}, d_{\mathfrak{g}})$ a dg Lie algebra. A Maurev-Cartan element in $\mathfrak{g}$ is some degree 1 $\alpha \in \mathfrak{g}_1$ such that

$$d_{\mathfrak{g}}(\alpha) + \frac{1}{2}[\alpha, \alpha] = 0.$$

Some kind of proof exists that like formal neighborhoods of a space in some moduli space are equivalent as ∞-categories to some category of dg Lie algebras. In field theory, we work with the moduli space $EL(S)$ of solutions to the Euler Lagrange equations. Hence we can locally model these solutions, up to ∞-categorical equivalence, by dg Lie algebras via taking the Maurev-Cartan locus or smt.

Proposition 3.3.15. *Let $(\mathfrak{g}, d_{\mathfrak{g}})$ a dgla. Then the (not Lie!) cohomology $H^\bullet(\mathfrak{g})$ is a graded Lie algebra.*

Proof. Let $\alpha \in H^i(\mathfrak{g})$ and $\beta \in H^j(\mathfrak{g})$. Then $[\alpha, \beta]$ by choosing a representative, computing the Lie bracket, and passing to cohomology. Note that the choice of representatives are closed (kernel mod image so everything in kernel). We need to check that the bracket of representatives is closed. This is immediate:

$$d[\widetilde{\alpha}, \widetilde{\beta}] = [d\widetilde{\alpha}, \widetilde{\beta}] + (-1)^{|\widetilde{\alpha}|}[\widetilde{\alpha}, d\widetilde{\beta}] = [0, \widetilde{\beta}] + (-1)^{|\alpha|}[\widetilde{\alpha}, 0] = 0.$$

Now we shw it is well-defined. This is easy to check using that the elements r closed. ■

Let $U \subseteq X = \mathbb{R}$. Consider the precosheaf of dglas

$$U \mapsto (\mathfrak{g} \otimes \Omega_c^\bullet(U), d_{\mathrm{dR}})$$

given by the compactly supported de Rham complex. Call this $\mathfrak{g}^{\mathbb{R}}(U)$. Ordinary de Rham cohomology is singular cohomology, and $\mathbb{R}^n$ is contractible so $H^i(\mathbb{R}) = \delta_{i0}\mathbb{R}$. Unfortunately, compactly supported cohomology is *not* homotopy invariant, so it turns out we will get

$$H_c^i(\mathbb{R}^n) = \delta_{in}\mathbb{R}.$$

Since $\mathfrak{g}^{\mathbb{R}}$ is just a cosheaf, we get

$$\mathfrak{g}^{\mathbb{R}}(U_1 \amalg U_2) \cong \mathfrak{g}^{\mathbb{R}}(U_1) \oplus \mathfrak{g}^{\mathbb{R}}(U_2).$$

If we take the symmetric algebra on that, it becomes a tensor product. So $U \mapsto C_\bullet(\mathfrak{g}^{\mathbb{R}}(U))$ is a prefactorization algebra in cochain complexes.

Proposition 3.3.16 (3.4.1)**.** *Let $\mathcal{H}$ denote the cohomology pFA of $C_\bullet(\mathfrak{g}^{\mathbb{R}})$ by assigning*

$$\mathcal{H}(U) = H^\bullet(C_\bullet(\mathfrak{g}^{\mathbb{R}}(U))).$$

This is a pFA valued in graded vector spaces. Then $\mathcal{H}$ is locally constant (corresponding to a graded associative algebra). The corresponding graded algebra is isomorphic to $\mathcal{U}(\mathfrak{g})$.

Proof. Let $I \hookrightarrow J$ an inclusion of open intervals. Then there is a quasi-isomorphism $\mathfrak{g} \otimes \Omega_c^\bullet(I) \simeq \mathfrak{g} \otimes \Omega_c^\bullet(J)$. Note that quasi-isomorphic dglas have quasi-isomorphic chevelay-eilenberg complexes (simple lemma). In particular the functor $\mathfrak{g} \mapsto C_\bullet(\mathfrak{g})$ is a homotopic functor. Hence there is $C_\bullet(\mathfrak{g}^{\mathbb{R}}(I)) \simeq C_\bullet(\mathfrak{g}^{\mathbb{R}}(J))$, yielding an isomorphism in cohomology. ■

Lecture 10: More constructing $\mathcal{U}(g)$

Fri 27 Feb 2026 14:01

The complex $\mathfrak{g}^{\mathbb{R}}$ is quasi-isomorphic to the abelian dgla consisting of $\mathfrak{g}$ in degree 1, and 0 elsewhere. Since the differential is 0 and the bracket is 0 (thats what we mean by abelian), both $d_{\mathfrak{g}}$ and d_{CE} are 0. Hence

$$C_\bullet(\mathfrak{g}^{\mathbb{R}}(U)) \cong \mathrm{Sym}(\mathfrak{g}[-1][1], d = 0) = \mathrm{Sym}(\mathfrak{g}).$$

Since $\mathfrak{g}$ lives in degree 0, this is a true symmetric algebra, giving us an associative algebra, which we call $A_{\mathfrak{g}}$. PBW says there is an increasing filtration on $\mathcal{U}(\mathfrak{g})$ and multiplication preserves this filtration. Hence $\mathcal{U}(\mathfrak{g}) \cong \mathrm{Sym}(\mathfrak{g})$ as vector spaces. We now only check the multiplicative structure. We construct a linear isomorphism $\phi : \mathfrak{g} \to A_{\mathfrak{g}}$ and check that it is a map of Lie algebras, using the commutator for $A_{\mathfrak{g}}$. Transposing under $\mathcal{U} \dashv \mathrm{Lie}$ yields a map of *algebras* $\mathcal{U}(\mathfrak{g}) \to A_{\mathfrak{g}}$.

Pick a bump function ε_i centered at $i \in \{\pm 1, 0\}$ with support $B_\delta(i)$, $\delta \ll \frac{1}{2}$, and integral 1. Let ε any compactly supported bump function with $\int_{\mathbb{R}} \varepsilon \mathrm{d}x = 1$. Define $\phi : \mathfrak{g} \to A_{\mathfrak{g}} \cong H^\bullet(C_\bullet(\mathfrak{g}^{\mathbb{R}}(\mathbb{R})))$ by $X \mapsto [X \otimes \varepsilon \mathrm{d}x]$. Since there are no brackets to take and since it is closed, this is a well-defined cohomology class under the differential $d_{\mathrm{CE}} + d_{\mathrm{dR}}$.

Recall multiplication in $A_{\mathfrak{g}}$ is done by choosing disjoint intervals I, J. Pick $\alpha, \beta \in A_{\mathfrak{g}}$, hence $\alpha \in \mathcal{H}(I), \beta \in \mathcal{H}(J)$. Then $\alpha\beta$ is the image of $\alpha \otimes \beta$ under the factorization structure map $\mathcal{H}(I) \otimes \mathcal{H}(J) \to \mathcal{H}(\mathbb{R}) = A_{\mathfrak{g}}$.

Now pick $X, Y \in \mathfrak{g}$. Then we pick compactly supported bump functions to write $\phi(X)\phi(Y) = [X \otimes \varepsilon_0 \mathrm{d}x][Y \otimes \varepsilon_1 \mathrm{d}x]$. Note that we choose the element multiplied on the left to be supported in an interval on the left. Thus $\phi(Y)\phi(X) = [Y \otimes \varepsilon_{-1}\mathrm{d}x][X \otimes \varepsilon_0 \mathrm{d}x]$. We also have

$$\phi[X, Y] = [[X, Y] \otimes \varepsilon_0 \mathrm{d}x].$$

We show that ϕ preserves the bracket: in $\mathcal{H}(\mathbb{R})$,

$$\phi(X)\phi(Y) - \phi(Y)\phi(X) = \phi[X, Y].$$

Remark 3.3.17. Since the CE *chain* complex is not a dg algebra, meaning the differential is not compatible with the multiplicative structure, just because multiplication commutes in the symmetric algebra does not mean that multiplication is the same in cohomology. Hence mutliplication of cohomology classes is not cohomology of mutliplication.

Note that

$$\int_{\mathbb{R}} \varepsilon_{-1}\mathrm{d}x - \varepsilon_1 \mathrm{d}x = 0$$

since they are compactly supported. Then there is $h \in \Omega^0_c(\mathbb{R})$ with $\mathrm{d}h = \varepsilon_{-1}\mathrm{d}x - \varepsilon_1\mathrm{d}x$, since in $\mathbb{R}$ exact forms are precisely $\mathrm{Ker} \int$ since $\mathbb{R} \simeq *$. Now consider

$$(d_{\mathrm{CE}} + d_{\mathrm{dR}})(X \otimes \varepsilon_0 \mathrm{d}x)(Y \otimes h).$$

I think this is the product minus the bracket under ϕ. The CE differntial gives $[X, Y] \otimes h\varepsilon_0 \mathrm{d}x$, and sufficiently close to 0 we have $h\varepsilon_0 = \varepsilon_0$. Finally, de Rham kills $\varepsilon_0 \mathrm{d}x$ and maps $h \mapsto \mathrm{d}h$, so we get

$$= [X, Y] \otimes \varepsilon_0 \mathrm{d}x + (X \otimes \varepsilon_0 \mathrm{d}x)(Y \otimes \mathrm{d}h) = [X, Y] \otimes \varepsilon_0 \mathrm{d}x + (X \otimes \varepsilon_0 \mathrm{d}x)(Y \otimes \varepsilon_1 \mathrm{d}x - Y \otimes \varepsilon_{-1}\mathrm{d}x).$$

Working out the algebra and using the fact that we may interchange things in Sym, this reduces to $\phi[X, Y] - \phi(X)\phi(Y) - \phi(Y)\phi(X)$. Since this is in the image of the differential, it vanishes in cohomology, so we are done.

Remark 3.3.18. Although the 1-forms are in degree 1 in de Rham, CE shifts degree by 1 so these objects do indeed commute, rather than anticommute.

3.4 (Pre)factorization envelopes

A sheaf $\mathcal{F}$ on a smooth manifold M is *fine* if it admits a partition of unity; i.e. for a locally finite cover $\{U_i\}_{i \in I}$ of M, there exist endomorphisms $\{\psi_i\}_{i \in I}$ of $\mathcal{F}$ such that $\sum_i \psi_i = 1$ and each ψ_i is supported in U_i. In particular, for any section $s \in \Gamma(U, \mathcal{F})$ there is a collection of sections $s_i \in U_i$ supported in U_i with $s = \sum_i s_i = \sum_i \psi_i(s)(?)$. For example, all smooth vector bundles over a smooth manifold.

Suppose we have a fine sheaf $\mathcal{L}$ on M valued in dglas, and $\mathcal{L}_c$ the cosheaf of its compactly supported sections. Fine-ness implies that $\mathcal{L}_c(U_1 \amalg U_2) = \mathcal{L}_c(U_1) \oplus \mathcal{L}_c(U_2)$. Define the pFA $\mathbb{U}\mathcal{L}$ as follows: for all open $V \subseteq M$, define

$$\mathbb{U}\mathcal{L} := C_\bullet(\mathcal{L}_c(V)).$$

Since Sym coherently takes $\oplus \to \otimes$, this is a preFA.

Examples: de Rham - $\mathcal{L}^{\mathrm{dR}}_{M,\mathfrak{g}} = (\mathfrak{g} \otimes \Omega^\bullet(M), d_{\mathrm{dR}})$. Dolbeaux ($X$ complex manifold) - $\mathcal{L}^{\mathrm{Dol}}_{X,\mathfrak{g}} = (\mathfrak{g} \otimes \Omega^{0,\bullet}_X, \overline{\partial}_X)$ (holomorphic evelope but not holomorpic sections - those aren't fine).

Recall in physics that we care about states up to global phase, and in particular up to scaling by normalization. Hence physics is really a projective space. Let $\mathfrak{g}$ a Lie algebra and $\mathfrak{h} \subseteq \mathfrak{g}$ a (Lie) ideal. For example, the center. The center of $\mathfrak{gl}(n, \mathbb{C})$, for example, is $\mathbb{C}1$. We can quotient by ideals in Lie$\mathcal{A}$lg.

A *central extension* of a Lie algebra $\mathfrak{g}$ is a Lie algebra $\hat{\mathfrak{g}}$ such that $\mathfrak{g} \cong \hat{\mathfrak{g}}/\mathfrak{a}$, for $\mathfrak{a}$ an abelian ideal contained in the center $\mathfrak{a} \subseteq Z(\hat{\mathfrak{g}})$. Equivalently, it is a short exact sequence

$$0 \longrightarrow \mathfrak{g} \longrightarrow \hat{\mathfrak{g}} \longrightarrow \mathfrak{a} \longrightarrow 0$$

with $\mathfrak{a}$ an abelian ideal.

Claim 3.4.1. *Central extensions of $\mathfrak{g}$ by $\mathfrak{a}$ are classified by a certain Lie algebra cohomology group $H^2(\mathfrak{g}, \mathfrak{a})$, where $\mathfrak{a}$ is the trivial module. I.e., every central extension of $\mathfrak{g}$ is described by a 2-cocycle $\Lambda^2\mathfrak{g} \to \mathfrak{a}$.*

Proof. Consider a central extension

$$0 \longrightarrow \mathfrak{g} \longrightarrow \hat{\mathfrak{g}} \longrightarrow \mathfrak{a} \longrightarrow 0.$$

Choose a split as vector spaces (i.e. a section j of the projection). Let $\omega \in \Lambda^2\mathfrak{g}^* \otimes \mathfrak{a} = \mathrm{Hom}(\Lambda^2\mathfrak{g}, \mathfrak{a})$ as

$$\omega(X, Y) = [j(X), j(Y)] - j[X, Y].$$

If j were a Lie algebra homomorphism, $\omega = 0$. But it need not be. We can show this cocycle vanishes in cohomology iff j is a split as a sequence of Lie algebras.

For the converse, let $\omega : \Lambda^2\mathfrak{g} \to \mathfrak{a}$ which dies in d_{CE}. Define $\hat{\mathfrak{g}}$ as $\mathfrak{g} \oplus \mathfrak{a}$ as vector spaces, with bracket

$$[(x, a), (y, b)] = ([x, y], \omega(x, y)).$$

∎

Lecture 11: Prefactorization envelopes (pFE's)

Tue 03 Mar 2026 14:01

Let $\mathcal{L}$ be a fine sheaf on a manifold M of dgla's, and $\mathcal{L}_c$ the corresponding cosheaf of compactly supported sections. The factorization envelope $\mathbb{U}\mathcal{L}$ is defined as the prefactorization algebra mapping $U \subseteq M$ to Chevelay chains $C_\bullet(\mathcal{L}_c(U))$. This is a preFA in cochain complexes, and the cohomology of this (which is the dgla homology) yields a preFA $H^\bullet(C_\bullet(\mathcal{L}_c(U)))$ in vector spaces.

It's often useful to consider *twisted* variants of this construction corresponding to central extensions of $\mathcal{L}$. Recall given a Lie algebra $\mathfrak{g}$, a central extension of $\mathfrak{g}$ is a Lie algebra $\hat{\mathfrak{g}}$ with a short exact sequence

$$0 \longrightarrow \mathfrak{a} \longrightarrow \hat{\mathfrak{g}} \longrightarrow \mathfrak{g} \longrightarrow 0,$$

where $\mathfrak{a} \subseteq Z(\hat{\mathfrak{g}})$ is an ideal contained inside the center. There is a correspondence of central extensions with $H^2(\mathfrak{g}, \mathfrak{a})$. Given a splitting of this sequence $j : \mathfrak{g} \to \hat{\mathfrak{g}}$, we define $\omega(x, y) = [j(x), j(y)] - j([x, y])$. We can show $d_{\mathrm{CE}}\omega = 0$. We can also show that if we choose a different splitting, then ω changes only by a coboundary. Write another split as $j' = j + \alpha$, then

$$\omega'(x, y) = [j(x) + \alpha(x), j(y) + \alpha(y)] - j[x, y] - \alpha[x, y].$$

We must have $\alpha : \mathfrak{g} \to \mathfrak{a}$, so $\alpha(x), \alpha(y)$ fall in the center and don't affect the bracket. So it is just

$$\omega' = [j, j] - j[\cdot, \cdot] - \alpha[\cdot, \cdot].$$

We see then that α is a coboundary since $x \wedge y \to \alpha[x, y]$ is $d_{\mathrm{CE}}\alpha$ in $C^\bullet(\mathfrak{g}, \mathfrak{a})$ apparently. So the cohomology class of ω is well-defined.

Example 3.4.2. Let $\mathfrak{g} = \mathbb{C}^2$ an abelian Lie algebra, and consider the 1-dimensional central extensions

$$0 \longrightarrow \mathbb{C} \longrightarrow \hat{\mathfrak{g}} \longrightarrow \mathbb{C}^2 \longrightarrow 0.$$

Then $H^2(\mathfrak{g}, \mathbb{C})$ looks like

$$0 \longrightarrow \mathfrak{g}^* \longrightarrow \Lambda^2\mathfrak{g}^* \longrightarrow \Lambda^3\mathfrak{g}^* \longrightarrow \cdots .$$

Given $\phi \in \Lambda^n\mathfrak{g}^*$ and $\{x_i\}_{1 \le i \le n+1} \subseteq \mathfrak{g}$, the differential is

$$(d_{\mathrm{CE}}\phi)(x_1 \wedge \cdots \wedge x_{n+1}) = \sum_{i,j} \pm\phi([x_i, x_j], x_1, \cdots, \hat{x}_i, \cdots, \hat{x}_j, \cdots, x_n).$$

The $\pm$ is determined by the parity of i, j. Since the brackets are all 0, $d_{\mathrm{CE}}\phi = 0$, so the homology is simply $H^2(\mathfrak{g}, \mathbb{C}) \cong \Lambda^2\mathfrak{g}^*$. Since this is 1-dimensional over $\mathbb{C}$ since $\mathfrak{g}^*$ is 2-dimensional, there is a unique central extension up to scaling. Thus $\hat{\mathfrak{g}} \cong \mathfrak{g} \oplus \mathbb{C}k$ for some $k \in \mathbb{C}$, and $\omega(x, y) = k$ (here x, y the coordinates of $\mathbb{C}^2$). There is then one bracket $[x, y] = k$. This is the *Heisenberg Lie algebra* in 3 dimensions. These can be thought of as upper-triangular 3 dimensional matrices of the form

$$\begin{pmatrix} 0 & x & k \\ 0 & 0 & y \\ 0 & 0 & 0 \end{pmatrix}.$$

Example 3.4.3 (Affine Ka(c?z?idk)-Moody algebras)**.** Let $\mathfrak{g}$ be a simple Lie $\mathbb{C}$-algebra. If A is a commutative $\mathbb{C}$-algebra, then $\mathfrak{g} \otimes_{\mathbb{C}} A$ is a Lie algebra $[x \otimes a, y \otimes b] = [x, y] \otimes ab$. This may fail in noncommutative rings due to failiure of Jacobi's identity. Take A either Laurent polynomials $\mathbb{C}[t^{-1}, t]$ or Laurent series $\mathbb{C}((t))$. In the first case, the *Loop algebra* is defined as

$$L\mathfrak{g} = \mathfrak{g} \otimes \mathbb{C}[t^{-1}, t] \cong \mathfrak{g}[t^{-1}, t].$$

If we identify t with $e^{i\theta}$ ($\mathbb{C}$-polynomials in z are polynomials in $e^{i\theta}$), then we may think of this as $C^\infty(S^1, \mathfrak{g})$, justifying the name.

Consider the cocycle $\langle -, - \rangle_{\mathfrak{g}}$, a symmetric $\mathfrak{g}$-invariant form (so all multiples of the killing form). We will define

$$\omega(x \otimes f(t), y \otimes g(t)) = \langle x, y \rangle_{\mathfrak{g}} \operatorname{Res}_{t=0} f \mathrm{d}g.$$

Skew-symmetry follows since the residue of $f\mathrm{d}g$ is negative of that of $g\mathrm{d}f$; easy to check. We claim $\omega \in H^2(\mathfrak{g}, \mathbb{C})$, and

$$\hat{\mathfrak{g}} = L\mathfrak{g} \otimes \mathbb{C}k,$$

$$[x \otimes f(t), y \otimes g(t)] = [x, y] \otimes f(t)g(t) + \omega(x \otimes f, y \otimes g)k.$$

Example 3.4.4 (Virasoro algebra)**.** Consider the Witt algebra of algebraic vector fields on $\mathbb{C}^*$; equivalently as vector fields on S^1. This is $\mathbb{C}[t, t^{-1}]\partial_t$. Then

$$[f\partial_t, g\partial_t] = (fg' - gf')\partial_t.$$

We present the Witt algebra as span $\left\{L_n = -t^{n+1}\partial_t\right\}$ with

$$[L_m, L_n] = (m-n)L_{m+n}.$$

Then $H^2(\mathrm{Witt}, \mathbb{C}) \cong \mathbb{C}$ has an interesting cocycle

$$\omega(L_m, L_n) = \frac{1}{12}(m^2 - m)\delta_{m(-n)}.$$

In string theory, we consider things called world sheets, and we are interested in gluing their boundaries (eg. spheres). Note that $\mathrm{Lie}(\mathrm{Diff}(S^1)) \cong \mathfrak{X}(S^1)$ – this should hold for all manifolds. The Witt algebra should act on this since it looks similar. QFT takes place in a complex Hilbert space $\mathcal{H}$, but the physical quantities only depend on the projective space $\mathbb{P}(\mathcal{H})$ since they don't depend on global phase. So the symmetries of a quantum system are actually $\mathrm{Aut}(\mathbb{P}(\mathcal{H})) =: \mathrm{PGL}(\mathcal{H})$, and we can write this as a central extension of groups

$$0 \longrightarrow \mathbb{C}^* \longrightarrow \mathrm{GL}(\mathcal{H}) \longrightarrow \mathrm{PGL}(\mathcal{H}) \longrightarrow 0.$$

Differentiating this yields a central extension of Lie algebras. A symmetry group G on a QFT is then a morphism $G \to \mathrm{PGL}(\mathcal{H})$. Since projective geometry is harder than working in $\mathcal{H}$, we would like to lift $G \to \mathrm{PGL}(\mathcal{H})$ to $\mathrm{GL}(\mathcal{H})$ as indicated by the dotted arrow:

$$\begin{tikzcd}
0 \arrow[r] & \mathbb{C}^* \arrow[r] & \mathrm{GL}(\mathcal{H}) \arrow[r] & \mathrm{PGL}(\mathcal{H}) \arrow[r] & 0 \\
0 \arrow[r, dashed] & \mathbb{C}^* \arrow[u, equal] \arrow[r, dashed] & \hat{G} \arrow[u, dashed] \arrow[r, dashed] & G \arrow[u] \arrow[ul, dotted] \arrow[r, dashed] & 0
\end{tikzcd}$$

This dotted arrow need not always exist, but we can always find the dashed arrows to obtain a central extension, allowing us to consider $\hat{G}$ instead. Differentiating this does the same thing for Lie algebras ($\mathrm{d} = T_1$). Interesting paper by Dan Freed about stuff like this for mroe physics and it uses categories.

Ok back to pFE's. Let $\mathcal{L} : \mathcal{O}\mathrm{pen}(M)^{\mathrm{op}} \to \mathcal{DG}\mathrm{Lie}$ a fine sheaf. An r-shifted central extension of $\mathcal{L}_c$is a cosheaf of dgla's $\widehat{\mathcal{L}}_c$ lifting into an exact sequence of dgla's

$$0 \longrightarrow \mathbb{C}[r] \longrightarrow \widehat{\mathcal{L}}_c \longrightarrow \mathcal{L}_c \longrightarrow 0.$$

In this case we can form the *twisted envelope* $\mathbb{U}\widetilde{\mathcal{L}} : U \to C_\bullet(\widehat{\mathcal{L}}_c(U))$.

For example, consider

$$\mathcal{L} = \mathfrak{g} \otimes \Omega^\bullet_{\mathbb{R}} = \mathfrak{g}^{\mathbb{R}}.$$

Pick a cocycle $H^2(\mathfrak{g}, \mathbb{C}) \ni \omega : \mathfrak{g} \otimes \mathfrak{g} \to \mathfrak{g}$ and write

$$\widehat{\mathcal{L}} = \mathfrak{g} \otimes \Omega^\bullet_{\mathbb{R}} \oplus \mathbb{C}[?]k.$$

To figure out what the shift is by, consider the thing we want the bracket to be

$$[x \otimes \alpha, y \otimes \beta] = [x, y] \otimes (\alpha \wedge \beta) + \left(\int_{\mathbb{R}} \omega(x, y)\alpha \wedge \beta\right) k.$$

One simply figures out what order everything has to be and determines that k must be degree 1, so we shift -1.

Chapter 4

Free field theories

4.1 The divergence complex of a measure

Lecture 12: Free field theories

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We now want to return to our free field theory motivation at the beginning of the course, but to lift them to pFAs in cochain complexes.

Let M an oriented n-manifold, and $\omega = e^{f/\hbar}\omega_0$ for $f : M \to \mathbb{C}$ and ω_0 a volume form. We think of this as some kind of generalized measure. Start with the de Rham complex $\Omega^\bullet(M)$. This is isomorphic to the divergence complex of polyvector fields $\Omega^\bullet(M) \cong (PV^\bullet, \mathrm{Div}_\hbar)$. We recall this looks like $\Gamma(M, \Lambda^{n-\bullet-1}TM)$. Although they are isomorphic as chain complexes, although dR is a dg algebra, $PV^\bullet$ is not. On the lowest degree,

$$\mathrm{Div}_\hbar = \lrcorner \hbar^{-1}\mathrm{d}f + \mathrm{Div}_{\omega_0}.$$

In the limit as $h \to 0$, we get $(PV^\bullet, \lrcorner \mathrm{d}f)$ somehow. We call $(M, PV^\bullet_M, \lrcorner \mathrm{d}f)$ the *derived critical locus* of f. We explain why.

Let $f : M \to \mathbb{R}$ some smooth map on a manifold. Recall

$$\mathrm{Crit}(f) = \{x \in M \mid \mathrm{d}f|_x = 0\}.$$

As before, we want to define observables to be functions on the critical locus. To formulate this precisely, it would be nice if $\mathrm{Crit}(f)$ was a manifold. Note that we can describe the critical locus as the interection of the graph $\Gamma(\mathrm{d}f)$ and the the zero section $M \times \{0\} \cong M$ in T^*M. Unfortunately, this need not be a manifold at all. We instead transition to the tools of algebraic geometry. Recall an intersection is a pullback, hence a limit. In particular, it is the limit in $\mathcal{S}\mathrm{et}$

$$\begin{tikzcd} M \cap \Gamma(\mathrm{d}f) \arrow[r] \arrow[d] & M \arrow[d] \\ \Gamma(\mathrm{d}f) \arrow[r] & T^*M. \end{tikzcd}$$

Of course the category of manifolds need not have pullbacks. However, we are not interested in $\mathrm{Crit}(f)$ itself, but rather functions $C^\infty(\mathrm{Crit}(f))$. So we obtain a notion of what $C^\infty(\mathrm{Crit}(f))$ should beby applying $C^\infty = \mathcal{O}$ and computing the pushout

$$\begin{tikzcd}
\mathcal{O}(\Gamma(\mathrm{d}f)) \otimes_{\mathcal{O}(T^*M)} \mathcal{O}(M) & \mathcal{O}(M) \arrow[l] \\
\mathcal{O}(\Gamma(\mathrm{d}f)) \arrow[u] & \mathcal{O}(T^*M) \arrow[l] \arrow[u]
\end{tikzcd}$$

Thus we would think of observables as elements of this tensor product.

However, observables should be some kind of function that gives us a number. We can reconcile this while at the same time solving another problem: this object is not well-behaved in general. So we apply the derived philosophy to rather consider the derived tensor $\otimes^{\mathbf{L}}_{\mathcal{O}(T^*M)}$. We remark later that we may define the derived tensor product of two modules $M \otimes^{\mathbf{L}}_R N$ as the complex $\mathrm{Tor}^R_\bullet(M, N)$. Hence we need to find a resolution of one of these modules by $\mathcal{O}(T^*M)$-modules.

We resolve $\mathcal{O}(\Gamma(\mathrm{d}f))$ by a Koszul-type complex. The resolution $K^\bullet$ is by

$$\begin{tikzcd}
\cdots \arrow[d] \\
\mathcal{O}(T^*M) \otimes_{\mathcal{O}(M)} \Lambda^2 TM \arrow[d] \\
\mathcal{O}(T^*M) \otimes_{\mathcal{O}(M)} \Lambda^1 TM \arrow[d] \\
\mathcal{O}(T^*M) \otimes_{\mathcal{O}(M)} \Lambda^0 TM \arrow[d] \\
\mathcal{O}(\Gamma(\mathrm{d}f)) \arrow[d] \\
0.
\end{tikzcd}$$

It is unclear what the differential is in this complex. The lowest degree map on the Koszul part is $1 \otimes X \mapsto X - X(f)$ apparently. Then clearly $\mathcal{O}(M) \otimes_{\mathcal{O}(T^*M)} K^\bullet \cong \Lambda^\bullet TM$, and we claim what survives of the differential is precisely contraction with $\mathrm{d}f$. Thus we have recovered the derived critical locus, which is the dg-manifold $(M, \mathrm{PV}^\bullet, \lrcorner \mathrm{d}f)$.

Unfortunately, this has not given us a notion of functions on the critical locus, as we originally hoped. With our current notation, M is the space of fields and f is the action functional. Thus the degree 0 term $C^\infty(M)$ is just functions on *all* fields. However, since the lowest degree map is contraction with $\mathrm{d}f$, the image $\operatorname{Im} \iota_{(-)}\mathrm{d}f$ consists of elements of the form $\mathrm{d}f(X)$, for X a vector field. We remark that this is defined as $X(f)$ and can be computed in coordinates easily.

In particular, the image $\operatorname{Im} \iota_{(-)}\mathrm{d}f$ is the ideal of $C^\infty(M)$ generated by the derivatives $\partial_i f$ of the action f. In order for $\mathrm{d}f$ to vanish identically, all of its partial derivatives must vanish, otherwise I could find a coordinate function x^i and a nonvanishing partial $\partial_i f$ and obtain $\mathrm{d}S(x^i) = \partial_i S \neq 0$. Moreover, if the partials all vanish, then $\mathrm{d}S$ vanishes. Thus quotienting out by the image $C^\infty(M)/\operatorname{Im} \iota_{(-)}\mathrm{d}f$ is universal with respect to identifying $\mathrm{d}f$ to zero. Thus, $\operatorname{Im} \iota_{(-)}\mathrm{d}f$ is precisely the kernel of the restriction morphism $C^\infty(M) \to C^\infty(\mathrm{Crit}(f))$ given by $f \mapsto f|\,\mathrm{Crit}(f)$. Hence the first isomorphism theorem identifies the quotient with $C^\infty(\mathrm{Crit}(f))$. Of course $C^\infty(\mathrm{Crit}(f))$ need not exist, so we treat this identification as a definition. Now simply recognize that $C^\infty(M)/\operatorname{Im}\iota_{(-)}\mathrm{d}f = H^0(\mathrm{PV}^\bullet)$. We thus see that the functions on the critical locus live inside of the homology complex of the derived critical locus, but the derived critical locus contains more information.

Toen has several survey papers on DAG that are quite readable.

Now free field theories. A free field theory is one where the action functional is quadratic in the fields. If $\mathcal{E}$ is the space of fields, for example scalar fields, then

$$S(\phi) = \int_{\mathbb{R}^n} \frac{1}{2}\phi\Delta\phi + \frac{1}{2}m^2\phi^2 \,\mathrm{d}V.$$

It is quadratic since $S(\lambda\phi) = \lambda^2 S(\phi)$. However something like

$$\widetilde{S}(\phi) = \int_{\mathbb{R}^n} \frac{1}{2}\phi\Delta\phi + \frac{1}{2}m^2\phi^2 + \frac{g}{4^n}\phi^4 \,\mathrm{d}V$$

is not an FFT. We would call that an interacting field theory. This is in opposition to FFTs since an FFT is one where particles don't interact. All particles are free. Physically these theories are not very interesting. The general idea of QFT is to view a theory as a perturbation of a FFT, since we are not quite good at solving interacting field theories.

Our goal is to construct pFAs of observables in FFTs as having values in cochain complexes, since this contains all higher order information as well as the observables which live in in degree 0. These will be twisted factorization envelopes of Heisenberg Lie algebras.

Consider a finite dimensional toy example of why quadratic functions yield Heisenberg algebras. Consider $q(x) = \frac{1}{2}x^t Ax$, A a symmetric matrix. Then (using summation notation)

$$\mathrm{d}q = \mathrm{d}\left(\frac{1}{2}x_i A_{ij} x_j\right) = x_i A_{ij}\mathrm{d}x_j.$$

We claim $\mathrm{d}q$ is a map $V \to V^*$. Choose $V = \mathbb{R}^2$ and $q(x,y) = x^2 - 3xy + 2y^2$. Then

$$\mathrm{d}q = 2x\mathrm{d}x - 3x\mathrm{d}y - 3y\mathrm{d}x + 3y\mathrm{d}y.$$

Let $x = (1,2)$ a tangent vector. Then $\mathrm{d}q$ and x somehow become

$$2\mathrm{d}x - 3\mathrm{d}y - 6\mathrm{d}x + 6\mathrm{d}y \in V^*.$$

This is the part where we use quadraticity of q. We do not know if this is contraction or not. We write

$$W : V \xrightarrow{\mathrm{d}q} V^*[-1].$$

We view this complex as an abelian dgla. Consider Chevalay-Eilenberg chains on this $C_\bullet(W) = \mathrm{Sym}(V[1] \to V^*)$ with differential $d_{\mathrm{CE}} + \mathrm{d}q$. Brackets are 0 so $d_{\mathrm{CE}} = 0$, so we just have the graded symmetric algebra with a derivation as the differential.

We have that the complex can be viewed as $\mathcal{O}(V) \otimes \Lambda^\bullet TV$; here $V \cong TV$. This is exactly polyvector fields on V, contracting with $\mathrm{d}q$. Hence this is the derived critical locus of q. This was the classical case, showing that Chevalley chains of W yield the derived critical locus of q. Can we correspondingly recover the quantum object (divergence complex) as a derived critical locus of something?

Yes. Consider a central extension of W. W has a graded skew form of cohomological degree $(\pm?)1$ by pairing V and V^*:

$$\langle (v,\phi),(v',\phi')\rangle = \phi(v') - \phi'(v).$$

Let $H_W = \mathbb{C}\hbar[-1] \oplus W$, with bracket given by

$$[w,w'] = \hbar \langle w, w'\rangle .$$

This is some kind of central extension of an abelian Lie algebra, also called a Heisenberg algebra.

Construction 4.1.1. I have decided to, after the fact, detail how to compute the derived tensor product $M \otimes_R^L N$ of two R-modules M, N. The category $\mathcal{C}\mathrm{h}(R)$ admits a model structure such that right exact functors are left Quillen, and thus the pointwise tensor product $- \otimes_R M$ is left Quillen. The left derived functor of a left Quillen functor $F : \mathcal{C} \to \mathcal{D}$ is defined as ([Hov99]):

$$\mathbf{L}F = \mathrm{Ho}\, F \circ \mathrm{Ho}\, Q,$$

for $Q : \mathcal{C} \to \mathcal{C}_c$ a cofibrant replacement. We now consider this for the case of the tensor product.

Let C be a chain complex. Passing to the homotopy category $\mathrm{Ho}\,\mathcal{C}\mathrm{h}(R) = \mathcal{D}(R)$ means we take the homotopy class $[C]$ of all quasi-isomorphic chain complexes. To compute the derived tensor product, we generally want to choose a representative $C \in [C]$ of C, meaning any quasi-isomorhic cochain complex. We then need to take $\mathrm{Ho}\, Q$, which means we pass to the homotopy class of a cofibrant replacement $Q(C)$ of C. However, we note that $Q : \mathcal{C}\mathrm{h}(R) \to \mathcal{C}\mathrm{h}(R)_c$ falls into the full subcategory of cofibrant objects, meaning that this is really a "cofibrant homotopy class" of only cofibrant objects. Essentially, we need to find a cofibrant replacement $Q(C) \simeq C$ which is quasi-isomorphic. This representst he homotopy class of C.

Note here that although Q is functorial, there is a natural trivial fibration (in particular, a natural quasi-isomorphism; [Hov99]) $Q \Rightarrow 1$, meaning that $Q(C) \simeq C$. Note here that when passing to homotopy, *any cofibrant* C' which is quasi-isomorphic to C is thus quasi-isomorphic to the proper cofibrant replacement $Q(C)$, and thus in homotopy they define the same (cofibrant) homotopy class. Thus we can choose *any* cofibrant replacement without worry that it be functorial.

So choose a cofibrant replacement $Q(C)$ for C. Then we simply compute the homotopy tensor product, meaning we simply compute the pointwise tensor with M and then take any quasi-isomorphic chain complex $X \simeq Q(C) \otimes_R M$ to be the derived tensor. It might seem simple to just say $Q(C) \otimes_R M$ is the answer, however in homological algebra, there is a more canonical choice. Note that for all complexes A, we have $A \simeq H_\bullet(A)$, meaning we can take its homology.

In the case that $C = N$ is an R-module concentrated in degree 0, then we note that any cochain complex bounded below of projective modules is cofibrant, and thus we simply need to find a bounded below complex of projective modules $P \simeq N$. This is precisely a projective resolution of N. Thus what we are doing is taking a projective resolution and taking the pointwise tensor with M. We then want to take the homology. Thus

$$(N \otimes_R^L M)_i = H_1(Q(N) \otimes_R M).$$

Note that this is preciely the ith derived functor of tensor, so

$$N \otimes_R^L M = \mathrm{Tor}_\bullet^R(N, M).$$

We recall that a bounded below resolution P of an R-module N is an exact cochain complex P that is bounded below such that $P \simeq N$, where we without loss of generality take P to be concentrated in nonnegative degree. If the quasi-isomorphism is $\varepsilon : P \simeq N$, then we can rewrite this as

$$\cdots \xrightarrow{d_3^P} P_2 \xrightarrow{d_2^P} P_1 \xrightarrow{d_1^P} P_0 \xrightarrow{\varepsilon_0} N \longrightarrow 0.$$

There is a standard result in homological algebra saying that the tensor $- \otimes_R M$ more generally preserves quasi-isomorphisms between *flat* modules, which need not be cofibrant. However, this means that we may choose a flat resolution instead of a projective resolution. Also note that all free modules are projective, and all projective modules over a PID are free, so we often use free resolutions if possible.

Intuition 4.1.2. We briefly explain the derived philosophy. Oftentimes a space X has a natural sheaf $\mathcal{O}_X$, often valued in rings, which encodes much of the important data of that space. For manifolds, the natural sheaf to use is C^∞. Thus to describe a manifold, especially if we only care about its homological data, it should suffice to describe the sheaf C^∞ for that manifold. The derived philosophy says that even if the pullback $M \times_{T^*M} \Gamma(\mathrm{d}f)$ is not a manifold, we can describe what its structure sheaf C^∞ *should* look like.

Lecture 13: Free field theories, continued

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Let us recall what is happening. Earlier in the course, we took a naïve approach to free field theories. We started with a vector bundle $E \to M$ on a Riemannian manifold M. The *fields* were then defined as the sheaf of sections $\mathcal{E}$ of E. The action functional was then defined as a map $S : \mathcal{E} \to \mathbb{R}$. Here we do not claim that S is a map of (pre)sheaves, and simply use this notation to indicate that for all open $U \subseteq M$, there is a map $S : \mathcal{E}(U) \to \mathbb{R}$. If one passes to compactly supported sections, then S is actually a map of cosheaves $S : \mathcal{E}_c \to \mathbb{R}$.

Physically, the equations of motion are extrema for the action functional. Thus we define the *Euler-Lagrange equations of motion* as the critical locus $\mathrm{Crit}(S) = \mathrm{d}S^{-1}(0)$. Here we recall that $\mathrm{d}S : T\mathcal{E}(U) \to \mathbb{R}$. Here we have viewed $\mathcal{E}(U)$ as a vector space, and thus a smooth manifold.

From here, there are two perspectives we can take: the classical perspective, and the quantum perspective. In the quantum perspective, we want to define correlation functions by integrating against the measure $e^{S/\hbar}\omega_0$, where ω_0 is a volume form. The quantum perspective is highly involved, so we continue for now with the classical perspective to build the necessary background.

In the classical perspective, observable quantities are functions $\mathrm{Crit}(S) \to \mathbb{R}$. We often work with polynomials for a important yet highly explicit example. Polynomials are informally understood to be defined as elements of the symmetric algebra of the dual space $\mathrm{Sym}(\mathrm{Crit}(S)^*)$.

Before we continue, there is a simplification we can provide. A dg manifold M is, intuitively, a manifold M with a nice sheaf of dg commutative algebras. For example, the derived critical locus $(M, \mathrm{PV}^\bullet)$ is a dg manifold. Here, "nice" means it is sufficiently close to a symmetric algebra.

We can establish some criterion for when such a dg manifold is *linear*. It is linear if $M = V$ is a vector space, and if the action is a quadratic form. In our case, the base manifold is the space of fields $C^\infty(M)$, which is a vector space, and the differential is indeed a quadratic form.

If M is a vector space, then TM is trivial, so sections are just maps $M \to M$. We note that $C^\infty(M, M)$ can be identified with $C^\infty(M) \otimes M$ by taking a smooth map $f : M \to M$, choosing a basis $\{e_i\}_{1 \le i \le n}$, and writing $f = \sum_i f_i e_i$. Then simply map $f \mapsto \sum_i f_i \otimes e_i$.

More generally, triviality of the tangent bundle yields that k-polyvector fields are maps $M \to M \times \Lambda^k M$, which are similarly identified with $C^\infty(M) \otimes \Lambda^k M$. We then use $\Lambda^\bullet M \cong \mathrm{Sym}(M[1])$ to obtain

$$\mathrm{PV}^\bullet \cong C^\infty(M) \otimes \mathrm{Sym}(M[1]).$$

We assume the situation is sufficiently nice such that $C^\infty(M) \cong \mathrm{Sym}(M^*)$. Then distributing Sym over tensor yields

$$\mathrm{PV}^\bullet \cong \mathrm{Sym}(M^* \oplus M[1]).$$

This is generated by the chain complex $M^* \oplus M[1]$, and we can take the graded dual $(M^* \oplus M[1])^\vee$ to obtain a cochain complex $M \oplus M^*[-1]$. We can go backwards, taking the symmetric algebra of the graded dual of any cochain complex, to obtain a linear dg manifold. Thus we can simplify the presentation of a linear dg manifold into a single cochain complex. We call such a complex the *derived space of solutions to the Euler-Lagrange equations.*

Discussion 4.1.3. Why do we call this complex the derived space of solutions to the Euler-Lagrange equations? First, we note that the graded vector space $M \oplus M^*[-1]$ without the structure of the cochain complex is referred to as the *derived space of fields.* Recall that we defined $\mathcal{O}(\mathrm{Crit}(S))$ as the Koszul complex $\mathrm{PV}^\bullet \cong C^\infty(M) \otimes \Lambda^\bullet V$. The derived critical locus is really the "functions" on the critical locus. However, it is unclear how to interpret the space $\mathrm{PV}^\bullet$ as functions on anything, so we define a derived notion of the underlying space $\mathrm{Crit}(S)$. Of course the term "derived critical locus" is already taken, so we define such a space to be the *derived space of solutions.*

The idea is that functions on such a derived space should be equivalent to the derived critical locus, which was defined to somewhat be functions on $\mathrm{Crit}(S)$. We intuitively understand functions on the space as the symmetric algebra of the graded dual, so we simply want a complex W whose symmetric algebra on its graded dual is the derived critical locus $PV^\bullet$. As we just saw, in the case where $M = V$ is a vector space, this is solved by $M \to M^*[-1]$. The differential $d : M \to M^*[-1]$ is given by $d(v) = \frac{1}{2}S(v, -)$, where $S(-,-)$ is the bilinear form determined by a quadratic action S on a vector space. This is explained in the following example.

So why do we call the underlying graded vector space of the derived space of solutions the *derived space of fields*? It will become more clear when we consider the example of abelian Yang-Mills, but for now we are content to simply say that the cohomology $H^0(W)$ of the derived space of solutions is the classical critical locus $\mathrm{Crit}(S)$. Hence adding in the differential corresponds in some way to working with the action functional, and the action functional acts on fields. Hence it makes sense to call the underlying graded vector space the *derived space of fields.*

Example 4.1.4. Let's consider a toy example. Let V be a vector space, and $q : V \to \mathbb{R}$ a quadratic function on V. Let ω_0 be the Lebesgue measure on V. We want to understand the divergence complex for the meaure $e^{q/\hbar}\omega_0$. We do not need to assume q is nondegenerate here.

We claim that the derived critical locus is described by the derived space of solutions, which we refer to as W:

$$V \xrightarrow{\partial/\partial q} V^*[-1]$$

which maps $v \in V$ to the linear functional $\frac{\partial}{\partial q}v = q(v, -)$. Here we abuse notation and identify q with its *polarization identity.* A quadratic form q uniquely determines a bilinear form B defined by $B(v, w) = q(v + w) - q(v) - q(w)$. We abuse notation and write $B = q$.

Note that W comes with a graded antisymmetric pairing $\langle -, - \rangle : W \otimes W \to \mathbb{R}$ of cohomological degree -1. Note that the fact that the pairing has cohomological degree -1 means that $\langle x, y\rangle$ only lands in $\mathbb{R}$, which is concentrated in degree 0, if $|x| + |y| = 1$. Hence we take $x = v \in V$ and $y = \alpha \in V^*[-1]$, and define

$$\langle v, \alpha \rangle = \alpha(v) = -\langle \alpha, v\rangle .$$

Now define the (shifted) *Heisenberg* dg Lie algebra $\mathcal{H}_W$ as the dg Lie algebra

$$\mathcal{H}_W := \mathbb{C} \cdot \hbar[-1] \oplus W,$$

where $\mathbb{C} \cdot \hbar$ is the 1-dimensional vector space with basis $\hbar$. The Lie bracket is given by

$$[w, w'] = \hbar \langle w, w' \rangle,$$

and it vanishes on $\mathbb{C} \cdot \hbar$.

Consider the Chevalley-Eilenberg chains on this dg Lie algebra $C_\bullet(\mathcal{H}_W)$. Recall that this is defined to be the symmetric algebra of the underlying graded vector space $\mathcal{H}_W[1] = W[1] \oplus \mathbb{C} \cdot \hbar$, whose differential is given by the Lie bracket acting as a coderivation. Note that

$$W^\vee = (V \oplus V^*[-1])^\vee = V^* \oplus V[1] = W[1]$$

We then obtain

$$C_\bullet(\mathcal{H}_W) \cong \mathrm{Sym}(W^\vee \oplus (\mathbb{C} \cdot \hbar)) \cong \mathrm{Sym}(W^\vee) \otimes \mathrm{Sym}(\mathbb{C} \cdot \hbar).$$

Since $\mathbb{C} \cdot \hbar$ is concentrated in degree 0, it symmetric algebra is just the polynomial ring $\mathbb{C}[\hbar]$, so tensoring with the polynomial ring just adjoins the variable $\hbar$ to the symmetric algebra, yielding

$$C_\bullet(\mathcal{H}_W) \cong \mathrm{Sym}(W^\vee)[\hbar].$$

We can show that $C_\bullet(\mathcal{H}_W) \cong \mathrm{PV}^\bullet(V)[\hbar]$ by showing that $\mathrm{Sym}(W^\vee) \cong \mathrm{PV}^\bullet(V)$. We have

$$\mathrm{Sym}(W^\vee) = \mathrm{Sym}(V^* \oplus V[1]) \cong \mathrm{Sym}(V^*) \otimes \mathrm{Sym}(V[1]) \cong \mathrm{Sym}(V^*) \otimes \Lambda^\bullet V,$$

where we have noted of course that since V^* is concentrated in degree 0, so is its symmetric algebra. Note obviously that $\mathrm{Sym}(V^*)$ consists of polynomials, and thus $\mathrm{Sym}(V^*) \otimes \Lambda^\bullet V$ can be identified with polyvector fields on V.

We see that if we can find the proper derived space of solutions, we should be able to obtain a graded antisymmetric pairing that induces a Heisenberg (dg) Lie algebra whose Chevalley-Eilenberg chains are the derived critical locus.

4.2 The prefactorization algebra of a free field theory

Example 4.2.1. Let's consider the more complex example of a basic scalar field theory, which takes place on a Riemannian manifold (M, g), has a space of fields $C^\infty(M)$, and action functional

$$S(\phi) = \int_M \phi \Delta \phi,$$

where Δ is the Laplacian. We integrate over $U \subseteq M$ when we want to define more local fields. This is much more complex since polyvector fields no longer look like $\Lambda^\bullet M$, but rather are sections of $\Lambda^\bullet TM$. Example 4.1.4 suggests that we should be able to define the derived critical locus as the Chevalley-Eilenberg chains of a Heisenberg algebra associated to a cochain complex with a graded antisymmetric pairing; in particular we should find a derived space of solutions to the equations of motion. Consider the sheaf $\mathcal{E}$ valued in cochain complexes

$$\mathcal{E}(U) = \Big(\; 0 \longrightarrow C^\infty(U) \xrightarrow{\Delta} C^\infty(U)[-1] \longrightarrow 0 \; \Big),$$

where the leftmost $C^\infty(U)$ is in degree 0 and the rightmost $C^\infty(U)$ is in degree 1. In the future, if V is a vector space, we will write $V[-n]$ to denote that V is in degree n. This notation is reconciled with the usual shift functor and the convention of taking vector spaces to be concentrated in degree

0 by simply taking cochain complexes to be valued in cochain complexes. We will also omit the trailing zeroes, and leave them implicit.

Earlier, our derived space of solutions was only one cochain complex. We could extend it to a sheaf of cochain complexes by restricting along open subsets, and the derived critical locus would then be the Chevalley-Eilenberg chains associated to its global sections. However, the derived critical locus here really means the global sections of the derived critical locus, since it is itself a sheaf. Hence we should be able to recover the derived critical locus sheaf by applying this construction pointwise. We proceed by considering the global sections, but there is nothing that prevents us from considering an open subset $U \subseteq M$ instead. Hence, we would like to define observables as the symmetric algebra of the graded dual of $\mathcal{E}$, possibly up to some shift. We can make this idea precise.

We work in a bit more generality than required for this particular problem. Given a graded vector bundle $E = \{E_i \to M\}_{i\in\mathbb{Z}}$ over a manifold M, define the graded vector bundle $E^!$ on M by taking the (graded) tensor product with the top exterior bundle $E^! := E^\vee \otimes_{\mathbb{R}} \Lambda^n T^*M$, where $\Lambda^n T^*M$ is concentrated in degree 0. We denote the sheaf of sections of E by $\mathcal{E}$, and the sheaf of sections of $E^!$ by $\mathcal{E}^!$. We note a theorem: the functor mapping vector bundles over M to their sheaves of sections preserves limits and colimits, and moreover one does not have to sheafify the colimits of presheaves. This holds in the graded case as well, where our sheaves now take values in graded vector spaces. This gives $\mathcal{E}^! \cong \mathcal{E}^\vee \otimes_{C^\infty} \Omega^n_M$. We denote by $\mathcal{E}_c$ the cosheaf of compactly supported sections of E. We denote by $E^\vee$ the graded dual $(E^\vee)_i = (E_{-i})^*$, and $\mathcal{E}^\vee$ its sheaf of sections.

There are a number of ways one could make precise the notion of functions on $\mathcal{E}(M)$. Of course, we are motivated to consider the symmetric algebra on the graded dual, but it will be helpful to consider other possibilities and to show their equivalence. One such possibility is the space of polynomials $P(\mathcal{E}(M))$. We intuitively expect the symmetric algebra of the dual to recover polynomials, so to see this intuitively, we start with a more intuitive definition of polynomials.

Costello and Gwilliam in [CG16] start with a definition of polynomials in terms of differentiable vector spaces. I think our class didn't go over this machinery, so I will skip to their identification

$$P_n(\mathcal{E}(M)) := \mathcal{D}_c\left(M^n, (E^!)^{\boxtimes n}\right)_{S_n}.$$

Here P_n denotes polynomials of degree n, $\mathcal{D}_c$ denotes compactly supported distributional sections of $(E^!)^{\boxtimes n}$, and $E \boxtimes E = E^{\boxtimes 2}$ denotes the second external tensor power of the bundle E, which is a $(\dim E)^2$-bundle over M^2 defined by, for $p, q \in M$,

$$E^{\boxtimes 2}_{(p,q)} = E_p \otimes E_q.$$

We then take S_n-coinvariants. Here the compactly supported distributional sections of M^n on $(E^!)^{\boxtimes n}$ are defined to be compactly supported continuous linear functionals on sections. For $n = 1$, this looks like $f : \Gamma^\infty(M, E) \to \mathbb{R}$. We do not go into detail about how this arises, and simply note that earlier we defined distributions as the dual to $C^\infty(M)$, which makes sense. Here we are trying to do something more general: define a dual to sections of an arbitrary vector bundle E. Apparently $\mathcal{D}(M, E)$ is defined as the linear functionals $\Gamma^\infty_c(M, E^!) \to \mathbb{R}$, and so considering compactly supported sections $\mathcal{D}_c(M, E^!)$ dualizes this definition in a sense.

We expect $P(\mathcal{E}(M)) := \bigoplus_n P_n(\mathcal{E}(M))$ to be the observables on a theory. However, there are a few other possible characterizations (we have already seen the derived critical locus). We continue going through them to show that they all make sense, and are all equivalent in this case. This will provide good intuition for how to formalize a free field theory axiomatically.

Since E is graded, the evaluation map $\langle f, v\rangle = f(v)$ taking a distributional section f and a section v must respect grading in the sense that $\langle f, v\rangle$ has degree $|f| + |v|$. Since $\mathbb{R}$ is concentrated in degree 0, this means $f(v)$ is nonzero precisely when $|f| = -|v|$. This makes sense since the graded dual flips the direction. Hence $\mathcal{D}_c$ produces a whole complex in this case with $f : \Gamma(M, E_i) \to \mathbb{R}$ in degree $-i$.

We see fit to expand a bit on external tensor powers. We use a graded external tensor power

$$(E^!)^{\boxtimes n}_k = \bigoplus_{d_1+\cdots+d_n=k} E^!_{d_1} \boxtimes \cdots \boxtimes E^!_{d_n}.$$

Compactly supported distributional sections of this bundle are compactly supported continuous linear functionals on sections $f : \Gamma^\infty(M^n, E^{\boxtimes n}) \to \mathbb{R}$. There is a canonical S_n-action by permuting the ordering of the base manifold. A section of $(E^!)^{\boxtimes n}$ can be represented as a tensor product of sections $s_1 \otimes \cdots \otimes s_n$ of $E^!$ in coordinates sufficiently close to a point. One simply permutes these by, for any $\sigma \in S_n$, mapping $s \mapsto \varepsilon(s, \sigma) s_{\sigma^{-1}(1)} \otimes \cdots \otimes s_{\sigma^{-1}(n)}$ for ε some Koszul sign rule. One then gets the dual action $\sigma \cdot f \coloneqq f \circ \sigma^{-1}$, where the σ^{-1} acts on sections. Hence

$$(\sigma \cdot f)(s) = f(\sigma^{-1} \cdot s).$$

Thus the S_n-coinvariants are symmetric linear functionals.

We explain why these are polynomials. Let V a vector space. Then there is a basis $\{e_i\}_{i \in I}$ of V, and a polynomial in e_i can be thought of as a linear combination in formal powers of linear functions $\varepsilon^i(e_j) = \delta_{ij}$. The intuition is that, for example if we conside $f(x, y) = x^2 + y$ and let $\chi(x) = 1$ and $\eta(y) = 1$ and otherwise zero, then to evaluate $(x, y) = (1, 2)$, we evaluate

$$\chi^2(1x + 2y) + \eta(1x + 2y) = \chi^2(x) + 2\eta(y) = 1^2 + 2.$$

Thus a polynomial can be thought of as an element of the symmetric algebra of the dual. This is just a graded generalization.

Now we return to our concrete example. Our derived space of fields can be realized as the sections of a graded vector bundle given by the trivial bundle $M \times \mathbb{R}$ concentrated in degrees 0 and 1. Write E_0 and E_1 for the 0th and 1st degrees of E, such that $E = E_0 \oplus E_1[-1]$ (we still make the degrees explicit). Then

$$E^! \coloneqq E^\vee \otimes \Lambda^n T^*M = (E_0^* \oplus E_1^*[1]) \otimes \Lambda^n T^*M \cong (E_0^* \otimes \Lambda^n T^*M) \oplus (E_1^* \otimes \Lambda^n T^*M)[1].$$

The book claims that $E^!$ is just two copies of the trivial bundle, one in degree -1 and one in degree 0. Since M is Riemannian, it has a canonical volume form, which yields a canonical basis for the 1-dimensional fibers of $\Lambda^n T^*M$. Thus we can identify the fibers $(E_1^* \otimes \Lambda^n T^*M)_p \cong (E_1^*)_p$ by the explicit isomorphism $f \otimes \mathrm{vol}_g \mapsto f$. Additionally, $\mathbb{R}$ has a canonical basis given by 1, so we can identify $\mathbb{R}^* \cong \mathbb{R}$, inducing an isomorphism $(M \times \mathbb{R})^* \cong M \times \mathbb{R}$ on the trivial bundles. This reconciles $E^!$ with the book's claims.

Note that since $P_1(\mathcal{E}(M))$ is just maps $\Gamma(M, E) \to \mathbb{R}$, and since as seen earlier it satisfies the same degree flipping as the graded dual, we obtain $P_1(\mathcal{E}(M)) \cong \mathcal{E}(M)^\vee$:

$$\mathcal{E}(M)^\vee = \left(\mathcal{D}_c(M)[1] \xrightarrow{\Delta} \mathcal{D}_c(M) \right).$$

The differential is still Δ since it is self-adjoint. Here we note that since monomials inherit a trivial S_n-action, passing to S_n-coinvariants is trivial, and the algebraic dual of $C^\infty(M)$ is just $\mathcal{D}_c(M)$.

One might think that an equivalent definition to a polynomial, and hence an observable, should be the symmetric algebra on the monomials $\mathcal{E}(M)^\vee$. This turns out to be a poor choice for quantization.

However, free field theories offer a very nice classification of observables. Note that our identification of functions and densities yields an isomorphism

$$\mathcal{E}^!_c(U) \cong \Big(\; C^\infty_c(U)[1] \xrightarrow{\;\Delta\;} C^\infty_c(U) \; \Big)$$

for compactly supported sections of $\mathcal{E}^!$. Recall this is defined by taking a compactly supported section $f \otimes \mathrm{vol}_g$ of $(U \times \mathbb{R}) \oplus \Lambda^n T^*U$ and identifying it with the compactly supported function $f : U \to \mathbb{R}$. This yields a natural map of cochain complexes $\mathcal{E}^!_c(M) \to \mathcal{E}^\vee$ by viewing a compactly supported function as a distribution (integrate against it). Here we note that the differentials on these complexes are the same; just trace definitions.

Lemma 4.2.2. *The map $\mathcal{E}^!_c(U) \to \mathcal{E}^\vee(U)$ is a homotopy equivalence of cochain complexes.*

The relevance of lemma 4.2.2 is that we are now working *homotopically.* As discussed earlier, we can recover the classical observables and the classical critical locus by taking cohomology of the relevant derived notions. Hence we expect observables to live in (the 0th) cohomology of some complex, meaning we only care about the derived notions up to homotopy. If observables in $\mathcal{E}(M)^\vee$ are closed distributional sections, then we can identify them with compactly supported smooth maps $f : U \to \mathbb{R}$ which are closed in $\mathcal{E}^!_c(U)$ in a homotopical sense. We think of the smooth observables as "smeared". For example, we intuitively replace the delta function δ_x at x with a bump function supported near x.

This gives us a nice notion of observables: *smeared observables*

$$\mathrm{Obs}^{cl}(U) \coloneqq \mathrm{Sym}(\mathcal{E}^!_c(U)) = \mathrm{Sym}\Big(\; C^\infty_c(U) \xrightarrow{\;\Delta\;} C^\infty_c(U) \; \Big)$$

by taking the symmetric algebra, just as we expected to take the symmetric algebra on $\mathcal{E}^\vee$. We claim that we may identify $\mathrm{Sym}^n(\mathcal{E}^!_c(U)) \cong C^\infty_c(U^n, (E^!)^{\otimes n})_{S_n}$, which can be identified with a subspace of $P_n(\mathcal{E}(U))$ defined by taking all distributions to have compact support (why have we gone $\boxtimes \to \otimes$?). Here we take the graded symmetric algebra with respect to a topologically completed tensor product $\otimes$. This does not change the algebraic machinery, and simply completes the dense subset $C^\infty(M) \otimes C^\infty(N) \subseteq C^\infty(M \times N)$. To summarize,

$$\mathrm{Sym}(\mathcal{E}^\vee(U)) \simeq \mathrm{Obs}^{cl}(U) \coloneqq \mathrm{Sym}(\mathcal{E}^!_c(U)) \subseteq P_n(\mathcal{E}(U)).$$

Lemma 4.2.3. *The map $\mathrm{Obs}^{cl}(U) \to P(\mathcal{E}(U))$ is a homotopy equivalence of cochain complexes.*

Proof. It suffices to show the map $\mathrm{Sym}^n(\mathcal{E}^!_c(U)) \subseteq P_n(\mathcal{E}(U))$ is a homotopy equivalence for all n. For this, it suffices to show the inclusion $C^\infty_c(U, (E^!)^{\otimes n}) \subseteq \mathcal{D}_c(U, (E^!)^{\otimes n})$ is an S_n-equivariant homotopy equivalence. Finish later. ■

Hence we have shown that all of our obvious notions of observables on this theory are equivalent, and distributional polynomial observables (given by integrating against some distribution on U^n) with smooth polynomial observables (given by integrating against a smooth function on U^n).

How do we intepret the observables we constructed in example 4.2.1? We start by describing $\mathrm{Obs}^{cl}(U)$ more explicitly. Recall that $\mathcal{E}^c_! \cong \mathcal{E}^\vee_c = C^\infty_c \oplus C^\infty_c[-1]$. Thus thee complex looks like

$$\mathrm{Sym}(\mathcal{E}_c(U)) = \mathrm{Sym}(C^\infty_c(U) \oplus C^\infty_c(U)[-1]) \cong \mathrm{Sym}\, C^\infty_c(U) \otimes \Lambda^\bullet C^\infty_c(U),$$

where $\Lambda^\bullet C^\infty_c(U)$ is concentrated in negative degees, and $\mathrm{Sym}\, C^\infty_c(U)$ is concentrated in degree 0. Diagramatically,

$$\mathrm{Obs}^{cl}(U) \cong \Big(\; \cdots \longrightarrow \Lambda^2 C^\infty_c(U) \otimes \mathrm{Sym}\, C^\infty_c(U) \longrightarrow C^\infty_c(U) \otimes \mathrm{Sym}\, C^\infty_c(U) \longrightarrow \mathrm{Sym}\, C^\infty_c(U) \; \Big).$$

All tensor products here are topologically completed tensor products.

By identifying $C_c^\infty(U) \subseteq \mathcal{D}(U) = (C^\infty(U))^*$, we may think of $\operatorname{Sym} C_c^\infty(U)$ as an algebra of polymomial functions on $C^\infty(U)$, using the motivation we explored earlier. We can actually obtain similar intuition from the other terms in the complex. Let $T_cC^\infty(U) \subseteq TC^\infty(U)$ be the subbundle of the tangent bundle given by the subspace $C_c^\infty(U)$. "An element of a fiber of $T_cC^\infty(U)$ is a first-order variation of a field which vanishes outside a compact set." Let's unpack that sentence from the authors of [CG16].

First, we view $T_cC^\infty(U)$ as a subbundle of $TC^\infty(U)$. Thus we are taking tangent vectors in the tangent bundle $TC^\infty(U)$, and restrictig them along $C_c^\infty(U)$. Since C^∞ is a sheaf of vector spaces, $C^\infty(U)$ has trivial tangent bundle, so we obtain $TC^\infty(U) \cong C^\infty(U) \times C^\infty(U)$. Thus fibers of $TC^\infty(U)$ are pairs $(\phi, \delta\phi)$ for $\phi \in C^\infty(U)$, and $\delta\phi$ a tangent vector to ϕ, which we refer to as a *first order variation* of ϕ, and define via the calculus of variations.

Now to find elements of $T_cC^\infty(U)$, we restrict $\delta\phi$ along $C_c^\infty(U)$, meaning that $T_cC^\infty(U) \cong C^\infty(U) \times C_c^\infty(U)$. Thus elements of $T_cC^\infty(U)$ are compactly supported first order variations. Apparently this defines an integrable foliation on $T_cC^\infty(U)$, and this foliation can be defined for any sheaf of spaces.

We can interpret an element of $C_c^\infty(U) \otimes \operatorname{Sym} C_c^\infty(U)$ as a polynomial section of $T_cC^\infty(U)$: a section of $T_cC^\infty(U)$ is a map $X : C^\infty(U) \to C^\infty(U) \times C_c^\infty(U)$ which is a section of the projection; equivalently it is a map $X : C^\infty(U) \to C_c^\infty(U) \subseteq \mathcal{D}(U)$. A *polyomial* in this sense means that component functions are polynomials, so a polynomial is an element of $C_c^\infty(U) \otimes \operatorname{Sym} \mathcal{D}(U)$ by defining $X(v) = \sum_i f_i(v) w_i$ for $f_i \in \operatorname{Sym} \mathcal{D}(U)$ and $w_i \in C_c^\infty(U)$. Since $C_c^\infty(U) \subseteq \mathcal{D}(U)$, we may interpret $C_c^\infty(U) \otimes \operatorname{Sym} C_c^\infty(U)$ as a space of polynomial sections of $T_cC^\infty(U)$. By the same logic, $\Lambda^k C_c^\infty(U) \otimes \operatorname{Sym} C_c^\infty(U)$ is the space of polynomial sections of $\Lambda^k C_c^\infty(U)$.

In particular, if $\mathrm{PV}_c(C^\infty(U))$ denotes polynomial polyvector fields on $C^\infty(U)$ along the subbundle $T_cC^\infty(U)$, then

$$\mathrm{Obs}^{cl}(U) \cong \mathrm{PV}_c(C^\infty(U))$$

as graded vector spaces.

Remark 4.2.4. We offer some motivating intuition for the tangent bundle to an infinite dimensional manifold. The definition of a tangent vector as a *derivation* diverges from the notion of a geometric tangent vector in infinite dimensions, since there are strictly more derivations than geometric tangent vectors. Instead, we use the definition via equivalence classes of smooth curves. Recall that given $F : M \to N$ a smooth map between smooth manifolds, $p \in M$, and $v \in T_pM$, we find $\mathrm{d}F_p(v) = (F \circ \gamma)'(0)$ for any open $0 \in J \subseteq \mathbb{R}$ and any smooth $\gamma : J \to M$ with $\gamma(0) = p$ and $\gamma'(0) = v$. Hence one could *define* a tangent vector at $p \in M$ as an equivalence class of smooth curves $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = 0$, where we define $\gamma_1 \sim \gamma_2$ iff, for any chart $(p \in U, \varphi)$,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\varphi \circ \gamma_1)(t) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\varphi \circ \gamma_2)(t).$$

The intuition is that this construction is just $\mathrm{d}\varphi_p(v)$, which is precisely the isomorphism $T_pM \cong \mathbb{R}^{\dim M}$. Thus computing this with two different smooth curves should give the same result.

We want to promote our identification $\mathrm{Obs}^{cl}(U) \cong \mathrm{PV}_c(C^\infty(U))$ to one of cochain complexes. Recall the action

$$S(\phi) = \frac{1}{2} \int_U \phi \Delta \phi$$

for Δ the Laplacian on the Riemannian manifold (M, g). Roughly, the differential on $\mathrm{PV}_c(C^\infty(U))$ should be given by contraction with $\mathrm{d}S$. We have to make this idea precise. Since S is quadratic in

ϕ, we expect that contraction with a tangent vector $\phi_0 \in C^\infty(U)$ should be the linear functional

$$(\phi_0 \lrcorner \mathrm{d}S)(\phi) = \frac{1}{2}\int_U \phi_0 \Delta\phi.$$

However, recall that this is not well-defined for all fields ϕ. However, it *does* make sense to say

$$\frac{\partial S}{\partial \phi_0}(\phi) = \frac{1}{2}\int_U \phi_0 \Delta\phi$$

for all fields ϕ, provided $\phi_0 \in C_c^\infty(U)$ is compactly supported. More precisely, the desired form $\mathrm{d}S$ doesn't make sense as a section of the tangent bundle, but it *does* make sense as a section of $T_c^* C^\infty(U)$, the dual of $T_c C^\infty(U)$. We may thus contract with this 1-form on $\mathrm{PV}_c(C^\infty(U))$, showing that this differential indeed agrees with that for $\mathrm{Obs}^{cl}(U)$.

Summary. We identify $C_c\infty(U) \subseteq \mathcal{D}(U) = (C^\infty(U))^*$ by viewing compactly supported maps $f : U \to \mathbb{R}$ as functionals

$$g \mapsto \int_U (f \cdot g)\mathrm{d}vol.$$

We start with the sheaf of sections $\mathcal{E}$ of the derived space of solutions E, given by the sheaf of cochain complexes $\Delta : C^\infty(U) \to C^\infty(U)[-1]$. We expect the symmetric dual of this to be polynomial polyvector fields, which are our observables. Thus we should have $\mathrm{Sym}\,\mathcal{E}^\vee \simeq \mathrm{Obs}^{cl}$ (we are working up to homotopy). We also expect (degree n) polynomials to be $\mathcal{D}_c\left(M^n, (E^!)^{\boxtimes n}\right)_{S_n}$ – that is, symmetric compactly supported distributional sections of the n-th exterior tensor power of $E^! = E^\vee \otimes \Omega^n_M$. Since Δ is self-adjoint and S is quadratic in the field ϕ, we find that $\mathcal{E}^!_c \cong (\Delta : C_c^\infty(U) \to C_c^\infty(U))$ has

$$\mathcal{E}^!_c(U) \simeq \mathcal{E}^\vee(U)$$

by simply viewing compactly supported sections as distributions. This yields

$$\mathrm{Obs}^{cl}(U) \cong \mathrm{Sym}\,\mathcal{E}^!_c(U) \simeq \mathrm{Sym}\,E^\vee(U) \simeq P(\mathcal{E}(U)).$$

Finally, we find that by viewing $\mathrm{Obs}^{cl}(U)$ as $\mathrm{Sym}\,\mathcal{E}^!_c(U)$, we may identify it with compactly supported polyvector fields $\mathrm{PV}_c(C^\infty(U))$, and we may take the differential to be contraction with the action $\mathrm{d}S$. Hence, we have shown that in the case of free field theories, provided we make the correct topological adjustments, the infinite-dimensional case echoes the finite dimensional case.

4.3 General free field theories

Definition 4.3.1. Let M be a manifold. A *free field theory* on M is the following data.

1. A graded vector bundle E on M, whose sheaf of sections is denoted $\mathcal{E}$, and whose compactly supported sections are denoted $\mathcal{E}_c$.

2. A morphism of sheaves of $\mathbb{R}$-vector spaces (meaning that it is not C^∞_M-linear!) $\mathrm{d} : \mathcal{E} \to \mathcal{E}$ of cohomological degree 1, making $\mathcal{E}(U)$ into a sheaf of cochain complexes. This must also be a differential operator, meaning locally it looks like a differential operator.

3. Let $E^! := E^\vee \otimes \mathrm{Dens}_M$, which on Riemannian manifolds is $E^\vee \otimes \Lambda^n T^*M$. Let $\mathcal{E}^!$ be the sections of $E^!$, and let $\mathrm{d}^!$ be the differential on $\mathcal{E}^!$ inherited from that of $\mathcal{E}$ as described in remark 4.3.3. Then there is a natural pairing between $\mathcal{E}_c(U)$ and $\mathcal{E}^!(U)$, which is compatible with differentials.

4. There is an isomorphism $E \cong E^![-1]$ which is compatible with differentials (I assume meaning the induced isomorphism of sheaves of graded vector spaces is an isomorphism of sheaves of cochain complexes), with the property that the induced pairing of cohomological degree -1 on each $\mathcal{E}_c(U)$ is graded anti-symmetric.

The complex $\mathcal{E}(U)$ is the derived space of solutions to the equations of motion of the theory on an open subset U. The rest of the data ensures that we can make sense of this field theory in the same way as for the free scalar field theory in the previous section. Hence we should (I hope???) be able to identify $\operatorname{Sym} \mathcal{E}^\vee \simeq \operatorname{Sym} \mathcal{E}_c^!(U)$ with the derived critical locus, and should be able to interpret it as polynomial polyvector fields on $\mathcal{E}_c^!(U)$ in some way.

Terminology 4.3.2. Let M be a manifold. To specify a free field theory on M, we usualy simply writ ethe sheaf of sections $\mathcal{E}$ of the graded vector bundle $\pi : E \to M$, and will refer to the tuple $(M, \mathcal{E})$ as a free field theory. This is an abuse of notation since it leaves the differential $\mathrm{d} : \mathcal{E} \to \mathcal{E}$ and the isomorphism $E \cong E^![-1]$ implicit, but is sufficient for our purposes.

The manifold M intuitively represents *spacetime*. As such, we can take M to be finite-dimensional, meaning the struggles of doing infinite dimensional differential geometry have been replaced with the language of sheaves of sections of vector bundles, and derived algebraic geometry. The sheaf of sections $\mathcal{E}$ is called the *derived space of fields*. When paired with the differential $(\mathcal{E}, \mathrm{d})$, we call it the *derived space of solutions to the Euler-Lagrange equations*. In practice, we rarely distinguish between th e two notions, and use them interchangeably. The intuition is that we will take an action functional S, extract the Euler-Lagrange equations of motion, and then encode it into the complex $(\mathcal{E}, \mathrm{d})$. For example, in the scalar field theory example, the action functional yielded the equations of motion $\Delta\phi = 0$, which we encoded into $\mathcal{E}$ by having $H^0(\mathcal{E}) = \operatorname{Ker} \Delta$ be the classical critical locus.

Remark 4.3.3. As noted previously, we assume that the graded vector bundle E is finite rank, such that we avoid annoying congergence issues. The pairing $\langle -, - \rangle : \mathcal{E}_c \otimes_{C^\infty_M} \mathcal{E}^! \to \mathbb{R}$ is defined as follows. Let $s \in \mathcal{E}_c(U)$, and $\alpha \otimes \omega \in \mathcal{E}_c^! = \mathcal{E}(U)^\vee \otimes_{C^\infty_M(U)} \Omega^n_M(U)$, where we assume $M = (M, g)$ is Riemmanian to make the presentation more simple. There are no changes in the fully general case. Then we have that $\alpha = \left\{\alpha^i : \mathcal{E}(U)^{-i} \to \mathbb{R}\right\}_{i\in\mathbb{Z}}$.

The direct sum perspective of the total space of a graded vector bundle is very useful here. Recall that $\mathcal{E}^\vee = \Gamma(-, E^\vee)$, and $E^\vee$ is defined fiberwise by

$$E_x^\vee = \underline{\operatorname{Hom}}_{\mathbb{R}}(E_x^{-i}, \mathbb{R}).$$

The direct sum perspective implies that

$$\alpha_x \in \bigoplus_{i\in\mathbb{Z}} (E_x^{-i})^*.$$

By writing this as an element of a direct sum, we conclude that α can be written fiberwise as a linear combination of maps

$$\alpha_x = \sum_{i\in\mathbb{Z}} \left(\alpha_x^i : E_x^{-i} \to \mathbb{R}\right),$$

where of course $\alpha_x^i \in (E_x^{-i})^*$. Thus we can act an element α of the graded dual $\mathcal{E}^\vee(U)$ on a section $s \in \mathcal{E}(U)$ to obtain a pairing $\mathrm{ev} : \mathcal{E}^\vee \otimes_{\mathbb{R}} \mathcal{E} \to \mathbb{R}$:

$$\mathrm{ev}(\alpha, s) = \alpha(s) = \sum_{i\in\mathbb{Z}} \alpha^{-i}(s^i).$$

Now we can define the pairing. If s is compactly supported, then $\alpha(s)$ is a compatly supported map $M \to \mathbb{R}$. This is easy to show by simply using $\alpha_x^i(0) = 0$. Thus, $\alpha(s)$ defines a derivation $(\alpha(s))^?$ by integrating against it. We have to integrate n-forms, so this determines a map $\Omega_M^n(U) \to \mathbb{R}$. This map is $C_M^\infty(U)$-linear:

$$(\alpha(s))^?(f\omega) = \int_U \alpha(s) f\omega = (f \cdot \alpha(s))^?(\omega).$$

Transposing under $- \otimes_{C_M^\infty} V \dashv \mathrm{Hom}_{C_M^\infty}(V, -)$ yields the pairing

$$\langle s, \alpha \otimes \omega \rangle_U = \int_U \alpha(s)\omega = \int_U \sum_{i \in \mathbb{Z}} \alpha^{-i}(s^i)\omega.$$

Now we show that this pairing is indeed a valid morphism in the category $\mathrm{Fun}(\mathcal{O}\mathrm{pen}(M), \mathcal{G}\mathcal{M}\mathrm{od}_{C_M^\infty})$. That is, we must show that for all open $U \subseteq M$, the map $\langle -, - \rangle_U$ can be defined degreewise as a map $\mathcal{E}_c \otimes_{C_M^\infty(U)} \mathcal{E}^! \to \mathbb{R}[0]$. Indeed, recall that if $\alpha^{-j} \in \mathcal{E}^\vee(U)^{-j}$ and $s^i \in \mathcal{E}(U)^i$, then by definition of the graded dual, $\alpha^{-j}(s^i) = \delta_{ij}\alpha^{-i}(s^i)$, for δ_{ij} the Kronecker delta. Thus the map vanishes unless $j = i$, meaning we obtain a map $\mathcal{E}_c^i \otimes_{C_M^\infty(U)} (\mathcal{E}^!)^{-i} \to \mathbb{R}$. Thus this map respects degrees.

We now define $\mathrm{d}^!$ as the (graded) adjoint of d with respect to this pairing. That is, $\mathrm{d}^!$ is the unique map $\mathcal{E}^! \to \mathcal{E}^!$ with the property that

$$\langle \mathrm{d}s, \sigma \rangle + (-1)^{|s|}(s, \mathrm{d}^!\sigma) = 0.$$

We can obtain an explicit description of $\mathrm{d}^!$. Start by assuming that s has even degree, so we can ignore the factor $(-1)^{|s|}$. Let $U \subseteq M$ be open, and let $s \in \mathcal{E}(U)^{i-1}$ and $\alpha \in \mathcal{E}^\vee(U)$. Then since d has cohomological degree 1, $\alpha(\mathrm{d}s)$ is defined. Thus we may consider the map

$$\langle \mathrm{d}s, \alpha \otimes \omega \rangle = \int_U \alpha(\mathrm{d}s)\omega$$

for a volume form ω. To explicitly describe $\mathrm{d}^!$, note that the pairing can be described as

$$\langle -, - \rangle = \int_U m \circ (\mathrm{ev} \otimes 1),$$

where $m : C_M^\infty(U) \otimes_{C_M^\infty(U)} \Omega_M^n(U) \to \Omega_M^n(U)$ is the multiplication map, $\mathrm{ev} : \mathcal{E}^\vee(U)^{-i} \otimes_\mathbb{R} \mathcal{E}_c^i \to C_M^\infty(U)$ is the evaluation map, and 1 is the identity map.

If we take $\mathrm{d}^!$ to be such that $\mathrm{d}_U^!(\alpha \otimes \omega)$ is the map $s \mapsto (\alpha(\mathrm{d}s) \otimes \omega)$, then d and $\mathrm{d}^!$ are necessarily adjoints when passing to the integral. We want to simply define $\mathrm{d}^! := \mathrm{d}^* \otimes 1$, for $(-)^*$ precomopsition. If we do this, then

$$\mathrm{d}^!(\alpha \otimes \omega) = (\alpha \circ \mathrm{d}) \otimes \omega.$$

The pairing is thus (up to a factor of $\pm$ depending on the degree):

$$\left\langle s, \mathrm{d}^!(\alpha \otimes \omega) \right\rangle = \int_U (\alpha \circ \mathrm{d})(s)\omega = \alpha(\mathrm{d}s)\omega = \langle \mathrm{d}s, \alpha \otimes \omega \rangle.$$

Unfortunately, $\alpha \circ \mathrm{d}$ need not be a section of the graded dual vector bundle, so this is in general ill-defined. However, a formal adjoint always exists, meaning we can always find an operator which equates to $\mathrm{d}^* \otimes 1$ when passing to the integral. Thus we have the best description of our map that we can obtain without passing to examples. Moreover, note that $\mathrm{d}^!(\alpha \otimes \omega) : \mathcal{E}^{i-1} \to \mathbb{R}$, and thus this is an element of homogeneous degree $-(i-1) = -i + 1$. Thus $\mathrm{d}^! : \mathcal{E}^! \to \mathcal{E}^![1]$.

Consider the scalar field theory. The graded vector bundle E was the trivial bundle ocncentrated in degrees 0 and 1, such that $\mathcal{E}$ was simply C^∞_M concentrated in degrees 0 and 1. The differential $\mathrm{d} : C^\infty_M \to C^\infty_M$ was $\Delta + m^2$, for Δ the Laplacian on the Riemannian manifold (M, g).

Note that we can identify $(\mathbb{R} \times M)^* = \mathbb{R}^* \otimes M$ with $\mathbb{R} \otimes M$ using the canonical basis for $\mathbb{R}$ given by the element 1. Thus we obtain

$$\mathcal{E}^\vee \cong \left(\; C^\infty_M[1] \xrightarrow{\Delta + m^2} C^\infty_M \; \right).$$

The isomorphism $\mathbb{R}^* \times M \cong \mathbb{R} \times M$ is given by evaluation, meaning a section α of $\mathbb{R}^* \times M$ is identified with a section of $\mathbb{R} \times M$ by evaluating fiberwise:

$$\alpha_x \mapsto \alpha_x(1).$$

Now note that $\Omega^n_M \cong \mathrm{span}_{C^\infty_M}(\mathrm{vol}_g)$. Thus we have another canonical identification $C^\infty_M \otimes_{C^\infty_M} \Omega^n_M$ by simply mapping $f \mapsto f \otimes \mathrm{vol}_g$. This yields

$$\mathcal{E}^! \cong \left(\; C^\infty_M[1] \xrightarrow{\Delta + m^2} C^\infty(M) \; \right)$$

where we identify $f \in C^\infty_M(U)$ with $\alpha \otimes \omega \in \mathcal{D}(U) \otimes \Omega^n_M(U)$ by defining $\omega = \mathrm{vol}_g$, and since $\alpha_x(1) = f(x)$, we identify $\alpha(g) = f \cdot g$. Thus the pairing is

$$\langle f, g \rangle_U = \int_U f g \mathrm{vol}_g.$$

We then recover that $\mathrm{d}^!$ is also Δ since the Laplacian is self-adjoint.

Remark 4.3.4. Condition (4) of definition 4.3.1 yields an isomorphism $E \cong E^![-1]$, which necessarily induces an isomorphism $\varphi : \mathcal{E} \cong \mathcal{E}^![-1]$ on sheaves of sections. We can define a canonical pairing $\omega : \mathcal{E}_c \otimes_{C^\infty_M} \mathcal{E}_c \to \mathbb{R}[0]$ is given by

$$\omega(s^i, s^j) := \left\langle s^i, \varphi(s^j) \right\rangle.$$

We investigate what this means. First, we assume that $s^i \in \mathcal{E}(U)^i$ and $s^j \in \mathcal{E}(U)^j$. Then $\varphi(s^j) \in (\mathcal{E}^!(U))^{j-1}$. Thus we can write $\varphi(s^j) = \alpha^{-j+1} \otimes \omega$, where $\alpha^{-j+1} : \mathcal{E}(U)^{-j+1} \to \mathbb{R}$. Hence we see that $\omega(s^i, s^j)$ is going to vanish unless $-j + 1 = i$. Hence we can write

$$\omega(s^i, s^{-i+1}) = \left\langle s^i, \varphi(s^{-i+1}) \right\rangle.$$

We also denote this by $\langle\langle -, - \rangle\rangle$ sometimes. We note that this is a map of cohomological degree -1, since $s^i \otimes s^{-i+1}$ has degree 1, and it maps into $\mathbb{R}[0]$.

Condition (4) also dictates that

$$\omega(s^i, s^j) = (-1)^{|s^i||s^j|} \omega(s^j, s^i).$$

Applying this to the case where ω is nonzero:

$$\begin{aligned} \omega(s^i, s^{-i+1}) &= -(-1)^{i(-i+1)} \omega(s^{-i+1}, s^i) = -(-1)^{-i^2+i} \omega(s^{-i+1}, s^i) \\ &= -(-1)^{-i+i} \omega(s^{-i+1}, s^i) = \omega(s^{-i-1}, s^i). \end{aligned}$$

Thus “graded antisymmetric” in this case means ”symmetric in the normal way that you would normally think of it”.

Remark 4.3.5. We remark on the actual meaning of the conditions in definition 4.3.1. Conditions (1) and (2) simply set up a system of linear differential equations, albeit using the advanced machinery of sheaf theory and homological algebra. However, they do not inherently carry any physical meaning. Sure, defining the differential to encode the equations of motion can write a physical theory as such a mathematical object, but the converse does not follow. We need more conditions in order for observables to make sense. This is what condition (4) does: it provides an identification that is necessary to obtain physical meaning. It is difficult to intuitively explain why this identification exists at this time, however. The thing that makes the theory *free* is the fact that the sheaf $\mathcal{E}$ is valued in cochain complexes of *vector spaces*. Interacting theories are usually modeled in L_∞-algebras.

For the next example, recall the following.

Definition 4.3.6. Let $\pi : E \to M$ be a vector bundle over a manifold M. A *connection* on E is an $\mathbb{R}$-linear map $\nabla : \Gamma(M, E) \to \Gamma(M, T^*M \otimes_{\mathbb{R}} E)$ which satisfies a product rule

$$\nabla(fs) = \mathrm{d}f \otimes s + f\nabla s$$

for all $f \in C^\infty(M)$ and $s \in \Gamma(M, E)$.

Definition 4.3.7. Let (M, g) be a pseudo-Riemmanian n-manifold. Then for all k, the *Hodge dual* of a k-form ζ is the $(n-k)$-form $\star\zeta$ which satisfies the property

$$\eta \wedge \star\zeta = g(\eta, \zeta)\mathrm{vol}_g$$

for all k-forms η.

Here we define $g(\eta, \zeta)$ in local coordinates as $\sum_{i,j} g^{ij}\eta_i\zeta_j$, where g^{ij} are the matrix components of the inverse matrix of $g_{ij} = g(\partial_i, \partial_j)$.

Apparently we can use the Hodge star to define the *coderivative* $\Omega^k_M \to \Omega^{k-1}_M$, given by $\delta = (-1)^k \star^{-1} \mathrm{d}\star$. The relevance of this map is that if $\langle\langle \eta, \zeta\rangle\rangle$ is the L^2 inner product

$$\langle\langle \eta, \zeta\rangle\rangle = \int_U \eta \wedge \star\zeta;$$

here $\eta, \zeta \in \Omega^k_M(U)$; then δ is the right adjoint of d under this inner product. Then the *Hodge Laplacian* $\mathrm{d} \star \mathrm{d}$ is defined as $\mathrm{d} \star \mathrm{d} \coloneqq \mathrm{d}\delta + \delta\mathrm{d} = (\mathrm{d} + \delta)^2$.

Example 4.3.8. We give an example of how to encode gauge symmetry into a free field theory. We describe Abelian Yang-Mills theory with gauge group $\mathbb{R}$. Let (M, g) a Riemannian 4-manifold. The "naive fields" are 1-forms on M ($A \in \Omega^1_M$), which we think of as connections on the trivial line bundle $\mathbb{R} \times M$ via using $\mathrm{d} + A$ as the connection corresponding to the 1-form A.

This is indeed a connection:

$$(\mathrm{d} + A)(f) = \mathrm{d}f + Af;$$

here we use the canonical isomorphism $TM \cong T^*M$ for Riemannian manifolds to define Af for 0-forms f. We then have $T^*M \otimes_{\mathbb{R}} (M \times \mathbb{R}) \cong T^*M$, so the product rule is immediate or something idk.

The abelian Yang-Mills action is

$$S_{\mathrm{YM}}(A) = -\frac{1}{2}\int_M \mathrm{d}A \wedge \star\mathrm{d}A = \frac{1}{2}\int_M A(\mathrm{d} \star \mathrm{d})A,$$

where $\star \mathrm{d}A$ is the Hodge star on $\mathrm{d}A$, and $\mathrm{d} \star \mathrm{d}$ is the *Hodge Laplacian on 1-forms (up to another factor of $\star$).* We can prove this equality:

$$\mathrm{d}(A \wedge \star \mathrm{d}A) = \mathrm{d}A \wedge \star \mathrm{d}A - A \wedge \mathrm{d}(\star \mathrm{d}A).$$

Integrating both sides, and applying Stokes' theorem to the left, and also assuming that M has no boundary yields

$$\int_M \mathrm{d}A \wedge \star \mathrm{d}A = \int_M A \wedge \mathrm{d}(\star \mathrm{d})A.$$

The notation $A(\mathrm{d} \star \mathrm{d})A$ is an abuse of notation for $A \wedge \mathrm{d}(\star \mathrm{d}A)$.

The Euler-Lagrange equations here are that $(\mathrm{d} \star \mathrm{d})A = 0$. The derivation here is mildly annoying, and we omit it. This theory has a symmetry, which is that if $X \in \Omega^0_M$, then $A \mapsto A + \mathrm{d}X$. This is a symmetry because $\mathrm{d}^2 X = 0$, so plugging in $A + \mathrm{d}X$ into the action yields

$$S_{\mathrm{YM}}(A + \mathrm{d}X) = S_{\mathrm{YM}}(A).$$

Hence these configurations would be considered *physically equivalent*, yielding a group of symmetries on the naïve fields of our theory.

The derived space of fields representing this situation is given by the complex

$$\mathcal{E} : \Omega^0(M)[1] \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}\star\mathrm{d}} \Omega^3(M)[-1] \xrightarrow{\mathrm{d}} \Omega^4(M)[-2]$$

We explain why. Note that similarly to the scalar field theory example, the map $\mathrm{d} \star \mathrm{d}$ going from degree 0 to degree 1 cuts out the classical critical locus as its kernel $\mathrm{Ker}(\mathrm{d} \star \mathrm{d})$. Hence if there was no degree -1 component (as there was in the scalar field theory), we could recover the classical space of solutions as $H^0(\mathcal{E})$. This justifies the term "derived space of solutions" or "derived space of fields" (which we use more or less interchangeably).

However, we also have our collection of symmetries which yield physically equivalent theories. Hence we should set A equal to $A + \mathrm{d}X$ when passing to cohomology. The way we do this is by adding a degree -1 component with image in $\Omega^1(M)$ given by $\mathrm{d}\Omega^0(M)$, which is exactly what we see in the derived space of fields. Thus we obtain

$$H^0(\mathcal{E}) = \frac{\text{Fields}}{\text{Gauge symmetries}}.$$

The last component will not be explained yet.

The idea we obtain from example 4.3.8 is that classical fields can be recovered as the homology of the derived space of fields. Additionally, the derived space of fields gives us the necessary machinery required to encode the notion of a *Gauge symmetry* into our field theory.

Definition 4.3.9. Let M have a free field theory on it with derived space of fields $\mathcal{E}$. Then the *classical observables* of the theory are defined as

$$\mathrm{Obs}^{cl}(U) = \mathrm{Sym}\left(\mathcal{E}^!_c(U)\right) \cong \mathrm{Sym}\left(\mathcal{E}_c(U)[1]\right).$$

We explore definition 4.3.9. First, we explore the isomorphism. If (M, g) is Riemannian, then

$$\mathcal{E}^!_c = \Gamma^\infty_c(-, E^\vee \otimes_{\mathbb{R}} \Lambda^n T^* M) \cong \Gamma^\infty_c(-, E^\vee) \otimes_{C^\infty_M} \Gamma^\infty_c(-, \Lambda^n T^* M) = \mathcal{E}^\vee_c \otimes_{C^\infty_M} (\Omega^n_M)_c.$$

Our intuition follows from the scalar free field theory example. The main reason that we were able to identify $\mathcal{E}^!_c$ with $\mathcal{E}^\vee(U)$ (which we expect to be observables) up to quasi-isomorphism is the isomorphism $\mathcal{E}^! \cong \mathcal{E}[-1]$. Since this aspect of a free field theory is part of the definition, we always have a notion of smeared observables on a free field theory.

Lemma 4.3.10. *Let $(M, \mathcal{E})$ be a free field theory on a manifold M. Then the classical observables Obs^{cl} form a prefactorization algebra valued in commutative differential graded $\mathbb{R}$-algebras.*

Proof. First, we recall that the symmetric algebra is by definition a commutative graded algebra. Additionally, the differential $\mathrm{d}^! : \mathcal{E}^! \to \mathcal{E}^!$ extends immediately by simply defining

$$\mathrm{d}^!(ab) = (\mathrm{d}^! a)b + (-1)^{|a|} a(\mathrm{d}b)$$

for all homogeneous elements a, b. Thus $\mathrm{Sym}\, \mathcal{E}^!_c$ is a precosheaf of commutative differential graded algebras. To be explicit, we describe the algebra homomorphisms $i^U_V : \mathrm{Obs}^{cl}(U) \to \mathrm{Obs}^{cl}(V)$ induced by inclusions $U \subseteq V$ of open sets. Let $s \in \mathcal{E}^!_c(U)$. Then one can define $s|(V \setminus U) := 0$, and the extension is still compactly supported. One then extends this to the entire symmetric algebra by simply defining

$$i^U_V(st) = i^U_V(s) i^U_V(t).$$

This is immediately seen to respect the degrees, so we need only show that it respects the differentials.

By the universal property of the symmetric algebra, it suffices to show this on the generators $\mathcal{E}^!_c(U)$. Let $s \in \mathcal{E}^!_c(U)^i$; then $\mathrm{d}^! s \in \mathcal{E}^!_c(U)^{i+1}$. Since extension morphisms act trivially on s on the original domain (in the sense that $(i^U_V s)|U = s$), it suffices to show that $\mathrm{d}^! i^U_V s$ and $i^U_V \mathrm{d}^! s$ agree on $V \setminus U$. Since $(i^U_V \mathrm{d}^! s)|(V \setminus U) = 0$, it suffices to show that $(\mathrm{d}^! i^U_V s)|(V \setminus U) = 0$.

We use the fact that $\mathrm{d}^!$ is a morphism of sheaves, meaning it commutes with the restriction maps. In particular, suppose that there is some open set $W \subseteq V$ such that $s|W = 0$. Then since the empty set is compact, $s|W$ is compactly supported, and moreovoer $(s|W)_x = s_x$ by the explicit structure of $\mathcal{E}^!$. We then obtain for any $x \in W$,

$$(\mathrm{d}^! s)_x = ((\mathrm{d}^! s)|W)_x = (\mathrm{d}^!(s|W))_x = (\mathrm{d}^! 0)_x = 0.$$

Thus if $s|W = 0$, then $(d^! s)|W = 0$.

Although $V \setminus U$ need not be an open set, we know that $V \setminus \mathrm{supp}\, s = V \cap (\mathrm{supp}\, s)^c$ is a finite intersection of open sets, and is thus open. Thus we can take $W = V \setminus \mathrm{supp}\, s = V \setminus \mathrm{supp}(i^U_V s)$, and use the fact that $(i^U_V s)|W = 0$ to obtain that $(\mathrm{d}^! i^U_V s)|W = 0$. Since $\mathrm{supp}\, s \subseteq U$, we obtain $V \setminus U \subseteq W$, proving the statement. Therefore, Obs^{cl} is a precosheaf valued in commutative differential graded algebras.

Now we prove Obs^{cl} is a prefactorization algebra. We recall that this means for every finite collection of pairwise disjoint open sets $\{U_i\}_{1 \le i \le n}$ and every open set V containing U_i for all i, there is a structure map $\bigotimes_i \mathrm{Obs}^{cl}(U_i) \to \mathrm{Obs}^{cl}(V)$ which satisfies the obvious coherence axiom. We define this map by defining the n-multilinear map

$$\begin{aligned} \prod_{1 \le i \le n} \mathrm{Obs}^{cl}(U_i) &\longrightarrow \mathrm{Obs}^{cl}(V) \\ (\alpha_1, \ldots, \alpha_n) &\longmapsto \prod_{1 \le i \le n} i^{U_i}_V \alpha_i, \end{aligned}$$

where the product $\prod_i i^{U_i}_V \alpha_i$ denotes the symmetric product in the symmetric algebra. Here we take each α_i to be a homogeneous element of $\mathrm{Obs}^{cl}(U)$ of arbitrary (but not necessarily the same!) degree. Universality allows this to descend to a map on the graded tensor product. Additionally, by definition of the differential on the product of cochain complexes, and by definition of the differential on the symmetric algebra, this is a morphism of cochain complexes, meaning the induced map is also a morphism of cochain complexes.

It is important to note that on fields $\phi \in \mathcal{E}(V)$, we define the action of $\prod_i i_V^{U_i} \alpha_i$ on ϕ as follows:

$$\left(\prod_{1 \leq i \leq n} i_V^{U_i} \alpha_i \right) (\phi) = \prod_{1 \leq i \leq n} \alpha_i(\phi | U_i),$$

where the product on the right is usual multiplication in $\mathbb{R}$. This makes physical sense, since one expects the product of the observables to be the... product of the observables. The coherence axiom is immediately checked by simply writing out the definitions. Hence classical observables form a prefactorization algebra valued in cochain complexes. ■

4.4 The one-dimensional case, in detail

We illustrate an example computing $H^0(\mathrm{Obs}^{cl})$ for a 1-dimensional free field theory on $\mathbb{R}$. We note that physically, this theory should reflect mechanics of a single particle in 1 dimension. Let $U \subseteq \mathbb{R}$ be an open subset. The action functional is

$$S_U(\phi) = \int_U \phi(\Delta + m^2)\phi \mathrm{d}x.$$

Here we note that $\mathrm{vol}_g = \mathrm{d}x$. We can write this in another form using interation by parts:

$$\int_U \phi \partial_x^2 \phi \mathrm{d}x = [\phi \partial_x \phi]_{\partial U} - \int_U (\partial_x \phi)(\partial_x \phi) \mathrm{d}x.$$

Recall that the action functional is generally ill-defined, and to recover mathematical intuition from it we use compactly supported sctions to extract the Euler-Lagrange equations. Hence for all mathematical purposes, we can take this to vanish on ∂U, meaning we can also take

$$S_U(\phi) = \int_U \left((\partial_x \phi)^2 + m^2 \phi^2 \right) \mathrm{d}x.$$

Our first important result in this section is the following.

Lemma 4.4.1. *If $U = (a, b) \subseteq \mathbb{R}$ is an open interval, then*

$$H^\bullet(\mathrm{Obs}^{cl}(U)) \cong \mathbb{R}[p, q]$$

is the polynomial algebra on two generators.

As we can see, we can interpret p as position and q as momentum, which makes physical sence since all observables can be written using position and momentum.

Proof. Recall that the derived space of fields for the scalar free field theory

$$\mathcal{E}(a, b) = \left(\; C^\infty(a, b) \xrightarrow{\Delta + m^2} C^\infty(a, b)[-1] \; \right).$$

Recall that up to quasi-isomorphism, we can identify

$$\mathcal{E}_c^!(a, b) = \left(\; C_c^\infty(a, b)[1] \xrightarrow{\Delta + m^2} C_c^\infty(a, b) \; \right).$$

We will show that the second complex is quasi-isomorphic to $\mathbb{R}^2$ concentrated in degree 0. The result then follows.

In the book, we use the convention

$$\Delta = -\left(\frac{\partial}{\partial x}\right)^2.$$

Thus the equations of motion say

$$\left(-\frac{\partial^2}{\partial x^2} + m^2\right)\phi = 0.$$

We can solve this differential equation because it is very easy. The solutions depend on if $m = 0$, $m > 0$, or $m < 0$. If $m = 0$, then ϕ is a linear function, so the solutions are spanned by 1 and x. If $m > 0$, then we want

$$\frac{\partial^2\phi}{\partial x^2} = m^2\phi.$$

The solutions are spanned by $e^{\pm mx}$. Similarly, if $m < 0$, then they are spanned by $e^{\pm imx}$.

We use a nice basis here. If $m = 0$, then define $\phi_q(x) = 1$ and $\phi_p(x) = x$. If $m > 0$, then choose the basis

$$\phi_p(x) = \frac{1}{2m}\left(e^{mx} - e^{-mx}\right), \qquad \phi_q(x) = \frac{1}{2}\left(e^{mx} + e^{-mx}\right).$$

The point of this choice is that the adjoint of the relations between position and momentum that are satisfied in physics are precisely satisied by ϕ_p as position and ϕ_q as momentum. For example,

$$\phi_p(0) = 1, \qquad \phi_q(0) = 0, \qquad \phi_q'(0) = 1, \qquad \phi_p' = \phi_q.$$

Without loss of generality, assume $(a, b) = (-1, 1)$. Define $\pi : \mathcal{E}_c(-1, 1) \to \operatorname{Span}(p, q) =: \mathbb{R}\{p, q\}$ by

$$g \mapsto \pi(g) = \left(\int_{\mathbb{R}} g\phi_q \mathrm{d}x\right) q + \left(\int_{\mathbb{R}} g\phi_p \mathrm{d}x\right) p.$$

We claim that this defines a cochain map

$$\begin{tikzcd}
C_c^\infty(-1,1) \arrow[r, "\Delta+m^2"] \arrow[d, "\pi^{-1}"'] & C_c^\infty(-1,1) \arrow[d, "\pi^0=\pi"] \\
0 \arrow[r] & \mathbb{R}^2\{p,q\}.
\end{tikzcd}$$

Indeed, we can use the fact that $\Delta + m^2$ is self adjoint to show that

$$\begin{aligned}
\pi((\Delta + m^2)(g)) &= q\int_{\mathbb{R}} (\Delta + m^2)g\phi_q \mathrm{d}x + p\int_{\mathbb{R}} (\Delta + m^2)g\phi_p \mathrm{d}x \\
&= q\int_{\mathbb{R}} g(\Delta + m^2)\phi_q \mathrm{d}x + p\int_{\mathbb{R}} g(\Delta + m^2)\phi_p \mathrm{d}x.
\end{aligned}$$

Since ϕ_p and ϕ_q solve $(\Delta + m^2)\phi = 0$, this is necessarily zero. Therefore this is a map of cochains. We claim that π is a quasi-isomorphism if $\operatorname{Ker}\pi$ is acyclic, meaning $H^\bullet(\operatorname{Ker}\pi) \cong 0$. To see this, write the cochain complex $0 \to \mathbb{R}^2\{p, q\}$ as D, and then surjectivity of π necessarily yields a short exact sequence of cochain complexes

$$0 \longrightarrow \operatorname{Ker}\pi \longrightarrow \mathcal{E}_c \longrightarrow D \longrightarrow 0.$$

Taking cohomology yields the long exact sequence

$$0 \longrightarrow H^{-1}(\operatorname{Ker}\pi) \longrightarrow H^{-1}(\mathcal{E}_c) \longrightarrow H^{-1}(D) \longrightarrow H^0(\operatorname{Ker}\pi) \longrightarrow H^0(\mathcal{E}_c) \longrightarrow H^0(D) \longrightarrow 0.$$

If we assume that $H^\bullet(\operatorname{Ker}\pi) \cong 0$, then this reduces to the two exact sequences

$$0 \longrightarrow H^{-1}(\mathcal{E}_c) \longrightarrow H^{-1}(D) \longrightarrow 0$$
$$0 \longrightarrow H^0(\mathcal{E}_c) \longrightarrow H^0(D) \longrightarrow 0,$$

and exactness implies these are isomorphisms. The definition of the long exact sequence tells us that these maps are precisely the homology of π, meaning that π lowers to an isomorphism in cohomology, and is thus a quasi-isomorphism. It suffices to show this.

It suffices to show that $\operatorname{Ker}\pi$ is contractible. Recall that if $\alpha, \beta : A \to B$ are cochain maps, then a homotopy $h : \alpha \Rightarrow \beta$ is a cochain map $h : A \to B[-1]$ such that $f - g = d_B \circ h + h \circ d_A$. In our case, we require that $1 - 0 = d_{\mathcal{E}} \circ h + h \circ d_{\operatorname{Ker}\pi}$ for 1 the identity. The map $\operatorname{Ker}\pi^{-1} \to \operatorname{Ker}\pi^0$ is just the restriction of $\Delta + m^2$. Note that since h goes backwards in degree, the only h that can be nonzero is the map $\operatorname{Ker}\pi^0 \to C_c^\infty(-1,1)[1]$. Thus we simply need a map $h : C_c^\infty(-1,1) \to C^\infty(-1,1)$ which takes degree 0 elements to degree -1 elements, and it must satisfy $1 = d_{\mathcal{E}} \circ h$, meaning

$$(\Delta + m^2)h(g) = g.$$

We recall the theory of Green's functions. Let L_x be a differential operator on a manifold M and $f : M \to \mathbb{R}$ compactly supported. A *Green's function* is a distribution (not a function because it could have singularities, so it only makes sense to integrate against compactly supported forms) $G : M \times M \to M$ with $L_x G(x,y) = \delta(x-y)$. What we mean by this, at least in the case of $M = \mathbb{R}$, is as follows.

$$L \int_{\mathbb{R}} G(x,y) g \mathrm{d}y = \int_{\mathbb{R}} LG(x,y) g \mathrm{d}y = \int_{\mathbb{R}} \delta(x-y) g(y) \mathrm{d}y = g(x).$$

Here we must commute the differential operator with the integral using some theorem from real analysis that does not always apply. If g is suffiiently nice, then it does.

We apply this to our case. Define $G(x,y) = \widetilde{G}(x-y)$, and define $\widetilde{G}(x) = \frac{m}{2} e^{-m|x|}$ when $m \neq 0$, and $\widetilde{G}(x) = -\frac{1}{2}|x|$ when $m = 0$. This is a Green's function for $\Delta + m^2$. for example, we check this when $m = 0$. We will choose a compactly supported test function g such that we can do integration by parts and the boundary terms will vanish. Additionally, copactly supported functions vanish at $\pm\infty$. We have

$$\begin{aligned}
\int_{\mathbb{R}} \Delta(\widetilde{G}) \cdot f \mathrm{d}x &= \int_{\mathbb{R}} \widetilde{G} \cdot \Delta f \mathrm{d}x \\
&= \int_{\mathbb{R}} -\frac{1}{2}|x| \left(-\frac{\partial^2 f}{\partial x^2}\right) \mathrm{d}x \\
&= -\frac{1}{2}\left(\int_{-\infty}^0 -x\left(-\frac{\partial^2 f}{\partial x^2}\right)\mathrm{d}x + \int_0^\infty x\left(-\frac{\partial^2 f}{\partial x^2}\right)\mathrm{d}x\right) \\
&= -\frac{1}{2}\left(\int_{-\infty}^0 x\frac{\partial^2 f}{\partial x^2}\mathrm{d}x + \int_\infty^0 x\frac{\partial^2 f}{\partial x^2}\mathrm{d}x\right) \\
&= -\frac{1}{2}\left(-\int_{-\infty}^0 \frac{\partial f}{\partial x}\mathrm{d}x - \int_\infty^0 \frac{\partial f}{\partial x}\mathrm{d}x\right) \\
&= \frac{1}{2} 2 f(0) = f(0).
\end{aligned}$$

Therefore there exists a Green's function for Δ, and π is indeed a quasi-isomorphism. ∎

Lecture 14: Poisson brackets and quantization

Tue 31 Mar 2026 14:01

4.5 The Poisson bracket

Definition 4.5.1. A Poisson R-algebra A is a commutative algebra together with a Lie bracket $\{-,-\} : A \otimes_R A \to A$ whihch acts as a derivation in each input:

$$\{ab, c\} = \{a, c\}\, b + \{b, c\}\, a,$$

$$\{a, bc\} = \{a, b\}\, c + \{a, c\}\, b.$$

Example 4.5.2. Let V a vector space with a skew form $\langle -,-\rangle : V \otimes V \to V$. Then there is a unique way to extend $\langle -,-\rangle$ to a Poisson bracket on the symmetric algebra $\mathrm{Sym}(V)$. Simply define $\{v, w\} = \langle v, w\rangle$, and inductively impose the derivation condition on higher degree elements.

Lemma 4.5.3. *Let $(M, \mathcal{E})$ be a free field theory on a manifold M. Let $\langle -,-\rangle : \mathcal{E}_c \otimes_{\mathbb{R}} \mathcal{E}_c \to \mathcal{E}_c$ be the skew pairing of cohomological degree 1 as described in remark 4.3.4. Then there is a unique smooth Poisson bracket on $\mathrm{Obs}^{cl}(U)$ of cohomological degree 1, which satisfies*

$$\{\alpha, \beta\} = \langle \alpha, \beta\rangle$$

for any observables $\alpha, \beta \in \mathcal{E}_c(U)[1]$.

Proof. We will not need the smooth part of this hypothesis, so the proof simly reduces to adapting example 4.5.2 to a graded situation. ■

Classical theories should carry a Poisson bracket, so this makes sense. Note that $\left\{ i_V^{U_1}\alpha, i_V^{U_2}\beta \right\} = 0$ whenever $U_1 \cap U_2 = \varnothing$ (not too hard to show, just recall definitions). Interpreting this physically means that observables on disjoint open subsets commute with respect to the Poisson bracket.

4.6 The quantum observables of a free field theory

In our toy model, we defined a Heisenberg dg Lie algebra $\mathcal{H}_W$ as a central extension of our compactly supported sections of $\mathcal{E}$, with Lie bracket determined by the canonical pairing. We follow this intuition in great detail to define Obs^q for an arbitrary free field theory $(M, \mathcal{E})$.

Construction 4.6.1. Recall that we have a graded antisymmetric pairing $\omega : \mathcal{E}_c \otimes_{\mathbb{R}} \mathcal{E}_c \to \mathbb{R}$. For the explicit construction, see remark 4.3.4. We want to construct a central extension $\widehat{\mathcal{E}}_c$ of $\mathcal{E}_c$. Such a construction is done pointwise, meaning for all open sets $U \subseteq M$, we have a central extension $\widehat{\mathcal{E}}_c(U)$ of $\mathcal{E}_c(U)$.

In order for such a central extension to make sense, we need $\mathcal{E}_c(U)$ to be a dg Lie algebra. Take $\mathcal{E}_c$ to be an abelian dg Lie algebra. Just like with Lie algebras, a central extension of a dg Lie algebra $\mathfrak{g}$ is a short exact sequence of dg Lie algebras

$$0 \longrightarrow \mathfrak{h} \longrightarrow \mathfrak{e} \longrightarrow \mathfrak{g} \longrightarrow 0.$$

Over a field, short exact sequences always split (since there is an underlying short exact sequence of graded vector space, which pointwise is a short exact sequence of vector spaces).

Recall that central extensions correspond to 2-cocycles. Since ω respects the differential, it is a 2-cocycle, meaning it defines a central extension. Since it has cohomological degree -1, and we want to define $[x, y] = \omega(x, y)$, we have to introduce a generator $\hbar$ of degree 1 to absorb the shift, ensuring that the resulting dg Lie bracket preserves degrees. We define $[x, y] = \hbar\omega(x, y)$, which now preserves degrees. Since $\hbar$ lives in degree 1, we find that $\mathcal{H}_{\mathcal{E}}$ arises as the central extension

$$0 \longrightarrow \mathbb{R}\hbar\,[-1] \longrightarrow \mathcal{H}_{\mathcal{E}} \longrightarrow \mathcal{E}_c \longrightarrow 0.$$

Since sequences split, $\mathcal{H}_{\mathcal{E}} \cong \mathbb{R}\hbar[-1] \oplus \mathcal{E}_c$. The dg Lie bracket is as defined: $[x, y] = \hbar\omega(x, y)$.

Now that we have a natural central Heisenberg extension of the dg Lie algebra, we define

$$\mathrm{Obs}^q(U) := C_\bullet(\mathcal{H}_{\mathcal{E}}(U)).$$

We define the Chevalley-Eilenberg chains using the completed tensor product. This yields isomorphisms of graded vector spaces.

$$\mathrm{Obs}^q(U) \cong \mathrm{Sym}\,(\mathcal{H}_{\mathcal{E}}(U)[1]) \cong \mathrm{Obs}^{cl}(U)[\hbar].$$

There is a differential here, which we now consider. The differential of the Chevalley-Eilenberg chains is simply $\mathrm{d}_{\mathcal{H}_{\mathcal{E}}} + \mathrm{d}_{\mathrm{CE}}$, where of coure d_{CE} acts by $\mathrm{d}_{\mathrm{CE}}(x \wedge y) = [x, y] = \hbar\omega(x, y)$, which extends to higher order terms by applying the Leibniz rule. The operator $\mathrm{d}_{\mathcal{H}_{\mathcal{E}}}$ is simply the operator $0 \oplus \mathrm{d}$, where d is the differential operator on $\mathcal{E}$, restricted to $\mathcal{E}_c$.

We saw previously that $\mathrm{Obs}^{cl}(U)$ can be viewed as $\mathrm{PV}_c^\bullet$, with differential $\lrcorner\mathrm{d}S$. Introducing the $\hbar$ factor and the Chevalley-Eilenberg differential yields the new differential $\lrcorner\mathrm{d}S + \hbar\mathrm{Div}$.

Definition 4.6.2. Let $(M, \mathcal{E})$ be a free field theory, and let $\mathcal{H}_{\mathcal{E}}$ be as constructed in **??** 4.6.1. Then

$$\mathrm{Obs}^q(U) := C_\bullet(\mathcal{H}_{\mathcal{E}}(U)).$$

As always, the quantum observables form a prefactorization algebra, and they agree with what we saw in chapter 2.

Chapter 5

Quantum mechanics and the Weyl algebra

Recall the scalar field theory $\mathcal{E}_c = \left(\Delta + m^2 : C_c^\infty[1] \to C_c^\infty\right)$ on a single dimension $\mathbb{R}$. In a single dimension, the quantum observables are extremely simple.

Lemma 5.0.1. *The prefactorization algebra* Obs^q *on the scalar free field theory on* $\mathbb{R}$ *is locally constant.*

We begin by reviewing some machinery that we will need.

Definition 5.0.2. Let $(C^\bullet, d)$ a cochain complex. An *increasing filtration* on $C^\bullet$ is a sequence

$$F^0C^\bullet \subseteq F^1C^\bullet \subseteq F^2C^\bullet \subseteq \cdots \subseteq C^\bullet$$

of subcomplexes, such that

$$\bigcup_{n\geq 0} F^nC^\bullet = C^\bullet,$$

and the differential preserves each stage, meaning $d(F^kC^\bullet) \subseteq F^kC^\bullet$. The *associated graded complex* $\mathrm{gr}(C^\bullet, d)$ is the complex

$$\mathrm{gr}(C^\bullet, d) \coloneqq \bigoplus_{k\geq 0} \frac{F^kC^\bullet}{F^{k-1}C^\bullet}.$$

Note that d defines a differential $\widetilde{d}$ on this object $[\alpha_k] \mapsto [d\alpha_k]$. This is well-defined because d factors through the quotient.

Proof of lemma 5.0.1. There is apparently an isomorphism $H^0(\mathrm{Obs}^q(U)) \cong H^0(\mathrm{Obs}^{cl}(U))[\hbar]$, so this follows by local constant-ness of $H^0(\mathrm{Obs}^{cl})$: we recall that all open intervals map to $\mathbb{R}[p, q]$. ∎

We now review the Weyl algebra. A polynomial differential operator on $\mathbb{R}^n$ is an expression $\sum_\alpha r_\alpha(x)\partial_\alpha$, where α is a multi-index $(\alpha_1, \ldots, \alpha_n)$, and p_α is a polynomial in $x_1, \ldots, x_n$, and

$$\partial_\alpha \coloneqq \left(\frac{\partial}{\partial x_1}\right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_n}\right)^{\alpha_n}.$$

Denote by $\mathscr{D}(\mathbb{R}^n)$ the $\mathbb{R}$-algebra of all such polynomial differential operators on $\mathbb{R}^n$. For example,

$$\Delta + m^2 = -\left(\sum_i \frac{\partial^2}{\partial x_i^2}\right) + m^2 \in \mathscr{D}(m^2).$$

We will show that there is a Heisenberg algebra $\mathcal{H}_{\mathbb{R}^n}$ such that $\mathscr{D}(\mathbb{R}^n) \cong U(\mathcal{H}_{\mathbb{R}^n})$. A natural definition for $\mathcal{H}_{\mathbb{R}^n}$ is to consider the algebra

$$\mathcal{H}_{\mathbb{R}^n} = \operatorname{span}\{x_1, \ldots, x_n, \partial_1, \ldots, \partial_n\}$$

with Lie bracket

$$\left[\frac{\partial}{\partial x_i}, x_j\right] = \delta_{ij}, \qquad [x_i, x_j] = 0, \qquad \left[\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}\right] = 0.$$

The idea here is if $f \in C^\infty(\mathbb{R}^n)$, then by the product rule, we see that $C^\infty(\mathbb{R}^n)$ is an $\mathcal{H}_{\mathbb{R}^n}$-module:

$$\left[\frac{\partial}{\partial x_i}, x_j\right] = \frac{\partial}{\partial x_i}(x_j f) - x_j\left(\frac{\partial}{\partial x_i} f\right) = \frac{\partial x_j}{\partial x_i} f + x_j \frac{\partial f}{\partial x_i} - x_j \frac{\partial f}{\partial x_i} = \delta_{ij} f.$$

Choose an ordering $\{x_i\}_{i \in I}$ for the generators of a Lie algebra $\mathfrak{g}$. We recall that the PBW yields a basis for $U\mathfrak{g}$ given by monomials of the form $x_{i_1}, \ldots, x_{i_r}$, where $i_1 \leq \cdots \leq i_r$. We apply this to $\mathcal{H}_{\mathbb{R}^n}$. Let $x_1 \leq \cdots \leq x_n \leq \partial_1 \leq \cdots \leq \partial_n$. Then the PBW implies

$$U(\mathcal{H}_{\mathbb{R}^n}) \cong \left\langle x_1^{\alpha_1} \cdots x_n^{\alpha_n} \partial_1^{\beta_1} \cdots \partial_n^{\beta_n} \right\rangle$$

for all $\alpha_i, \beta_i \in \mathbb{N}$. Note that this is precisely $\mathscr{D}(\mathbb{R}^n)$ as a vector space – the x_is turn into the polynomial coefficients and the derivatives turn into the... derivatives. A fascinating fact is that due to the definition of the Lie bracket, we see that this promotes to an isomorphism of associative algebras. I think.

Now we follow this process more concretely. Recall that if (ψ_{-1}, ψ_0) and (ψ'_{-1}, ψ'_0) are compactly supported sections of the graded vector bundle E (not necessarily homogeneous elements!), then we defined our pairing

$$\left\langle (\psi_{-1}, \psi_0), (\psi'_{-1}, \psi'_0) \right\rangle = \int_{\mathbb{R}} \left(\psi'_{-1}\psi_0 - \psi'_0 \psi_{-1}\right) \mathrm{d}x.$$

We will show that $C_\bullet(\mathcal{H}_{\mathbb{R}})$ yields the quantum observables $\mathrm{Obs}^q(U)$ for the scalar field theory in 1 dimension.

We want to show the pFA $H^0(\mathrm{Obs}^q)$ valued in vector spaces is locally constant. We will filter the Chevalley-Eilenberg chains $C_\bullet(\mathcal{H}_{\mathcal{E}_c})$ by polynomial degree, and check that the associated graded is locally constant.

Lecture 15: The Weil algebra as the observables of the scalar field theory on $\mathbb{R}$

Fri 10 Apr 2026 14:00

The plan to finish this course is as follows. First, we will finish discussing FFTs. This includes Abelian Chern-Simons and correlation functions. Then, we will discuss holomorphic field theories and vertex algebras, showing that 1-complex dimensional CFT can be extracted from factorization algebras.

We now finish our discussion of scalar field theory on $\mathbb{R}$, and the Weil algebra. Recall that the Weil algebra on $\mathbb{R}^n$ is the associative $\mathbb{R}$-algebra of polynomial differential operators on $\mathbb{R}^n$, and is

$$\mathcal{W}_n \cong \left\langle x_1^{\alpha_1} \cdots x_n^{\alpha_n} \partial_1^{\beta_1} \cdots \partial_n^{\beta_n} \right\rangle.$$

Today, we work in $n = 1$. We recall that we have a sheaf of cochain complexes

$$\mathcal{E}(a,b): \qquad C^\infty(a,b) \xrightarrow{\Delta + m^2} C^\infty(a,b).$$

We get a corresponding cosheaf of compactly supported sections of the relevant graded vector bundle

$$\mathcal{E}_c(a,b): \qquad C_c^\infty(a,b) \xrightarrow{\Delta+m^2} C_c^\infty(a,b).$$

Then we showed that classical observables arise as

$$\mathrm{Obs}^{cl}(a,b) \cong \mathrm{Sym}\left(\mathcal{E}_c(a,b), \Delta+m^2\right).$$

Here we recall that $\Delta = -\partial_x^2$. Then we had that the physical classical observables arise from homology

$$H^0(\mathrm{Obs}^{cl}(a,b)) \cong \mathbb{R}[p,q].$$

There is a pairing on $\mathcal{E}_c$ which yieled the Heisenberg algebra

$$\mathcal{H}(a,b) = \mathcal{E}_c(a,b) \oplus (\mathbb{C}\hbar)[-1],$$

with bracket $[a,b] = \hbar\langle a,b\rangle$. Then we argued that the quantum observables arise as

$$\mathrm{Obs}^q(U) \cong C_\bullet(\mathcal{H}(U)) \cong \mathrm{Sym}\left(\mathcal{H}(U)[1], d_{\mathrm{CE}} + (\Delta+m^2)\right).$$

We then showed that $H^0(\mathrm{Obs}^q)$ is a locally constant pre factorization algebra valued in vector spaces, by constructing a filtration on Obs^q by symmetric degree, looked at the associated graded, and identified $\mathrm{gr}(\mathrm{Obs}^q(U)) \cong \mathrm{Obs}^{cl}(U)[\hbar]$. The point here is that the quantum case has an extra Chevalley-Eilenberg differential which encodes the $\hbar$-interaction. This then showed

$$H^0(\mathrm{Obs}^q(U)) \cong H^0(\mathrm{Obs}^{cl}(U))[\hbar].$$

Then to check local constancy, we need to check that inclusions of open intervals induce isomorphisms in cohomology. Since we aleady know this happens in classical observables, the statement follows.

Recall that locally constant factorization algebras on $\mathbb{R}$ correspond to associative algebras. We now consider what the relevant associative algebra we obtain is. To consider this, we first talk about derivations on prefactorization algebras. This is important since the algebras *should* be parameterized by mass, but the algebras turn out to all be isomorphic, regardless of m. To recover mass, we see that it shows up in the Hamiltonian, via derivations.

Definition 5.0.3. Let $\mathcal{F}$ be a prefactorization algebra on a manifold M valued in a concrete symmetric monoidal category. Then a *derivation* D of $\mathcal{F}$ is a collection of maps $D_U : \mathcal{F}(U) \to \mathcal{F}(U)$ for all open sets U satisfying the following property: For a structure map $m : \mathcal{F}(U_1) \otimes \cdots \otimes \mathcal{F}(U_r) \to \mathcal{F}(V)$,

$$D_V(m)(\alpha_1,\ldots,\alpha_r) = \sum_{j=1}^r (-1)^{\sum_{1\le i<j}|\alpha_i|} m\left(\alpha_1,\ldots D_{U_j}(\alpha_j),\ldots,\alpha_r\right).$$

In our FFT on $\mathbb{R}$, note that $[\partial_x, \Delta+m^2] = 0$. We claim that that this extend to a pFA derivation as follows: if $f_i \in C_c^\infty(U_1)$ and $g_i \in C_c^\infty(U_2)$. Let $\alpha = \prod_i f_i$ and $\beta = \prod_i g_i$. Then if $U_1, U_2 \subseteq V$ and $U_1 \cap U_2 = \varnothing$, then the derivation property implies that

$$\frac{\partial}{\partial x}(\alpha\beta) = \alpha\frac{\partial}{\partial x}\beta + \left(\frac{\partial}{\partial x}\alpha\right)\beta.$$

Thus ∂_x is a derivation on the prefactorization algebra $H^0(\mathrm{Obs}^q)$. It will be more convenient to work with the derivation $-\partial_x$ to get some signs right.

Now denote the associative algebra of $H^0(\mathrm{Obs}^q)$ by A_m. We will show that the derivation $-\partial_x$ on $H^0(\mathrm{Obs}^q)$ induces a derivation of the form $H_m = \frac{1}{2\hbar}(p^2 - m^2q^2)$.

Terminology 5.0.4. An *inner* derivation d on an associative R-algebra A is a map of the form $d = [x, -]$ for some $x \in A$, and $[-,-]$ the commutator. We usually write $d = D_x$.

Consider the following identity with the *commutator* on an associative algebra.

$$D_x(a_1 \cdots a_n) = [x, a_1 \cdots a_n] = \sum_{j=1}^{n} a_1 \cdots [x, a_j] \cdots a_n.$$

This shows that all inner derivations are derivations.

Proposition 5.0.5. *The associative algebra A_m is the Weil algebra $\mathcal{W}$ generated by $p, q, \hbar$, with the relation that $[p, q] = \hbar$, and all other brackets are 0. Additionally, the derivation corresponding to $-\partial_x$ is the inner derivation*

$$H_m = \frac{1}{2\hbar}\left(p^2 - m^2 q^2\right).$$

I believe there is a factor of i being eaten in q, hence the sign.

Recall that if $\mathfrak{g}$ is a Lie algebra, then for any open interval (a, b), we have a dgla $\mathfrak{g} \otimes \Omega_c^\bullet(a, b)$. Then we had a prefactorization algebra $\mathfrak{g}^{\mathbb{R}}$ mapping $(a, b) \mapsto C_\bullet(\mathfrak{g} \otimes \Omega_c^\bullet(a, b), d_{\mathrm{CE}} + \mathrm{d}_{\mathrm{dR}})$. We showed the cohomology of this was a locally constant prefactorization lgebra, and that the corresponding associative algebra was $U\mathfrak{g}$. In the proof, we assigned to each $x \in \mathfrak{g}$ a bump function $x \otimes f\mathrm{d}t$, and what we had to verify was that in cohomology, the commutators agreed with that of $U\mathfrak{g}$. In particular, $[x, y]_{\mathfrak{g}} \otimes f\mathrm{d}t = [x \otimes f\mathrm{d}t, y \otimes f\mathrm{d}t]_{U\mathfrak{g}}$. We did this by choosing 3 disjoint intervals, and working with their relations.

Our proof proceeds similarly. Recall

$$\mathrm{Obs}^q = \mathrm{Sym}\left(\; C_c^\infty(a, b) \xrightarrow{\Delta + m^2} C_c^\infty(a, b)[\hbar] \; \right),$$

with differential $d_{\mathrm{CE}} + (\Delta + m^2)$. We defined functions

$$\phi_q(x) = \frac{1}{2}\left(e^{mx} + e^{-mx}\right), \qquad \phi_p(x) = \frac{1}{2m}\left(e^{mx} - e^{-mx}\right),$$

both of which are killed by $\Delta + m^2$. We have $\partial_x \phi_p = \phi_q$. We note that ϕ_q is an even function and ϕ_p is odd. Choose an even function $f \in C_c^\infty(-1/2, 1/2)$ that satisfies

$$\int_{\mathbb{R}} f \phi_q \mathrm{d}x = 1.$$

It follows that

$$\int_{\mathbb{R}} f \phi_p \mathrm{d}x = 0.$$

Let $f_t(x) = f(x - t)$ be defined over $C_c^\infty(t - 1/2, t + 1/2)$. Then define observables

$$Q_t = f_t, \qquad P_t = -\partial_x f_t.$$

We are taking Q_t, P_t to be degree 0, meaning they sit in $C_c^\infty(a, b)[\hbar]$. To show these are observables, they need to be cocycles in the symmetric algebra. To see this, d_{CE} kills this since it takes brackets, and idk? (maybe since they only have 1 term?) But there is no higher differential in the original chain complex, so they die here as well. So these descend to cohomology classes.

Recall that the cohomology classes $[P_0]$ and $[Q_0]$ generate $\mathbb{R}[p, q]$. Since Obs^{cl} is the associated graded of Obs^q, it's also true that $[P_0], [Q_0]$ generate $H^0(\mathrm{Obs}^q(U))$. The book then proves $[[P_0], [Q_0]] = \hbar$. The proof is similar to that of $U\mathfrak{g}$.

Now we need to identify the Hamiltonian

$$H_m = \left[\frac{1}{2}[P_0]^2 - m^2[Q_0]^2, -\right]$$

with $-\partial_x$. To see this,

$$-\partial_x Q_t = -\partial_x f_t =: P_t = \frac{\mathrm{d}}{\mathrm{d}t} f(x-t) = \frac{\mathrm{d}}{\mathrm{d}t} Q_t.$$

The non-obvious step probably comes from even-ness or odd-ness, so something like $\partial_x f(x-t) = -\partial_t f(x-t)$. Maybe write $x(t)$ and $t(x)$ to show this. Also $-\partial_x P_t = \partial_x^2 f_t$.

Recall that in cohomology, the image of $\Delta + m^2$ is trivial (use $d_{CE} = 0$ on the relevant degree which i think is true here). We get that

$$(\Delta + m^2)[f_t] = 0 \implies -\partial_x^2[f_t] + m^2[f_t] = 0 \implies \partial_x^2[f_t] = m^2[f_t].$$

Thus

$$\frac{\mathrm{d}}{\mathrm{d}t}[P_t] = [\partial_x^2 f_t] = [m^2 f_t].$$

We define H_m to be the map

$$H_m[P_0] = m^2[Q_0], \qquad H_m[Q_0] = [P_0].$$

This is a definition, since $[P_0], [Q_0]$ are generators of $H^0(\mathrm{Obs}^q)$. If we assume part (1) [commutator identities], then we can show this is implemented precisely by the Hamiltonian H_m we found. It's enough to check

$$\frac{1}{2\hbar}\left[[P_0]^2 - m^2[Q_0]^2, [Q_0]\right] = [P_0],$$
$$\frac{1}{2\hbar}\left[[P_0]^2 - m^2[Q_0]^2, [P_0]\right] = m^2[Q_0], .$$

This is immediate by the commutator identities and $[x, x] = 0$. We check one, and use the identity for commutators on associative algebras:

$$\frac{1}{2\hbar}\left[[P_0]^2, [Q_0]\right] = \frac{1}{2\hbar}\left([P_0][[P_0], [Q_0]] + [P_0][[P_0], [Q_0]]\right) = \frac{1}{2\hbar} 2\hbar[P_0] = [P_0].$$

Using the idea that $-\partial_x = \partial_t$, we see that the Hamiltonian kinda corresponds to the generator of time evolution.

Lecture 16: Pushforwards of prefactorization algebras, Abelian Chern-Simons

Tue 14 Apr 2026 14:04

Remark 5.0.6. We identify p with ∂_x and q with x to recover $U\mathcal{W}$ from $H^0(\mathrm{Obs}^q)$.

Chapter 6

Pushforwards and canonical quantization

Definition 6.0.1. Let $\mathcal{F}$ be a prefactorization algebra on a manifold M, and $\pi : M \to N$ a smooth map between smooth manifolds. We define a prefactorization algebra $\pi_*\mathcal{F}$ on N the same way pushforwards of sheaves are defined: $(\pi_*\mathcal{F})(U) := \mathcal{F}(\pi^{-1}(U))$.

Lemma 6.0.2. *Withnotation as in definition 6.0.1, then $\pi_*\mathcal{F}$ is a prefactorization algebra.*

Proof. All we have to check is that structure maps exist. Since $\pi^{-1}(U) \subseteq \pi^{-1}(V)$ implies $U \subseteq V$, and $\pi^{-1}(U_1) \cap \pi^{-1}(U_2) = \varnothing$ implies $U_1 \cap U_2 = \varnothing$, the structure maps are precisely those of $\mathcal{F}$ on the open sets given by the preimages. ∎

Intuition 6.0.3. If $\mathcal{F}$ describes quantum observables on a theory M, then $\pi_*\mathcal{F}$ describes the observables of the theory on N compactified ($\leftarrow$ physics term, not math) along the fibers of π. Usually π is some kind of fibration here.

We will show that dimensional reduction to $\mathbb{R}$ amounts to canonical quantization.

Example 6.0.4. Let M a manifold and $Q = M \times \mathbb{R}$ the cylinder; the idea is $\mathbb{R}$ is time and M is space I think. Let $\pi : M \times \mathbb{R} \to \mathbb{R}$ be the projection. Suppose $M = (M, g)$ is compact and Riemannian. Consider the scalar field theory $\Delta + m^2 : C^\infty_{Q,c} \to C^\infty_{Q,c}[-1]$, and let $\{e_i\}_{i \in I \subseteq \mathbb{R}}$ be an orthonormal basis of eigenvectors of $\Delta + m^2$ on C^∞_M with eigenvalues $\lambda_i \geq 0$. We claim the eigenspaces are orthogonal with respect to the L^2 inner product. This follows since $\Delta + m^2$ is self-adjoint, and is not immediate.

There is a theorem saying that the direct sums of the eigenspaces $\bigoplus_i \mathbb{R}e_i$ is dense in C^∞_M. Writing A_m as the Weyl algebra with inner derivation depending on m, we consider

$$\mathbb{A}_M = A_{\sqrt{\lambda_1}} \otimes_{\mathbb{R}[\hbar]} A_{\sqrt{\lambda_2}} \otimes_{\mathbb{R}[\hbar]} \cdots .$$

To define such an infinite tensor product, we can take the complete tensor product—in all but finitely many slots, entries in a tensor are the identity 1 of the relevant unital $\mathbb{R}[\hbar]$-algebra.

Proposition 6.0.5. *There is a dense sub-prefactorization algebra of $\pi_* \operatorname{Obs}^q$ which is locally constant, and whose cohomology is the infinite tensor product $\mathbb{A}_M$ of Weyl algebras.*

Intuitively, this means scalar field theory is essentially composed of a bunch of non-interacting harmonic oscillators. Here, we have assumed compactness, meaning we can use less harmonic oscillators by assigning one to each oscillation mode, but on non-compact space we would need one in each point of spacetime since the Laplacian no longer has a discrete collection of eigenvalues. Of course, none of these interact.

Sketch of proof. First, recall $\pi_* \operatorname{Obs}^q$ assigns to each open $U \subseteq \mathbb{R}$ the Chevalley-Eilenberg chains of the Heisenberg central extension of $\mathcal{E}_c$, and a dense subcomplex is given by

$$\left(\bigoplus_{i \in I} \Delta + \lambda_i\right) : \bigoplus_{i \in I} C_c^\infty(U) e_i \to \bigoplus_{i \in I} C_c^\infty(U) e_i.$$

Let $\mathcal{F}_i$ be the prefactorization on $\mathbb{R}$ associated to quantum mechanics with (unknown word) $\sqrt{\lambda_i}$. This corresponds to $\mathcal{E}_c : C_c^\infty(U) e_i \to C_c^\infty(U) e_i[-1]$.

Prefactorization algebras can be tensored pointwise: Let $\mathcal{F}, \mathcal{G} : M \to (\mathcal{C}, \otimes)$; then $\mathcal{F} \otimes \mathcal{G}$ is given by $(\mathcal{F} \otimes \mathcal{G})(U) \coloneqq \mathcal{F}(U) \otimes \mathcal{G}(U)$. Consider $\mathcal{F} = \mathcal{F}_{\sqrt{\lambda_1}} \otimes \mathcal{F}_{\sqrt{\lambda_2}} \otimes \cdots$. Since we can define infinite tensor products pointwise, this exists. Then $\mathcal{F}$ can be viewed as the Heisenberg prefactorization algebra associate to the direct sum. There is a dense embedding $\mathcal{F} \hookrightarrow \pi_* \operatorname{Obs}^q$. Cohomology agrees also with the statement, and moreover the embedding in cohomology is still dense. ■

Chapter 7

Abelian Chern-Simons theory

Let M be a compact oriented 3-manifold. The space of fields $\mathcal{E} = \Omega^\bullet_M[1]$ with de Rham differential, and the action functional is

$$S_U(A) = \int_U A \wedge \mathrm{d}A.$$

For this to be meaningful, we need A to be a 1-form. General Chern-Simons takes values in an arbitrary Lie algebra

$$S(A) = \int_M \mathrm{tr}\left(A \wedge \mathrm{d}A + \frac{2}{3} A \wedge A \wedge A\right).$$

General Chern-Simons is not quadratic since the term $A \wedge A \wedge A$ scales with λ^3 under $A \to \lambda A$. But if we take the Lie algebra to be abelian, then $A \wedge A \wedge A = A \wedge [A, A] = 0$.

Abelian Chern-Simons is both a free and a topological field theory. Recall that a TQFT has no metric-dependence in the action functional. What we have here is $U(1)$-Chern-Simons using forms valued in $\mathbb{C}$, which is the Lie algebra of $U(1) \otimes_{\mathbb{R}} \mathbb{C}$.

The pairing of cohomological degree -1 on $\mathcal{E}_c$ is

$$\langle \alpha, \beta \rangle = (-1)^{|\beta|} \int_M \alpha \wedge \beta.$$

We as always view 1-forms as connections on the trivial bundle via $A \to \mathrm{d} + A$. Then $S(A) = \langle A, \mathrm{d}A \rangle$. The equations of motion are that $\mathrm{d}A = 0$. This is as always computed by taking

$$\frac{\delta S}{\delta \alpha}(\beta) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_{\varepsilon=0} \int_U (\alpha + \varepsilon\beta) \wedge (\mathrm{d}\alpha + \varepsilon \mathrm{d}\beta) = \int_M \beta \wedge \mathrm{d}\alpha + \alpha \wedge \mathrm{d}\beta$$

$$= 2\int_M \beta \wedge \mathrm{d}\alpha - \mathrm{d}(\alpha \wedge \beta) = 2\int_M \beta \wedge \mathrm{d}\alpha = 0 \implies \mathrm{d}\alpha = 0.$$

Now note that $\Omega^1_M \to \Omega^2_M \to \cdots$ is an exact sequence of sheaves by Poincare, and if we want to extend it to the left keeping it exact then we would add in $\Omega^1_{M,\text{closed}}$. However, we are using a *gauge theory* where transformations $A \to A + \mathrm{d}f$ leave the action functional invariant:

$$S(A + \mathrm{d}f) = \langle A + \mathrm{d}f, \mathrm{d}(A + \mathrm{d}f) \rangle = \langle A, \mathrm{d}A \rangle + \langle \mathrm{d}f, \mathrm{d}A \rangle = S(A) + \int_M \mathrm{d}f \wedge \mathrm{d}A$$

$$= S(A) + \int_M \mathrm{d}(f + \mathrm{d}A) = 0.$$

Thus our symmetry is encoded by Ω^0_M, motivating our space of fields $\Omega^\bullet_M[1]$. (again, the shift is so that the physical fields live in H^0). Thus we have our derived space of fields

$$\Omega^0_M[1] \xrightarrow{\mathrm{d}} \Omega^1_M \xrightarrow{\mathrm{d}} \Omega^2_M[-1] \xrightarrow{\mathrm{d}} \Omega^3_M[2].$$

The classical observables are (the

$$\mathrm{Obs}^{cl} = \mathrm{Sym}(\mathcal{E}_c[1]),$$

with de Rham differential. Sym commutes with cohomology, so

$$H^\bullet \,\mathrm{Sym}(\mathcal{E}_c[1]) = \mathrm{Sym}\, H^\bullet(\mathcal{E}_c[1]) = \mathrm{Sym}(H^\bullet_{\mathrm{dR},c}(U)[2]).$$

Globally, since M is compact, $H^\bullet \,\mathrm{Obs}^{cl}(M) = \mathrm{Sym}(H^\bullet_{\mathrm{dR}}(M)[2])$.

Lecture 17: Continuing abelian Chern-Simons?

Fri 17 Apr 2026 13:57

Now we look at what happens on the level of quantum observables

$$\mathrm{Obs}^q := C_\bullet(\mathcal{H}_\mathcal{E}).$$

To construct the Heisenberg extension, we can use the skew pairing $\langle -,-\rangle : \mathcal{E}_c \otimes_\mathbb{R} \mathcal{E}_c \to \mathbb{R}$ is

$$\langle \alpha, \beta\rangle_U = \int_U \alpha \wedge \beta.$$

Of course, this vanishes if $\alpha \wedge \beta \notin \Omega^3_M(U)$. Due to our degree shifting, this has cohomological degree -1. This also agrees with the canonical skew pairing.

We use this to construct the Heisenberg extension $\mathcal{H}_\mathcal{E} := \mathcal{E}_c \oplus \mathbb{C}\hbar[-1]$. We can filter $C_\bullet(\mathcal{E}\oplus\mathbb{C}\hbar[-1])$ by symmetric degree to obtain a filtration. The cohomology of a filtered complex can be computed by a spectral sequence, which proceeds via a series of approximations (called *pages*). In good situations, this gives the associated graded of the cohomology of the complex.

To see how this arises, let $(C^\bullet, d)$ be a cochain complex of tensors of a Lie algebra $\mathfrak{g}$, like with our situation. with ascending filtration $F^\bullet C$. We will take the filtration to start at $F^{-1}C \cong 0$. We define $F^0 C \cong \mathbb{C}$ (the tensors spanned by 0 or fewer terms in a Lie algebra are constants). Then $F^1 C \cong \mathbb{C} + \mathfrak{g}$, and so on. This induces a filtration $F^\bullet H^\bullet(C^\bullet)$ on the cohomology. The spectral sequence provides a sequence of complexes $E^{n\geq 0}C^\bullet$, where $E^i C^\bullet$ is the cohomology $H^\bullet(E^{i-1}C^\bullet)$. Assuming $C^\bullet$ is sufficiently bounded, we find that $E^\infty \cong \mathrm{gr}(H^\bullet(C^\bullet))$. This final object is not canonically $H^\bullet(C^\bullet)$, but if we are working with vector spaces over a field, then they are non-canonically isomorphic. This at least allows us to extract dimension as an integer invariant. If we work with coefficients in something other than a field, it can be difficult to lift information from the associated graded to the original cohomology. This arises often in algebraic topology.

We will apply this here to compute $H^\bullet(C_\bullet(\mathcal{E}_c \oplus \mathbb{C}\hbar[-1]))$. Define a filtration $E_0 = C_\bullet(\mathcal{H}_\mathcal{E})$ with differential $\mathrm{d}_{\mathrm{CE}} + \mathrm{d}_{\mathrm{dR}}$. The first pageis $E_1 = H^\bullet(\mathrm{gr}(E_0))$ with whatever survives the differential. Since the Chevalley-Eilenberg differential always lowers symmetric degree, it vanishes at the level of the associated graded, so we recover $H^\bullet(\mathrm{Sym}(\mathcal{E}_c)[\hbar]) \cong H^\bullet(\mathrm{Obs}^{cl}[\hbar])$ with the de Rham differential. Then E_2 is $H^\bullet(\mathrm{Obs}^{cl}[\hbar])$, with differential induced by the Chevalley-Eilenberg differential via the cup product. We find that the induced differential is

$$\mathrm{d}([\alpha],[\beta]) = \hbar \left\langle [\alpha],[\beta]\right\rangle_{H^\bullet}.$$

In this good case, since the filtration has very trivial cohomology, $H^\bullet(\mathrm{Obs}^q(M)) \cong \mathbb{R}[\hbar][1 - \dim H^2(M)]$ is canonical. The first bracket is polynomials and the second is a shift. For more details, see the book.

7.1 Compactifying along a closed surface

Consider abelian Chern-Simons on a 3-manifold $\Sigma \times \mathbb{R}$, where Σ is a genus g closed surface (2-manifold). We will argue that $H^\bullet(\pi_* \operatorname{Obs}^q)$ is a type of Weyl algebra modeled on $H^\bullet(\Sigma, \mathbb{R})[1]$. First, note that $H^\bullet(\Sigma, \mathbb{R})$ is a graded symplectic vector space. It has no more than three nonvanishing cohomology groups, $H^{0,1,2}$. The bottom and top cohomologies are $\mathbb{R}$, and H^1 is $\mathbb{R}^{2g}$. To see this, note that each hole determines two 1-cycles (think of the two canonical homotopy classes of loops in $S^1 \times S^1$). For the bottom homology, we simply note path-connectedness. There is a nondegenerate skew-pairing (symplectic form) on $H^1(\Sigma)$ (I think by multiplying in the cochains?), and since we shift down by 1, this extends to a graded symplectic form (on the odd degrees, it means symmetric). Choose a basis $v = 1 \in H^0(\Sigma, \mathbb{R})$ and $\mu = [\Sigma] \in H^2(\Sigma, \mathbb{R})$. The cup product turns $H^\bullet(\Sigma, \mathbb{R})[1]$ into a graded symplectic vector space.

If W is graded symplectic, then $W \oplus \mathbb{C}\hbar[-1]$ is a Lie algebra with bracket given by $\hbar$ times the symplectic form $[u, v] = \hbar \langle u, v\rangle$. We can thus associate a Weyl algebra $\operatorname{Weyl}(W) := U(W \oplus \mathbb{C}\hbar[-1])$ to the graded symplectic space. Note how this echoes what we have been doing with the Heisenberg central extensions.

The classical Weyl algebra $\mathbb{R}\partial_x \oplus \mathbb{R}x$ with $\langle \partial_x, x\rangle = 1$ is a special case of this by playing with indicies and taking $\hbar = 1$.

Proposition 7.1.1. *The prefactorization algebra $H^\bullet(\pi_* \operatorname{Obs}^q))$ on $\mathbb{R}$ is locally constant, and corresponds to the Weyl algebra A_Σ modeled on $H^\bullet(\Sigma, \mathbb{R})[1]$. This Weyl algebra has brackets*

$$[v, \mu] = \hbar, \qquad [\alpha_i, \beta_j] = \hbar\delta_{ij}.$$

Proof. Start on $\Sigma \times \mathbb{R}$. The complex of fields is $\mathcal{E} = (\Omega^\bullet_M[1], \mathrm{d})$. To push forward, we can push forward the complex of fields and then find the quantum observables thereof. Let $\mathcal{L} = \pi_*\Omega^\bullet_M[1]$, which is an abelian dgla. Then $\pi_* \operatorname{Obs}^{cl} \cong C_\bullet * \mathcal{L}_c)$, and $\pi_* \operatorname{Obs}^q \cong C_\bullet(\widehat{\mathcal{L}}_c)$, where $\widehat{\mathcal{L}}$ denotes the Heisenberg central extension.

The pushforward $\mathcal{L}_c$ is defined on $U \subseteq \mathbb{R}$ by taking $\Omega^\bullet_{M,c}(\pi^{-1}(U))[1]$. Using $\pi^{-1}(U) \cong \Sigma \times U$, we find

$$\Omega^\bullet_M(\pi^{-1}(U)) \cong \pi^\bullet_\Sigma(M) \otimes \Omega^\bullet_\mathbb{R}(U).$$

The tensor product here is completed. Since Σ is a compact Riemann surface (when did we say that?), which is Kähler, it has been proven that $\Omega^\bullet_\Sigma(\Sigma)$ is *formal*. This means $\Omega^\bullet_\Sigma(\Sigma)$ is quasi-isomorphic as a commutative differential graded algebra to its singular cohomology. Thus up to quasi-isomorphism,

$$\mathcal{L}(U) \simeq \Omega^\bullet_\mathbb{R} \otimes H^\bullet(\Sigma, \mathbb{R})[1].$$

Compactly supported sections are thus

$$\mathcal{L}_c(U) \simeq \Omega^\bullet_{\mathbb{R},c} \otimes H^\bullet(\Sigma, \mathbb{R})[1].$$

Thus

$$\operatorname{Obs}^{cl} \simeq C_\bullet(\Omega^\bullet_{\mathbb{R},c} \otimes H^\bullet(\Sigma, \mathbb{R})[1]).$$

The quantum observables are given by taking a central extension of the inside of the chains. We have seen this earlier in the course:

$$H^\bullet(\Omega^\bullet_M \otimes \mathfrak{g}) \cong U\mathfrak{g}.$$

Doing this here gives us the universal enveloping algebra on the Weyl algebra $H^\bullet(\Sigma, \mathbb{R})[1]$. ∎

Intuitively, this tells us that pushing down a theory to $\mathbb{R}$ gives us some notion of quantum mechanics, which always gives us some type of Weyl algebra since it is rigid. Usually the Weyl algebra is modeled on the cohomology of the above space.

Remark 7.1.2. More general Chern-Simons theory has many applications to low dimensional topology and knot theory. Abelian Chern-Simons can show you some of this information by showing us the linking number of a knot. Recall that a knot is an embedding $K : S^1 \hookrightarrow \mathbb{R}^3$ (or sometimes more general 3-folds, like S^3). Givn two disjoint knots κ, κ', their *linking number* $\ell(\kappa, \kappa')$ measures how many times κ' winds around κ. The way of rigorizing this notion is as follows. Consider the product of these two knots $\kappa \times \kappa' \cong S^1 \times S^1$. Since the torus embeds into $\mathbb{R}^3$, we obtain an embedding $\kappa \times \kappa' \subseteq \mathbb{R}^3$. We also have a smooth map $S^1 \times S^1 \to S^2$ given by the Gauss map $(x, y) \mapsto \frac{x-y}{|x-y|}$ in coordinates. Then the linking number $\ell(\kappa, \kappa')$ is the *degree* of the Gauss map $\kappa \times \kappa' \to S^2$. More precisely, we find that G is multiplication by a number, and that number is the linking number.

To compute this geometrically, we can find a filler

$$\begin{tikzcd} S^1 \arrow[r, "\kappa"] \arrow[d, hook] & \mathbb{R}^3 \\ D \arrow[ur, "\widetilde{\kappa}"'] & \end{tikzcd}$$

for a knot since $\mathbb{R}^3$ is a CW complex. We then find the number of times κ' intersects the filler $\widetilde{\kappa}$. This can actually be found by looking at the factorization algebra.

Lecture 18: stuff

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the pairing on sctions is compatible wih the differential: $\int_M \langle \mathrm{d}f, g\rangle = \int_M -\langle f, \mathrm{d}g\rangle$.

Recall $\mathrm{Obs}^{cl}(U) = \mathrm{Sym}(\mathcal{E}_c(U)[1])$. We claim this is valued in $\mathbb{P}_0$-algebras (better than cdgas): there is a bracket $\{-,-\}$ of degree 1 s.t. Jacobi is satisfied and it is a biderivation for the product (think Poisson). Let $\alpha, \beta \in \mathcal{E}(U)[1]$. Define $\{\alpha, \beta\} = \int_U \langle \alpha, \beta\rangle$. This is the bracket on the Heisenberg central extension

Recall the quantum observables Obs^q are the quantization of Obs^{cl} in the BV sense:

$$\mathrm{Obs}^q(U) = (\mathrm{Sym}(\mathcal{E}_c(U)[1])[\hbar], d_{cl} + \hbar\Delta)\,.$$

Here d_{cl} is the differential for classical observables, and $\Delta|\,\mathrm{Sym}^1 = 0$, $\Delta(\alpha\beta) = \{\alpha, \beta\}$. This is called the BV Laplacian. This extends by acting as a sort of derivation (not fully since it kills single observables): for any O_1, O_2:

$$\Delta(O_1 O_2) = \Delta(O_1)O_2 + (-1)^{|O_1|}O_1\Delta(O_2) + \{O_1, O_2\}\,.$$

Good exercise to check that this is $\mathrm{d_{CE}}$. There is a dequantization map of FAs $\mathrm{Obs}^q \to \mathrm{Obs}^{cl}$ via $\hbar \mapsto 0$. This is "BV quantization" which is different from most types of quantization, and basically amounts to defining Δ.

We now interpret quantization in a more physical way. To motivate this, consider the scalar free field theory on a Riemannian manifold (M, g). If M is compact, then the cohomology $H^\bullet(\mathcal{E}(M), \Delta)$ is concentrated in degree 0. Harmonic functions on a compact manifold are constant functions. If the differential is $\Delta + m^2$ then the cohomology fully vanishes.

To see this, let $(\Delta + m^2)u = 0$. Then

$$0 = \int_M u(\Delta + m^2)u\mathrm{vol}_g = \int_M u\Delta u\mathrm{vol}_g + m^2 \int_M u^2\mathrm{vol}_g = \int_M |\nabla u|^2 \mathrm{vol}_g + m^2 \int_M u^2\mathrm{vol}_g.$$

Here ∇ is the Levi-Civita connection. These functions are both always nonnegative. If $m \neq 0$ then there is no way of killing u^2 unless $u = 0$. This proves that cohomology is nonvanishing iff $m = 0$.

A Green's function for Δ is $G \in \mathcal{D}(M \times M)$ s.t. $(\Delta \otimes 1)G(x, y) = \delta(x - y)$. If we had mass, it would be $(\Delta \otimes 1)G + m^2G = \delta_{\mathrm{diag}}$. These exist.

We want to generalize a Green's function to fre field theories $(\mathcal{E}, \mathrm{d}, \langle -, - \rangle)$. We define it to be a map

$$G : \mathcal{E}_c[1]^{\otimes 2} \to \mathbb{R}$$

which is graded symmetric, and satisfies

$$\mathrm{d}G(\phi_1 \otimes \phi_2) = \int_M \langle \phi_1, \phi_2 \rangle .$$

To see why this is useful, consider the *contraction* operator $\partial_G : \mathrm{Sym}(\mathcal{E}_c(U)[1]) \to \mathrm{Sym}(\mathcal{E}_c(U)[1])$ given by summing over all possible ways of evaluating G on two terms in a tensor. Note that G has degree 0, thus so does ∂_G. Thus it's not really a differential. We can extend this to a map on $\mathrm{Sym}(\mathcal{E}_c(U)[1])[\hbar]$ which preserves $\hbar$, and then we define the invertible operator $e^{\hbar\partial_G}$. First, note that $[\partial_G, d] = \Delta_{\mathrm{BV}}$. Thus $[\partial_G, \Delta_{\mathrm{BV}}] = 0$. Consider $\alpha \in \mathrm{Obs}^q(U)$ a homogeneous element, and

$$(\mathrm{d} + \hbar\Delta)e^{\hbar\partial_G}(\alpha) = e^{\hbar\partial_G} \left(e^{-\hbar\partial_G} (\mathrm{d} + \hbar\Delta)\, e^{\hbar\partial_G} \right)(\alpha).$$

Using the commutators, this is

$$e^{\hbar\partial_G}(\mathrm{d}\alpha).$$

Ok, what does this tell us? Well, it tells us that we can construct a cochain isomorphism

$$\mathrm{Obs}^{cl}[\hbar] \to \mathrm{Obs}^q .$$

We just checked that it respects the differentials $(\mathrm{d} + \hbar\Delta_{\mathrm{BV}}) \circ e^{\hbar\partial_G} = e^{\hbar\partial_G} \circ \mathrm{d}$. However, this *does not preserve* the factorization algebra structure *or* the $\mathbb{P}_0$-structure. However, since we have a pointwise map of dg modules, we can try and move the factorization product from Obs^{cl} to Obs^q.

First, define it on $\mathrm{Obs}^{cl}[\hbar]$. Let $\alpha \in \mathrm{Obs}^{cl}(U)[\hbar]$ and $\beta \in \mathrm{Obs}^{cl}(V)[\hbar]$ with $U \cap V = \varnothing$. We define the factorization product

$$\alpha \star_\hbar \beta := e^{-\hbar\partial_G} \left(e^{\hbar\partial_G}(\alpha) e^{\hbar\partial_G}(\beta) \right).$$

The idea is that we send this up to Obs^q, compute the factorization product, and pull it back. This generalizes to k-fold product. We claim

Theorem 7.1.3. *There is an isomorphism of factorization algebras* $(\mathrm{Obs}^{cl}[\hbar], \mathrm{d}, \star_\hbar) \cong \mathrm{Obs}^q$.

This is a new approach to quantization which does not involve the BV Laplacian. We consider an example. Let $\mathcal{E}$ be the field theory for abelian Chern-Simons $\mathcal{E} = \Omega^\bullet_{\mathbb{R}^3}[1]$. The pairing is $\langle \alpha, \beta \rangle = \alpha \wedge \beta$. Consider a map $(\mathbb{R}^3 \times \mathbb{R}^3) \setminus \langle (x, x) \rangle \to S^2$ given by $(x, y) \mapsto \frac{x-y}{|x-y|}$. Call this map ψ, and let $\omega := \psi^\alpha \mathrm{vol}_{S^2}$. Check the picture – the superscript is the coordinate.

It's a 2-form which defines a distribution $G_\omega : \Omega^\bullet_c(\mathbb{R}^3)^{\otimes 2} \to \mathbb{R}$ given by

$$\alpha \otimes \beta \mapsto \int_{\mathbb{R}^3 \times \mathbb{R}^3} \alpha(x)\beta(y)\omega(x, y).$$

(useful to note the diagonal has measure 0). This is a Green's function for d_{dR}.

Let K, K' be knots in $\mathbb{R}^3$. Let T, T' be open tubular (diffeo to $S^1 \times \mathbb{R}^2$) neighborhoods of K, K'. Consider CS observables $\mu_K \in \mathrm{Obs}^{cl}(T)$ and $\mu_{K'} \in \mathrm{Obs}^{cl}(T')$. These are defined by

$$\mu_K(\alpha) = \int_K \alpha, \qquad \mu_{K'}(\alpha) = \int_{K'} \alpha.$$

(Here we have $\mathcal{E}_c \simeq \overline{\mathcal{E}_c} \cong \mathcal{E}^\vee$?) Next time we will see

$$\mu_K \star_\hbar \mu_{K'} = \mu_K \mu_{K'} + \hbar \ell(K, K')$$

for ℓ the linking number.

Lecture 19: More stuff

Fri 24 Apr 2026 13:58

Let $K : S^1 \to \mathbb{R}^3$ be a knot. A very basic type of observable in CS is defined by

$$O_K : \Omega^1_{\mathbb{R}^3}(\mathbb{R}^3) \ni A \mapsto \int_{K(S^1)} K^* A.$$

Recall here that observables in CS are 1-forms. This observable is supported along $K(S^1)$, which is a closed submanifold of $\mathbb{R}^3$. Factorization algebras use open submanifolds, so to do this we extend it to an open subset. Additionally, recall

$$\mathrm{Obs}^{cl}(U) \cong \mathrm{Sym}\left(\Omega^\bullet_c(U)[2], \mathrm{d}\right).$$

These aren't really observables in the sense of physics. We need to find a way to turn these into some kind of functions on the fields. To fix this, we recall that $\mathrm{Obs}^{cl} := \mathrm{Sym}\,\mathcal{E}^\vee$. This is a perfectly fine definition. However, this does not admit an easy quantization: the BV laplacian is distributional (wedge fields and integrate over a diagonal), so it's kind of hard to act distributions on distributions. However, since $\mathcal{E}(U) = \Gamma(U, E)$, then we can think of $\mathcal{E}(U)^\vee$ as distributional sections $\overline{\Gamma}_c(U, E^*)$ with compact support. Then $\Gamma_c(U, E^!) \subseteq \overline{\Gamma}_c(U, E^!)$. As seen previously, this is a quasi-isomorphism when $(\mathcal{E}, \mathrm{d})$ is elliptic. This is more properly how to do observables in QFT. This follows by $E^! \cong E[1]$ by the BV pairing. Hence this promotes to

$$\mathrm{Obs}^{cl} := \mathrm{Sym}\,\mathcal{E}^\vee \simeq \mathrm{Sym}(\mathcal{E}_c[1]).$$

It should be clear that this quasi-isomorphism is given by integrating against it.

Remark 7.1.4. If the field theory is not free, then there is another term in the differential on $\mathrm{Sym}\,\mathcal{E}^\vee$. If $\langle -, - \rangle$ is the BV pairing, then the action is $S = \langle \phi, \mathrm{d}\phi \rangle$. Then $S \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E}) \subseteq \mathcal{O}(\mathcal{E}_c)$. However if the theory is *interacting* , then there is another local functional which is at least cubic, meaning

$$S = \langle \phi, \mathrm{d}\phi \rangle + I(\phi) \implies I \in \mathcal{O}^{\geq 3}_{\mathrm{loc}}(\mathcal{E}).$$

The true differential then requires the BV bracket $\mathrm{d}_{\mathrm{free}} + \{I, -\}$. It is hard to then translate $\{I, -\}$ to $\mathrm{Sym}(\mathcal{E}_c[1])$, so we can't really appeal to smeared observables, even in the classical sense. "Interactions don't preserve smearedness."

Back to knots. Write Obs_s for smeared observables and Obs^{cl} for normal observables. If U is an open set with $K(S^1) \subseteq U$, then we have that $O_K \in \mathcal{O}^{cl}(U)$. The observable is closed by Stokes' theorem. Thus we should be able to find a representative as a smeared observable. The systematic way to do this is to start with a torus $T := S^1 \times \mathbb{R}^2$. We then view $K(S^1) \times 0 \subseteq S^1 \times \mathbb{R}$ by embedding $T \mapsto \mathbb{R}^3$. Let κ be the embedding and π the projection to $\mathbb{R}^2$. Fix a compactly supported 2-form $\mu \in \Omega^2_c(\mathbb{R}^2)$ such that $0 \in \operatorname{supp} \mu = 1$ and $\int_{\mathbb{R}^2} \mu = 1$. Then $[\mu]$ generates $H^2_c(\mathbb{R}^2) \cong \mathbb{R}^2$ (Poincare lemma).

Let $\mu_T = \pi^*\mu$. So we are pulling a bump function back into the torus. This gives us a smoothed observable O_μ which takes the gauge field $A \in \Omega^1$, and integrates

$$A \mapsto \int_{\mathbb{R}^3} A \wedge \mu_T.$$

We claim this is a representative of O_K as a smeared observable. This is how we make sense of the quasi-isomorphism. Then since μ_T is compactly supported, it pushes forward to $\mu_K := \kappa_* \mu_T$.

All of these things have degree 0, so we can look at the cohomology class of a product of observables

$$[\mu_K \mu_{K'}] \in H^0(\mathrm{Obs}^q(\mathbb{R}^3)) \cong \mathbb{R}[\hbar].$$

Thus essentially this is just a number times $\hbar$. We claim that it is represented by $\hbar \ell(K, K')$.

We will show this using the factorization product $\star_\hbar$ we found last time. We will use the Green's function for $\mathbb{R}^3$ minus the diagonal. To construct this, let

$$j : \mathbb{R}^3 \times \mathbb{R}^3 \to S^2$$

map

$$(x, y) \mapsto \frac{x - y}{|x - y|}.$$

Define

$$G(\alpha \otimes \beta) = \lim_{\varepsilon \to 0} \int_U \alpha(x)\beta(y)\omega(x, y), \qquad \omega = j^* \mathrm{vol}_{S^2}.$$

This is a Green's function for d_{dR} when $|x - y| > \varepsilon$. ω is kinda like the residue of the 2-sphere. Then the linking number is the cohomology class of the factorization product

$$[\mu_K \star \mu_{K'}] = \mu_K \mu_{K'} + \hbar \int_{\mathbb{R}^3 \times \mathbb{R}^3} \mu_K(x)\mu_{K'}(x)\omega(x, y).$$

Note that $\mu_K \mu_{K'}$ is 0 in cohomology, so we just have the integral.

We will now discuss *translation invariant* factorization algebras on $\mathbb{R}^n$. This, unlike locally constantness, is more physically interesting. First we use a "discrete version" which does not fully capture the info. Let $x \in \mathbb{R}^n$ and $U \subseteq \mathbb{R}^n$ open. We can translate U by x by mapping

$$U \mapsto \tau_x U := \{y \in \mathbb{R}^n \mid y - x \in U\} = U + x.$$

Definition 7.1.5. A prefactorization algebra $\mathcal{F}$ on $\mathbb{R}^n$ is *discretely translation invariant* if $\mathcal{F}(U) \cong \mathcal{F}(\tau_x U)$ for all $x \in \mathbb{R}^n$ and all open sets $U \subseteq \mathbb{R}^n$. These isomorphisms should be compatible with the factorization products.

Remark 7.1.6. Every pFA that we have seen so far is discretely translation invariant.

This isn't quite right. We also want the factorization *products* to be controlled in a nice way under translations. To do this, we consider a derivation of the prefactorization algebra. Recall a degree k dervation is a map $D : \mathcal{F}(U) \to \mathcal{F}(U)[k]$ for all U which acts as a derivation on the structure maps. Then $\operatorname{Der}(\mathcal{F})$ is a dgla with bracket given by commutator.

Note that if $\mathfrak{g}$ is a dg Lie algebra, then $\mathfrak{g}$ can act on a pFA by $\mathfrak{g} \to \operatorname{Der}(\mathcal{F})$.

Definition 7.1.7. A *smooth* translation invariant prefactorization algebra $\mathcal{F}$ on $\mathbb{R}^n$ is a discrete prefactorization algebra together with a representation

$$\rho : \mathbb{R}^n_{\text{trans}} \to \operatorname{Der}(\mathcal{F}).$$

Here note that the Lie algebra of (infintesimal?) translations $\mathbb{R}^n_{\text{trans}}$ is abelian n-dimensional, concentrated in degree 0.

Remark 7.1.8. If $\mathcal{F}$ is a pFA valued in symmetric monoidal category, then we look at Lie algebras in that symmetric monoidal category. For example, valued in spaces.

For example, CS provides that the Lie derivative $\mathcal{L}_{\partial_x} : \mathbb{R}^3_{\text{trans}} \to \operatorname{Der}\operatorname{Obs}^q_{CS}$. Translation invariant pFAs can often be easier to describe. In particular, they are algebras over a much simpler colored operad (built from disks of various radii) than usual pFAs.

Lecture 20: Translation invariance

Tue 28 Apr 2026 14:03

Let $\{U_i \subseteq V\}_{1 \le i \le k}$ be disjoint open subsets of an open set V in M. Define $\mathcal{D}\mathrm{isj}(U_1, \ldots, U_k; V)$ to be the set of k-tuples $(x_1, \ldots, x_k) \in (\mathbb{R}^n)^k$ such that $\tau_{x_1}(U_1), \ldots, \tau_{x_k}(U_k)$ are still disjoint and are contained in V. If the closures of the U_i's are still disjoint, then $\mathcal{D}\mathrm{isj}(U_1, \ldots, U_k; V)$ is an open subset of $(\mathbb{R}^n)^k$.

Let $\mathcal{F}$ be a discretely translation invariant prefactorization algebra. Then for any $(x_1, \ldots, x_k) \in \mathcal{D}\mathrm{isj}(U_1, \ldots, U_k; V)$, there is a multilinear map

$$m_{(x_1,\ldots,x_k)} : \bigotimes_{1 \le i \le k} \mathcal{F}(U_i) \to \bigotimes_{1 \le i \le k} \mathcal{F}(\tau_{x_i} U_i).$$

There is some level of sophistication required to address some analytical issues that we are ignoring.

Definition 7.1.9. A discretely translation invariant prefactorization algebra $\mathcal{F}$ is *smoothly translation invariant* if

1. The map above depends smoothly on $(x_1, \ldots, x_k) \in \mathcal{D}\mathrm{isj}(U_1, \ldots, U_k; V)$ (this requires DVS machinery).

2. Since $\mathbb{R}^n$ is its own (abelian) Lie algebra, $\mathcal{F}$ has an action $\rho : \mathbb{R}^n \to \operatorname{Der}(\mathcal{F})$. For all $v \in \mathbb{R}^n$, denote by $\frac{\mathrm{d}}{\mathrm{d}v}$ the corresponding derivation, which acts $\mathcal{F}(U) \to \mathcal{F}(U)$.

3. The infinitesimal action of $\mathbb{R}^n$ is given by differentiating the global action. More precisely, if $v \in \mathbb{R}^n$, let $v_i = (0, \ldots, v_i, \ldots, 0)$ in $(\mathbb{R}^n)^k$. We want

$$\frac{\mathrm{d}}{\mathrm{d}v} m_{(x_1,\ldots,x_k)}(\alpha_1, \ldots, \alpha_k) = m_{x_1,\ldots,x_k)} \left(\alpha_1, \ldots, \frac{\mathrm{d}}{\mathrm{d}v_i} \alpha_i, \ldots, \alpha_k \right).$$

For example, for the frfee scalar field theory on $\mathbb{R}$, we have $\mathbb{R}$ acting on itself by $x \mapsto \frac{\mathrm{d}}{\mathrm{d}x} \mapsto [H, -]$ for H the Hamiltonian. This makes sense since the Hamiltonian generates time evolution in qm: $U = \exp(iH/\hbar)$ or smt.

This is where the operadic perspective becomes useful. Define $\mathcal{D}\mathrm{isj}$ to be the colored operad with colors $\mathcal{O}\mathrm{pen}(M)$, and n-multimorphisms as defined earlier.

We now define the little disks operad, which has one color. Define $\mathcal{O}_k(n)$ to be all possible configurations of n different ordered k-balls with disjoint closures inside of the unit k-ball. Composition can be given by shrinking down one of the unit ball and inserting it into one of the inner balls. Note that unlike arbitrary factorization algebras, this operad $\mathcal{O}$ does not care about the size of things. The ones that satisfied this are the locally constant factorization algebras. Now the following is not adequate if we care about some kind of weak equivalence i think. BUT. Locally constant prefactorization algebras on $\mathbb{R}^n$ are algebras for the little n-disks operad $\mathcal{O}_n$. This is called an E_n-algebra.

Theorem 7.1.10 (Stasheff, May)**.** *A space X has the homotopy type of the n-fold loop space of a manifold L^nM if and only if X is an algebra over the little n-disk operad.*

The n-fold loop space is $\mathcal{T}\mathrm{op}(S^n, M)$. Not hard to see this if we note that $S^n = B_n/\partial B_n$, B_n an n-ball.

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